Volume 23 • October/December 2017 • Number 4 • ISSN (print) 1512-5866
ISSN (online) 2232-8

MEDICAL JOURNAL
MEDICINSKI ŽURNAL

Journal of the Institute for Research and Development
Clinical Center University of Sarajevo
Online first - www.kcus.ba



KLINIČKI CENTAR UNIVERZITETA SARAJEVO

INSTITUT ZA NAUČNO ISTRAŽIVAČKI RAD I RAZVOJ

Bosnia and Herzegovina was the fourth country in Europe that developed National version of HeartScore program !

Bosna i Hercegovina je bila četvrta zemlja u Evropi koja je razvila Nacionalnu verziju HeartScore programa !



Bosnia and Herzegovina version of HeartScore is developed on the languages of the people of Bosnia and Herzegovina i.e. Bosnian, Serbian and Croatian! Program is easy to use and accessible at www.heartscore.org/eu !

Verzija za Bosnu i Hercegovinu razvijena je na jezicima naroda Bosne i Hercegovine, bosanskom, srpskom i hrvatskom! Program je jednostavan za upotrebu preko web stranice www.heartscore.org/eu !

Bosnia Herzegovina	France	Russian Federation
Croatia	Germany *	Spain *
Cyprus *	Greece *	Sweden *
Czech Republic *	Poland *	Slovakia *
Estonia	Romania	Turkey





Novi Centralni medicinski blok - Klinički centar Univerziteta u Sarajevu
New Central Medical Building - Clinical Center University of Sarajevo





Novi Evropski vodič za prevenciju tromboembolizma kod A Fib

CHA_2DS_2-VASc skor za procjenu rizika od tromboembolizma kod A Fib!

Risk factor-based point-based scoring system - CHA_2DS_2 -VASc

Risk factor	Score
Congestive heart failure/LV dysfunction	1
Hypertension	1
Age ≥75	2
Diabetes mellitus	1
Stroke/TIA/thrombo-embolism	2
Vascular disease*	1
Age 65–74	1
Sex category (i.e. female sex)	1
Maximum score	**9**

*Prior myocardial infarction, peripheral artery disease, aortic plaque. Actual rates of stroke in contemporary cohorts may vary from these estimates.

EUROPEAN SOCIETY OF CARDIOLOGY®

Major i non-major riziko fakori za procjenu tromboembolizma kod A Fib!

Risk factors for stroke and thrombo-embolism in non-valvular AF

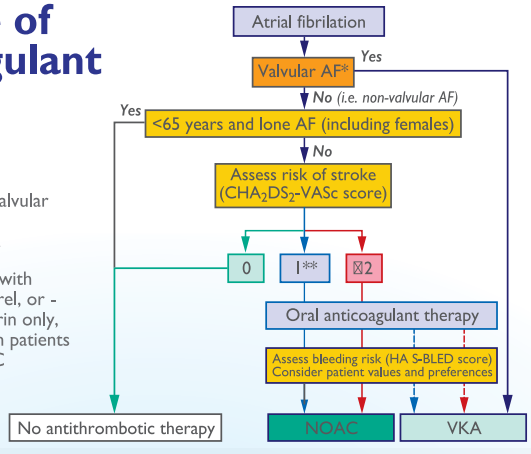
Major risk factors	Clinically relevant non-major risk factors
Previous stroke	CHF or moderate to severe LV systolic dysfunction [e.g. LV EF ☒ 40%]
TIA or systemic embolism	Hypertension
Age ☒75 years	Diabetes mellitus
	Age 65-74 years
	Female sex
	Vascular disease

AF = atrial fibrilation; EF = ejection fraction (as documented by echocardiography, radio nuclide ventriculography, cardiac catheterization, cardiac magnetic resonance imaging, etc.); LV = left venticular; TIA = trasient ischaemic attack.

EUROPEAN SOCIETY OF CARDIOLOGY®

Algoritam antikoagulantne terapije nakon procjene CHA_2DS_2VASc i major risk faktora!

Choice of Anti-coagulant



* Includes rheumatic valvular AF, hypertrophic cardiomyopathy, etc.

** Antiplatelet therapy with aspirin plus clopidogrel, or - less effectively - aspirin only, may be considered in patients who refuse any OAC

NOAC - Novel Oral Anticoagulants, VKA - Vitamin K Antagonists

PUBLISHER:
Discipline for Research and Development
Clinical Center University of Sarajevo
71000 Sarajevo, Bolnička 25
Bosnia and Herzegovina

For publisher:
Sebija Izetbegović, MD, PhD
General Manager
CCUS

Publishing editor:
Mirza Dilić, MD, PhD

AIMS AND SCOPE

The Medical Journal is the official quarterly journal of the Discipline for Research and Development of the Clinical Center University of Sarajevo and has been published regularly since 1994. It is published in the languages of the people of Bosnia and Herzegovina i.e. Bosnian, Croatian and Serbian as well as in English.

The Medical Journal aims to publish the highest quality materials, both clinical and scientific, on all aspects of clinical medicine. It offers the reader a collection of contemporary, original, peer-reviewed papers, professional articles, review articles, editorials, along with special articles and case reports.

Copyright: the full text of the articles published in the Medical Journal can be used for educational and personal aims i.e. references cited upon the authors' permission. If the basic aim is commercial no parts of the published materials may be used or reproduced without the permission of the publisher. Special permission is available for educational and non-profit educational classroom use. Electronic storage or usage: except as outlined above, no parts of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means without prior written permission from the Publisher.
All rights reserved©2017. Discipline for Research and Development, CCUS.

Notice: the authors, editor and publisher do not accept responsibility for any loss or damage arising from actions or decisions based on information contained in this publication; ultimate responsibility for the treatment of patients and interpretation of published materials lies with the medical practitioner. The opinions expressed are those of the authors and the inclusion in this publication of materials relating to a specific product, method or technique does not amount to an endorsement of its value or quality, or of the claims made by its manufacturer.

EDITORIAL OFFICE

Address:
Medical Journal, Discipline for Research and Development
Clinical Center University of Sarajevo,
71000 Sarajevo,
Bolnička 25,
Bosnia and Herzegovina,
Phone: +387 33 668 415 +387 33 297 264
Email: institutnir@bih.net.ba
Web. www.kcus.ba
Technical secretariat: svjetlana.barosevcic@kcus.ba

SUBSCRIPTION

Annual subscription rates: Bosnia and Herzegovina € 50; Europe € 80; and other € 100.

SUPPLEMENTS, REPRINTS AND CORPORATE SALES

For requests from industry and companies regarding supplements, bulk articles reprints, sponsored subscriptions, translation opportunities for previously published material, and corporate online opportunities, please contact;
Email: institutnir@bih.net.ba

PRINT

Blicdruk, Sarajevo
Printed on acid-free paper.

TECHNICAL DIRECTOR

Blicdruk, Sarajevo

CIRCULATION

500 copies

Editor-in-Chief
Sebija Izetbegović, MD, PhD

Editorial Board

Mirza Dilić, Senija Rašić,
Svjetlana Radović, Asja Prohić,
Enra Suljić-Mehmedika,
Amela Begić, Semra Čavaljuga,
Dželaludin Junuzović, Semir Bešlija

International Advisory Board

Kenan Arnautović (USA), Raffaele Bugiardini (Italy), Erol Ćetin (Turkey), Maria Dorobantu (Romania), Oktay Ergene (Turkey), Zlatko Fras (Slovenia), Dan Gaita (Romania), Mario Ivanuša (Croatia), Steen Dalby Kristensen (Denmark), Mimoza Lezhe (Albania), Mario Marzilli (Italy), Milica Medić-Stojanovska (Serbia), Herman Haller (Croatia), Fausto Pinto (Portugal), Mihailo Popovici (Moldova), Marcella Rietschele (Germany), Nadan Rustemović (Croatia), Georges Saade (Lebanon), Peter Seferović (Serbia), Dragan Stanisavljević (Slovenia), Panos Vardas (Greece), Gordan Vujanić (UK), Jose Zamorano (Spain)

English language revision
Svjetlana Barošević

Medical Journal is Indexed in

EBSCO publishing
USA
www.ebscohost.com

Member of National Journals Networks of the European Society of Cardiology

Original article

Serum concentration of tumor necrosis-alpha (TNF-α) in patients with alopecia universalis 115
Emina Kasumagić-Halilović

Assertiveness and stress coping strategies in students of Faculty of Medicine of Sarajevo University 119
Alma Džubur-Kulenović, Maja Muhić

Social adaptation in patients with anxious and depressive disorders 125
Mira Spremo, Nada Veselić, Zihnet Selimbašić

Immune parameters as reliable biomarkers during the treatment of patients with renal complications infected with beta-hemolytic streptococci 131
Mevludin Mekić, Đemo Subašić, Edmira Isak, Majda Hadžiabulić,
Alen Džubur, Sanja Mišeljić

Symptoms of depression in patients with coronary heart diseases following myocardial revascularization 136
Dženana Hrustemović, Ermina Mujičić, Alma Džubur-Kulenović

Incidence of Surgical Site Infection (SSI) in a six month sample of patients treated at Clinic of General and Abdominal Surgery of the Clinical Center University of Sarajevo 141
Salem Bajramagić, Adi Mulabdić, Edin Hodžić, Samir Muhović,
Jusuf Šabanović, Adnan Kulo

Professional article

Complication of chickenpox in immunocompetent patients hospitalized at the Clinic of Infectious Diseases of the Clinical Center University of Sarajevo 147
Rusmir Baljić, Hadžan Konjo, Bekir Rovčanin, Mirsada Hukić

Evaluation of patients with life-threatening injuries at Clinic of Emergency Medicine of the Clinical Center University of Sarajevo 150
Amela Tuco, Zoran Hadžiahmetović

Instructions to authors 155

Uputstva autorima 157

Medical Journal (2017) **Vol. 23, No 4**, 115 - 118 **Original article**

Serum concentration of tumor necrosis-alpha (TNF-α) in patients with alopecia universalis

Serumske koncentracije faktora nekroze tumora-alfa kod pacijenata sa univerzalnom alopeciom

Emina Kasumagić-Halilović*

Clinic of Dermatovenereology, Clinical Center University of Sarajevo, Bolnička 25, 71000, Sarajevo, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

Alopecia universalis (AU) is an uncommon form of alopecia areata that involves the loss of all hair and body hair. The etiology is unknown, although the evidence suggests that AU is an immunologically mediated disease. In the pathogenesis of AU, Th1 immune response is predominant. A special cytokine profile is created by Th1 cells which disturbs the natural balance of the cytokine networks and leads to inflammatory reaction and follicle damage. In this prospective study, we investigated the serum level of tumor necrosis factor-alpha (TNF-α) using an enzyme-linked immunosorbent assay (ELISA) in 22 patients with AU and 20 healthy controls. We also examined a possible association between serum levels of TNF-α and the duration of the disease. The serum concentration of TNF-α in patients with AU was significantly higher than in the control group ($p < 0.0014$). No correlations were found between the duration of disease and the serum levels of TNF-α ($p= 0.7361$). Our results have demonstrated the importance of determining TNF-α concentrations in serum of patients with AU. This research could contribute to the interpretation of insufficiently well-known views of the pathogenesis role and significance of TNF-α in AU.

Keywords: alopecia universalis, cytokines, tumor necrosis factor alpha

SAŽETAK

Alopecia universalis (AU) je rijetka forma alopecie areate koja označava potpuni gubitak kose i dlaka na svim dijelovima tijela. Etiologija je nepoznata, mada dokazi upućuju da je AU imunološki posredovana bolest. U patogenezi AA preovladava imunološki odgovor helper Th1 ćelija. Ćelije Th1 stvaraju poseban citokinski profil koji remeti prirodnu ravnotežu u citokinskoj mreži, što uzrokuje upalnu reakciju i oštećenje folikula. U ovom prospektivnom istraživanju odredili smo koncentraciju faktora nekroze tumora-alfa (TNF-α) u uzorcima seruma 22 pacijenta sa AU i kontrolnoj grupi 20 zdravih ispitanika pomoću imunoenzimskog ELISA testa. Nadalje, poređene su koncentracije TNF-α među grupama pacijenata sa različitom dužinom bolesti. Serumske vrijednosti TNF-α su značajno veće kod pacijenata sa AU u odnosu na uzorke kontrolne grupe ($p<0.0014$). Nije utvrđena povezanost između dužine trjanja bolesti i serumske vrijednosti TNF-α ($p=0.7361$). Naši rezultati ukazuju na značaj određivanja koncentracije TNF-α u serumu pacijenata sa AU. Istraživanje može pružiti doprinos razjašnjenju nedovoljno poznate uloge i značaja TNF-α u patopgenezi AU.

Ključne riječi: alopecia univerzalis, citokini, faktor nekroze tumora alfa

INTRODUCTION

Alopecia areata (AA) is common cause of reversible hair loss. Although it usually presents as asymptomatic localized hair loss, it is a disese of very broad spectrum. Alopecia universalis (AU) is an uncommon form of AA that involves the loss of all haed and body hair and is estimated to account for 7 to 30% of all alopecia cases (1). The cause of disease is unknown, although there is evidence to suggest that the link between lymphocytic infiltration of the follicle and the disruption of the hair follicle cycle in AA may be provided by a combination of factors, including cytokine release, cytotoxic T-cell activity, and apoptosis (2, 3). It is also considered that disequilibrium in the production of cytokines, with a relative excess of proinflammatory and Th1 types, vs. anti-inflammatory cytokines may be involved in the persistence of AA lesions, as shown in human scalp biopsies (4). Evidence for the role of cytokines and T-cells in AU includes the beneficial effects of systemic steroids and calcineurin inhibitors.

Tumor necrosis factor-alpha (TNF-α)) is a multifunctional proinflammatory cytokine wich has been implicated in the pathogenesis of several chronic inflammatory disorders with an autoimmune component. This cytokine is synthesized in epidermal keratinocytes along with several other cytokines and is known to be a very potent inhibitor of proliferation (5). The changes in serum TNF-α levels we found in many

diseases, such us psoriasis (6), lichen planus (7) and vitiligo (8). In some of these diseases, serum TNF-α concentration corelated with activity and intensity of the disease, and may be used as a prognostic factor.

Recent progress in the understanding of AU has shown that the regulation of local and systemic cytokines plays an important role in its pathogenesis. Therefore, the aim of our study was to evaluate serum concentrations of TNF-α in patients with AU and healthy subjects and also to asses a possible association between TNF-α and duration of the disease.

MATERIALS AND METHODS

This is a case-control study of serum concentration of TNF-α in AU patients. The study was conducted in the Clinic of dermatovenereology at University Clinical Center Sarajevo. The study included 22 patients with AU (10 female and 12 male). A detailed history and examination were taken in all study subjects, including patients age, age at onset and duration of disease. The diagnosis of AU was made on clinical grounds. None of the patients had used any systemic medications for AU treatment for at least 6 months before the study. We excluded the patients with other types of illnesses, such as autoimmune diseases that could affect the outcome of the study.

Control group consisted of 20 generally healthy subjects (11 females and 9 males). They did not have any scalp lesions in their personal history or on clinical examination.

All subjects gave their informed consent in accordance with the requirements of the Institutional Ethics Committee. The study was conducted in accordance with the principles of the Declaration of Helsinki.

Serum cytokine determination

Serum concentrations of TNF-α were measured by enzyme-linked immunosorbent assay (ELISA) technique, using Quantikine Human TNF-α Immunoassay (R&D System, Minneapolis, USA), in accordance with the manufacturer's instructions.

Briefly, a monoclonal antibody specific for TNF-α has been precoated onto a microplate. Standards and samples are pipetted into the wells and any IFN-α present is bound by the immobilized antibody. After washing away any unbound substances, an enzyme-linked polyclonal antibody specific for TNF-α is added to the wells. Following a wash to remove any unbound antibody-enzyme reagent, a substrate solution is added to the wells and colour develops in proportion to the amount of TNF-α bound in the initial step. The colour development was stopped and the intensity of the colour was measured at 450 nm with a photometar (Rider Biotek Elx800).

Statistical analysis

Statistical analyses were performed using MedCalc Statistical Software version 15.2.2. (MedCalc Software bvba, Ostend, Belgium). Statistical comparisons were performed using T test and Mann Whitney U test for independent samples. We used Spearman correlation coefficient rho for calculate relationship between duration of disease and serum levels of cytokines. Data were considered statistically significance at $p<0.01$.

RESULTS

The study group composed of 22 patients with AU (10 females and 12 males; the mean age of the patients was 33.96 years, ranging from 5 to 60 years), and 20 healthy controls (11 females and 9 males; the mean age 32.55 years, ranging from 6 to 63 years). There were no significant difference in age and female/male ratio between the patients and controls ($p>0.05$).The mean duration of AU was 27.59±29.95 (range 2-108 months). Dermographic data of patients and controls are shown in Table 1.

Table 1 Demographic characteristics of patients and healthy controls.

	A Alopecia universalis (n=22)	Healthy Controls (n=20)	p
Age (mean±SD)	33.96±15.65	32.55±16.12	0.889*
Range	5-60	6-63	
Sex (male/female)	12/10	9/11	0.7574**
Duration of disease (months) (mean±SD)	27.59±29.95	/	
Range	2-108	/	

** T test*
*** Chi-squared test*

The serum concentration of TNF-α in patients with AU was significantly higher than that in the control group (11.000 pg/ml vs 9.600 pg/ml, respectively) (Table 2, Figure 1).Patients with longer duration of the disease had higher concentration of TNF-α, but not significantly (Table 3).

Table 2 Serum concentrations of TNF-α in patients and healthy controls.

	Patients (pg/ml)	Controls (pg/ml)	Mann-Whitney	Z	p*
TNF-α (med)	11 000	9600	93.50	3.187	0.0014***
Range**	7.700-17.000	8.400-11.300			

** Mann-Whitney U test*
*** Range – min-max values*
**** Statistical significance ($p<0.01$)*

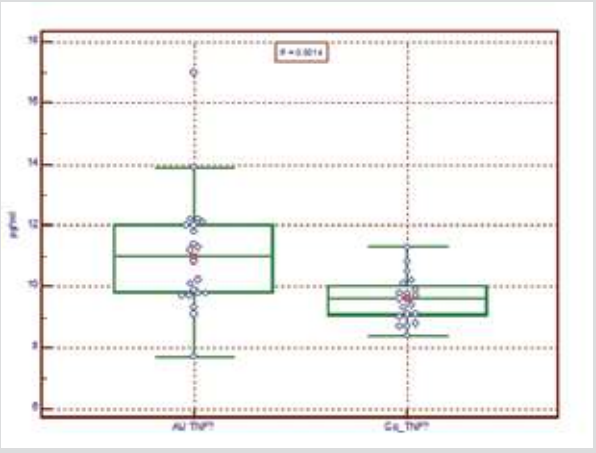


Figure 1 Median, minimum-maximum values of serum levels TNF-α in alopecia universalis (AU) patients and healthy controls (CO).

Table 3 Relationship between duration of disease and serum levels of TNF-α (Spearman rank test).

	Duration of disease			
	n	rho	95% CI	p
TNF-α	22	0.0762	-0.357-0.482	0.7361

** Statistical significance (p<0.01)*

DISCUSSION

Although the cause of the disease is at present unknown, several studies have shown that within cascade of pathogenesis of AA, cytokines play a crucial role (3, 9). Hair loss may occur because proinflammatory cytokines interfere with the hair cycle, leading to premature arrest of hair cycling with cessation of hair growth. This concept may explain typical clinical features of AA such as a progression pattern in centrifugal waves and spontaneous hair regrowth in concentric rings (10), suggesting the presence of soluble mediators within affected areas of the scalp.

TNF-α is a multifunctional proinflammatory cytokine which has been implicated in the pathogenesis of many infections and inflammatory disorders. However, this cytokine not only acts as mediator of immunity and inflammation, but also affects not-immune responses within tissues such as cell proliferation and differentiation (11). In vitro studies have shown that TNF-α, along with IL-1α and IL-β, causes vacuolation of matrix cells, abnormal keratinization of the follicle bulb and inner root sheath, as well as disruption of follicular melanocytes and the presence of melanin granules within the dermal papilla (12). Experiments in cultured human hair follicles by Hoffmann et al. showed that TNF-α completely abrogated hair growth (13). Additionally, TNF-α induced the formation of a club-like hair follicle, similar to catagen morphology of the hair bulb. A study by Thein et al. examined cytokine profiles of infiltrating activated T-cells from the margin of involved AA lesions (14). It was found that T-cell clones from involved lesions inhibited the proliferation of neonatal keratinocytes. In examining the cytokine profiles and relating them to regulatory capacity, the authors found that T-cell clones that released high amounts of IFN-γ and/or TNF-α inhibited keratinocyte growth. Recent study has shown that TNF-α level in lesional biopsies of patients was higher than that of controls' biopsies (15).

In addition, increased serum levels of TNF-α in patients with AU compared with normal controls has been reported, further suggesting a role for this cytokine (16-18). The results presented in our study demonstrate that the mean serum levels of TNF-α were significantly elevated in AU patients in comparison to healthy subjects. No correlations were found between duration of the disease and the serum levels of TNF-α. In contrast to our results, Teraki et al. (19), reported that serum levels of TNF-α in patients with localized AA were significantly higher than those in AU. They said that these findings could be interpreted as an indication that Th1 type cytokines might be critical for the progression to the extensive form and that Th2 type cytokines may exert a more subtle influence on the inhibition of a cell-mediated attak on hair follicles.

In the study of Koubanova and Gadjlgoroeva, serum levels of TNF-α in patients with AA did not differ from that in controls (20). However, TNF-α was lower in patients with severe form of AA than in patients with mild form. They hypothesized that similar levels of TNF-α in patients with both forms of AA and controls may indirectly indicate the absence of systemic immunopathological reactions in patients with AA, and the lowering of TNF-α level in the mild form may indicate the tendency to formation of immunodeficiency in patients with severe AA. In addition, Lis et al. found that serum levels of sTNF-α receptor type I were significantly elevated in patients with AA in comparison with healthy subjects (21). As they conclude, these results indicate that immune mechanisms in AA are characterized by activation of T-cells and other cells, possibly keratinocytes.

CONCLUSION

TNF-α seems to be a useful indicator of the activity of AU and that it may play an important role in the development of this disease. Further investigations are required to clarify the pathogenic role and clinical significance of TNF-α, and these findings may provide important clues to assist in the development of new therapeutic strategies for patients with AU.

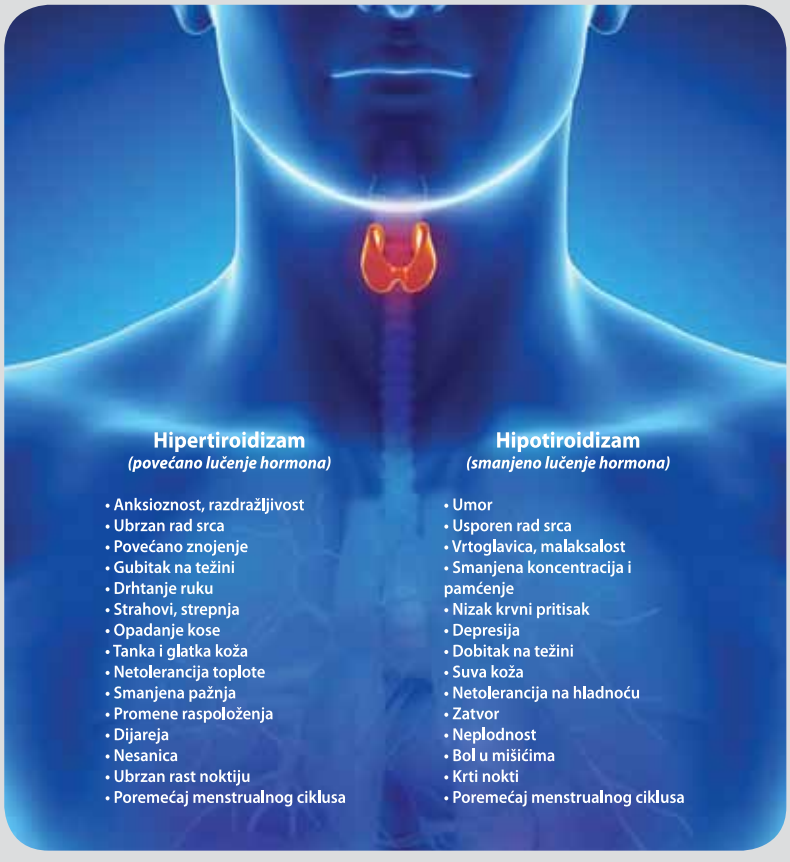
Conflict of interest: none declared.

REFERENCES

1. Hunt N, McHale S. The psychological impact of alopecia. BMJ.2005;331:951-953.
2. Ito T. Recent advances in the pathogenesis of autoimmune hair loss disease alopecia areata. Clin Dev Immunol. 2013:ID348546.
3. Ito T, Tokura Y. The role of cytokines and chemokines in the T-cell-mediated autoimmune process in alopecia areata. Exp Dermatol. 2014;23:787-791.
4. Bodemer C, Peuchmaur M, Fraitag S, Chatenoud L, Brousse N, De Prost Y. Role of cytotoxic T cells in chronic alopecia areata. J Invest Dermatol. 2000;114:112-116.
5. Symington FW. Lymphotoxin, tumor necrosis factor, and gamma interferon are cytostatic for normal human keratinocytes. J Invest Dermatol. 1989;92:798-805.
6. Sereflican B, Goksugur N, Bugdayci G, Polat M, Haydar Parlak A. Serum visfatin, adiponectin, and tumor necrosis factor alpha (TNF-α) levels in patients with psoriasis and their correlation with disease severity. Acta Dermatovenerol Croat. 2016;24(1):13-19.
7. Alpinar Kara Y. The measurement of serum TNF-α levels in patients with lichen planus. Acta Dermatovenereol Alp, Panonica Adriat. 2017;26:85-88.
8. Camara-Lemarroy CR. Salas-Alanis JC. The role of TNF-α in the pathogenesis of vitiligo. Am J Clin Dermatol. 2013;14(5):343-350.
9. Giordano CN, Sinha AA. Cytokine pathways and interactions in alopecia areata. Eur J Dermatol. 2013;23(3):308-318.
10. del Rio E. Targetoid hair regrowth in alopecia areata. The wave theory. Arch Derm. 1998;134:142.
11. Whicher JT, Evans SW. Cytokines in disease. Clin Chem. 1990;36:1269-1281.
12. Philpott MP, Sanders DA, Bowen J, Kealey T. Effects interleukins, colony-stimulating factor and tumor necrosis factor-α in alopecia areata. Br J Dermatol. 1996;135:942-948.
13. Hoffmann R, Eicheler W, Huth A, Wenzel E, Happle R. Cytokines and growth factors influence hair growth in vitro: Possible implications for the pathogenesis and treatment of alopecia areata. Arch Dermatol Res. 1996;288:153-156.
14. Thein C, Strange P, Hansen EB, Baadsgaard O. Lesional alopecia areata T lymphocytes downregulate epithelial cell proliferation. Arch Dermatol Res. 1997;289: 384-388.
15. Gohary YM, Abdel Fattah DS. Detection of tumor necrosis factor-alpha in nonlesional tissues of alopecia areta patients: a pruve for a systemic disease. Int J Trichology. 2017;9(4):154-159.
16. Rossi A, Cantisani C, Carlesimo M, Scali E, Mari E, Garelli V, et al. Serum concentrations of IL-2, IL-6, IL-12 and TNF-α in patients with alopecia areata. Int J Immunopathol Pharmacol. 2012;25(3):781-788.
17. Bilgic O, Sivrikaya A, Unlu A, Altinyazar HC. Serum cytokine and chemokine profiles in patients with alopecia areata. J Dermatolog Treat. 2016; 27(3):260-263.

18. Atwa MA, Youssef N, Bayoumy NM. T-helper 17 cytokines (interleukins 17, 21, 22, and 6, and tumor necrosis factor-α 9 in patients with alopecia areata: association with clinical type and severity. Int J Dermatol. 2016;55(6):666-672.
19. Teraki Y, Imanishi K, Shiohara T. Cytokines in alopecia areata: contrasting cytokine profiles in localized form and extensive form (alopecia universalis). Acta Derm Venereol (Stockh). 1996;76:421-423.
20. Koubanova A, Gadjlgoroeva A. On the problem of pathogenetic heterogeneity of alopecia areata. EHRS Brussles, Conference abstracts, 2002. p 13.
21. Lis A, Pierzchala E, Brzezinska-Weislo L. The role of cell-mediated immune response in pathogenesis of alopecia areata. Wiad Lek. 2001;54:159-63.

Reprint requests and correspondence:
Emina Kasumagić-Halilović, MD, PhD
Clinic of Dermatovenereology
Clinical Center University of Sarajevo
Bolnička 25, 71 000 Sarajevo
Bosnia and Herzegovina
Email: eminakahalilovic@gmail.com



Assertiveness and stress coping strategies in students of Faculty of Medicine of Sarajevo University

Asertivnost i strategije suočavanja sa stresom kod studenata Medicinskog fakulteta Univerziteta u Sarajevu

Alma Džubur-Kulenović [1]*, Maja Muhić[2]

[1]Psychiatric Clinic, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina
[2]Discipline for Research and Development, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

Introduction: enrolling at a university represents a significant life change and an overall challenge for adaptive potentials of older adolescents. Coping skills represent a central role in successfully adapting to these stressful events of a new environment, new friendships, and fear of failure. The students need to find more functional ways of coping by using adaptive coping strategies that they already possess and those that they can yet learn. Failing to complete the year represents additional major stress, for students who, as a rule, have a previous history of outstanding achievements in education. This stress may reduce their self-esteem and further compromise academic performance. Aim: to assess possible relationships and differences between assertiveness and coping strategies in the first-year medical students. Materials and methods: the study included 225 first-year medical students of the Sarajevo Faculty of Medicine. They were divided into two groups: those who enrolled for the first time (n=133) and those who failed the first year and re-enrolled (n=92). We tried to examine the association between assertiveness and coping strategies on the entire sample, and also to determine possible differences related to assertiveness and coping strategies between female and male students, and between students who changed residence during the study and those who did not. The study used a coping strategy scales (Folkman & Lazarus, 1988 adapted by Arcel-Tata and Ljubotina, 1994), Rathus Assertiveness Schedule, and a socio-demographic questionnaire developed for this study. Results: our results show significant differences between assertiveness measured by Rathus scale and coping strategies assessed by Folkman & Lazarus scale between two groups of students. Significant differences on the two scales were also found between male and female students and students who moved from the place of residence to Sarajevo in order to enroll at university and those who did not need to change their residence. We found a significant positive correlation between assertiveness measured by the Rathus Assertiveness Schedule and the Folkman & Lazarus sub-scales of confrontation and solution planning on the level of $p<0.01$. Assertiveness was found to significantly correlate with acceptance of responsibility and avoidance/escape on the level of $p<0.05$. Conclusion: our results may provide further evidence for the idea of providing non-stigmatizing counseling and psychoeducational services and assertiveness training for the first-year medical students within the Faculty. Implications for further studies have been discussed.

Keywords: assertiveness, coping strategies, adolescents

SAŽETAK

Uvod: upisivanje fakulteta je značajna životna promjena koja predstavlja izazov za adaptivne potencijale pojedinca. Strategije suočavanja sa stresom predstavljaju centralnu ulogu u prilagodbi na takve stresne događaje, nov okruženje, nova prijateljstva i strah od neuspjeha. Neophodno je naći što bolje načine suočavanja, a u tome nam može pomoći koncept asertivnosti, interpersonalni aspekt kojeg je moguće naučiti, iako ga ne posjedujemo. Neuspjeh za dovršetak godine predstavlja daljnji veliki stres, za učenike koji po pravilu imaju prethodnu povijest izvanrednih postignuća u obrazovanju. Ovaj stres može smanjiti njihovo samopoštovanje i dodatno kompromitiraju akademske rezultate. Cilj: procijeniti mogući odnos i razlike između asertivnosti i strategija kod studenata prve godine medicinskog fakulteta. Materijali i metode: uzorak je obuhvatio 225 ispitanika, studenata prve godine Medicinskog fakulteta Univerziteta u Sarajevu. Podijeljeni su u dvije skupine: one koji su se prvi put upisali (n = 133) i oni koji su obnovili godinu - ponovno upisani (n = 92). Nastojali smo ispitati povezanost između asertivnosti i strategija suočavanja na cjelokupnom uzorku, te odrediti moguće razlike u vezi s asertivnošću i strategijama suočavanja između muškaraca i žena, te između studenata koji su promijenili boravak tijekom studija i onih koji to nisu učinili. U istraživanju je korištena skala suočavanja (Folkman & Lazarus, 1988 adaptirana Arcel-Tata i Ljubotina, 1994), Rathusova skala asertivnosti-samoprocjena, te sociodemografski upitnik izrađen za potrebe ovog rada. Rezultati: naši rezultati pokazuju značajne razlike između asertivnosti mjerene Rathusovom skalom i strategija suočavanja mjerenih Folkman & Lazarus skalom između dvije skupine studenata. Značajne razlike u dvjema mjerilima pronađene su i između muških i ženskih studenata i studenata koji su se preselili iz mjesta porijekla u Sarajevo radi upisa na fakultet i onih koji nisu trebali mijenjati prebivalište. Pokazali smo pozitivnu korelaciju između asertivnosti mjerene Rathusovom skalom i subskalama konfrontacija i planskog rješavanja, na Folkman & Lazarus skali na razini $p<0,01$. Utvrđeno je da asertivnost korelira s prihvaćanjem odgovornosti i izbjegavanjem / bijegom na razini $p<0,05$. Zaključak: naši rezultati mogu pružiti dodatne dokaze o ideji pružanja ne-stigmatizirajućih savjetodavnih i psihoedukacijskih usluga i treninga asertivnosti za studente prve godine unutar Medicinskog akulteta. Razmatrane su implikacije za daljnja istraživanja.

Ključne riječi: asertivnost, strategije suočavanja, adolescent

INTRODUCTION

School enrollment is a significant life-style change that is a challenge for adaptive potentials of older adolescents. Stress-facing strategies represent the central method of adapting to such stressful events (1,2). It is necessary to find the best ways of coping, and it can help us with the concept of assertiveness, the interpersonal aspect that can be learned, even though we do not possess it. In the academic environment we can distinguish several different, potential sources of stress.

Students may be under pressure to meet academic requirements, adjust to the new environment, or develop new friendships. Strenuous reactions are especially prone to students of the first year of study (3,4). Most of them are forced to adapt to a change in the middle of life, independent living or living with people who are not members of their family.

Transition to college is a significant but stressful life event as it poses a series of new demands to young people e.g. new educational tasks, new friendships, new living conditions, etc.) (5). In adolescents who leave their place of residence and parental home due to studies, there are also problems with the separation from close persons.

Previous studies have shown that stress management strategies play a central role in adapting to stressful events in the new environment (6). They are a specific kind of effort that an individual drives to overcome, tolerate, or reduce the level of stressful events (7). By striving to find a unique and easy way to help students overcome stress, we have tried to find a comprehensive, easily-taught concept that is related to coping strategies.

The answer was found in a research based on a sample of Teheran University of Medical Sciences, one of the largest and most prestigious in Iran. Results showed that 60% of students suffer from a lack of assertiveness, and that this disadvantage had a negative impact on their learning and success. On the other hand, students with a high level of assertiveness had fewer problems with adaptation and less suffering from loneliness (8). Assertiveness is an interpersonal aspect that can be learned, even though we do not possess it.

MATERIALS AND METHODS

The study included students from the Faculty of Medicine, University of Sarajevo, N = 233, who agreed to fill in the survey questionnaires. Participants were first year students in the 2015/2016 academic year (students of the integrated study program "Medicine" at the Faculty of Medicine, University of Sarajevo, first time enrollement N=131 and students of the integrated study program "Medicine" at the Faculty of Medicine, University of Sarajevo, re-entered N=92). The survey used a confrontation scale (Folkman & Lazarus, 1988 adapted Arcel-Tata and Ljubotina, 1994). The scale consists of 39 particles, divided into 8 subscales: acceptance of responsibility, avoidance, confrontation, distancing, self-control, search for social support, problem solving, positive assessment. The response format was a Likert type of 3 degrees (0-never, 1-sometimes, 2-mostly, 3-always) scale. Rathus Assertiveness-Self-Assessment Scale, made up of 30 particles. Respondents should be marked by the number of claims they describe (1 - very characteristic for me, 2 quite characteristic for me, 3 - somewhat characteristic for me; - 1 - a bit uncharacteristic for me, - 2 - quite uncharacteristic for me, very uncharacteristic for me). Socio-demographic questionnaire prepared for the purpose of this paper consisted of twenty-one questions.

RESULTS

Firstly, descriptive statistics were produced for the results of Rathus's Assertiveness-Self-Assessment Scale and Scale Facing (Folkman & Lazarus, 1988 adapted by Arcel-Tata and Ljubotina, 1994). The results are shown in Table 2. It should be emphasized that during the statistical processing we made two interventions in the Rathus scale. The first related to recoding of the original scoring (1 very characteristic for me; 2 pretty characteristic for me; 3 somewhat characteristic for me; -1 a bit non-characteristic for me; -2 quite non-characteristic for me; -3 very non-characteristic for me) at a value of 1 to 6 (-3 = 1; -2 = 2; -1 = 3; 3 = 4; 2 = 5; 1 = 6). Subsequently, we reversed the particle which lower Rathus scale implies higher assertiveness (1, 2, 4, 5, 9, 11, 12, 13, 14, 15, 16, 17, 19, 23, 24, 26, 30). Both procedures were conducted to give a clearer assertiveness measure on a scale from 1 to 6, where a higher number implies a higher order, that is, a higher measured assertiveness according to the Rathus scale.

Table 1 Descriptive statistics for the entire sample.

Scale	M	St. dev.
Rathus scale	3.59	0.663
Acceptance of responsibility	3.02	0.630
Escape-avoidance	2.08	0.625
Confrontation	2.77	0.548
Distancing	2.50	0.512
Self-control	2.69	0.449
Social support	2.36	0.491
Planned problem solving	2.84	0.520
Positive assessment	3.08	0.525

Although the results of this study do not seem to have major problems with normality, we also applied Pearson's coefficient of correlation to examine the relationship between assertiveness and individual stress-strain strategies. The values calculated using Spearman's correlation coefficient are presented in Table 2.

Table 2 Coping strategies.

Rathus scale			
Subscale	P	N	P
Acceptance of responsibility	-.170	223	<0.05
Escape-avoidance	-.157	223	<0.05
Confrontation	.376	224	<0.01
Distancing	-.235	224	<0.01
Self-control	-.276	224	<0.01
Social support	.055	224	>0.05
Planned problem solving	.206	223	<0.01
Positive assessment	-.038	224	>0.05

Assertiveness significantly correlates positively with confrontation and problem-solving, at the level of 1%, while it significantly negatively correlates with distancing and self-control at the level of 1%, with accepting responsibility and avoidance, at the level of 5%.

Gender

The next analysis relates to gender. Descriptive statistics are oresented in Table 3.

Table 3 Descriptive statistics in respect of male and female gender.

	Male			Female		
	M	St. dev.	N	M	St. dev.	N
Rathus scale	3.7740	.64700	61	3.5279	.65797	163
Acceptance of responsibility	2.9426	.63475	61	3.0525	.62788	162
Escape-avoidance	2.0475	.59227	61	2.0960	.63813	162
Confrontation	2.8825	.54844	61	2.7336	.54424	163
Distancing	2.5855	.52834	61	2.4730	.50336	163
Self-control	2.6902	.41229	61	2.6913	.46365	163
Social support	2.3992	.49561	61	2.3444	.48963	163
Planned problem solving	2.7393	.49439	61	2.8846	.52483	162
Positive assessment	2.9612	.45802	61	3.1194	.54236	163

After descriptive statistics, as with the entire sample, we calculated Pearson's correlation coefficient for gender. In the male sex we obtained two significant correlations. The first was between assertiveness and confrontation at the level of 5% (r = 0.302), and the second between assertiveness and planned problem solving at the level of 1% (r = 0.470) which was also the highest.

When it comes to female sex, we obtained more significant correlations with assertiveness (all at 1%). There were four negative correlations with acceptance of responsibility (r = -0.249), avoidance (r = -0.214), distancing (r = -0.362), and self-control. We also obtained a positive correlation with the assertiveness and this was in the confrontation subscale (r=0.414). Interestingly, we neglected the correlation between assertiveness and the problem-solving subscale (r=0.151), while the male gender was the highest (r=0.470).

Re-enrollment after failing the first year of the Faculty of Medicine

After sex analysis, we conducted the status analysis, for those who re-enter the year and those who did not. Table 4 shows descriptive statistics for respondents who re-entered the year (Have you re-entered the year = yes) and those who did not (Have you re-entered the year? = No).

Table 4 Descriptive statistics in respect of the study status.

	Faild			First time		
	M	St. dev.	N	M	St. dev.	N
Rathus scale	3.5713	.66329	91	3.6006	.65563	132
Acceptance of responsibility	3.0540	.61781	91	3.0025	.64217	131
Escape-avoidance	2.0841	.62347	91	2.0855	.62936	131
Confrontation	2.7363	.56931	91	2.7967	.53446	132
Distancing	2.4989	.51185	91	2.5082	.51499	132
Self-control	2.7092	.44786	91	2.6818	.45177	132
Social support	2.3581	.52201	91	2.3614	.47178	132
Planned problem solving	2.7621	.49634	91	2.8966	.52792	131
Positive assessment	3.0304	.58425	91	3.0198	.48038	132

After the descriptive statistics for the repetition of the year (Did you repeat the year?), we calculated Pearson's correlation coefficient. For respondents who re-entered the year (Did you re-enter the year? = Yes), at the level of 1%, we obtained four significant correlations, two positive and two negative.

Positive correlations were established between assertiveness and confrontation subscale (r = 0.313), and between assertiveness and problem solving subscale (r = 0.290). Negative correlations were obtained between assertiveness and the distance subscale (r = -0.327), and between assertiveness and self-control subscale (r = -0.323).

For those who did not re-enter the year (Have you re-enter the year? = No), we obtained four significant correlations; two at a significance level of 1% and two at 5%. Three correlations were negative and three positive.

At the significance level of 1% we obtained a correlation between assertiveness and confrontation subscale (r = 0.451), and between assertiveness and self-control subscale (r = -0.273). At the level of significance of 5%, the correlation between assertiveness and acceptance subscale (r = -0.220), and between assertiveness and avoidance subscale (r= -0.200).

Change of residence

After analyzing the year's re-entry, we conducted analysis for students who changed and not changed their place of residence during the study. Table 5 shows descriptive statistics for respondents who changed the city (Did change residence when enrolled at college? = Yes), and for those who did not (Did you change residence when enrolled at college? = No).

Table 5 Descriptive statistics in respect of change of residence.

	change of city			Not change city		
	M	St. dev.	N	M	St. dev.	N
Rathus scale	3.4269	.64485	127	3.8058	.61954	96
Acceptance of responsibility	3.0899	.59811	126	2.9288	.66348	96
Escape-avoidance	2.1321	.59010	126	2.0188	.66867	96
Confrontation	2.7507	.59489	127	2.8056	.48405	96
Distancing	2.5647	.51511	127	2.4264	.50015	96
Self-control	2.7154	.46919	127	2.6660	.41894	96
Social support	2.3432	.51304	127	2.3823	.46379	96
Planned problem solving	2.8329	.52768	126	2.8651	.51181	96
Positive assessment	3.1241	.54340	127	3.0191	.49427	96

With regard to change of residence, we calculated Pearson's correlation coefficient. For students who changed residence (Did you changed residence when enrolling at college? = Yes) we obtained three significant correlations, two positive and one negative. The first was a positive correlation between the assertiveness and the subcalculating comparison at a significance level of 1% (r = 0.381). Other negative correlation was 5%, between assertiveness and self-control subscale (r = -0.224). The third significant, positive correlation was also obtained at the level of 5%, between assertiveness and the problem solving subcalculation (r = 0.205).

For students who did changed residence (Did you changed residence when enrolling at college? = No) we obtained five significant correlations with assertiveness. There were three correlations at a significance level of 1% and two at the level of 5%. We also had three positive and two negative correlations.

Table 6 Significant correlation matrices for students who did not change residence.

	Rathus scale	Confrontation subscale	Distancing subscale	Self control subscale	Social support sub.	Planned problem solving sub.
Rathus scale	1	.441**	-.295**	-.387**	.202*	.229*
Confrontation subscale		1	-.045	-.268**	.196	.358**
Distancing subscale			1	.407**	.148	.181
Self control subscale				1	-.195	-.012
Social support sub.					1	.390**
Planned problem solving sub.						1
*p<0.05 **p<0.01						

DISCUSSION

Differences at the level of entire sample

Firstly, we investigated general level of assertiveness measured on Rathus's assertiveness scale, which proved moderate (3.59 for the entire sample). We also showed that there was no statistically significant gender difference, year re-entry, and change of residence which was in line with our expectations based on earlier studies.

On all Folkman & Lazarus (Acceptance of Responsibility, Avoidance, Conflict, Distancing, Self-Control, Social Assistance, Problem-solving, Positive Assessment) sub-scales we obtained moderate results for the entire sample. We did not show any differences in the scores on these subscales with regard to gender, re-entry or change of residence. However, when it comes to linking the two scales used across the entire sample, we obtained a significant positive correlation of assertiveness measured on the Rathus scale with subscales of confrontation and planned problem solving at the level of 1%. Assertiveness significantly negatively correlated with distancing and self-control at the level of 1%, with accepting responsibility and avoidance, at a level of 5%. In the study of Živčić-Bećirević et al., (9), there were differences in adaptation to study with regard to gender, change of residence, and year re-entry. The study conducted on a sample of 845 students at the University of Rijeka investigated academic, social and emotional adaptation. It was shown that girls, for example, achieved better academic (compliance with study requirements) and social adaptation (learning in college environments and interaction with colleagues at the study), but less emotional adaptation (the presence of various signs of emotional discomfort and physical symptoms) compared to young men. Such findings indicate a different way of wearing stressors, which can be assumed to be the use of different coping strategies. Furthermore, this would mean that there will be differences in the correlation of strategies with other constructs (9).

Re-entry of year and change of residence

Although we have showen that the baseline on the Rathus scale and Folkman & Lazarus scale do not differ by sex, year re-entry and change of residence, we still have differences in the correlation of these two scales within the same groups. First of all, it was shown that the assertiveness and stress-coping strategies were different in the case of students re-entering the year and those who did not confirm one of our hypotheses (H3-1). For respondents who were asked "Did you re-enter the year?" and who answered "Yes", in 1% of them we obtained four significant correlations, two positive and two negative. Assertiveness and sub-scalar correlations had positive correlation (r = 0.313), as well as the assertiveness and problem-solving subscales (r = 0.290). Regarding negative correlations, we had correlation between assertiveness and sub-scaling (r = -0.327), and between assertiveness and self-control subscale (r = -0.323). In respondents who answered the same question with "No", we also obtained four significant correlations. Two were at a significance level of 1% and two at 5%, three negative and one positive. However, here we showed the difference. Assertiveness and confrontation subscales positively correlated at 1%, but the coefficient of correlation was higher (r=0.451), whereas assertiveness and subcontents of self-control at the same level correlated slightly less significant with students who re-entered the year (r = -0.273). Interestingly, in this group, the assertiveness significantly correlated with other two subscales at the significance level of 5%. The correlation between assertiveness and subscale acceptance (r = -0.220), and between assertiveness and avoidance subscale (r = -0.200), was demonstrated. Similarly to the year re-entry, for those who changed the place of residence and those who did not, there were no significant differences in the age of assertiveness or coping strategies. But here too there was a difference in correlation, which proved another hypothesis we set (H4-1). In those respondents who provided affirmative answer to the question "Did you change residence when enrolling at college?", we obtained three significant correlations, two positive and one negative. At the significance level of 1%, we obtained a significantly positive correlation between assertiveness and the subconscious of confrontation (r = 0.381). The second positive correlation was r=0.205 at the level of 5% between assertiveness and planned problem solving. The third correlation, also at the significance level of 5%, was a negative correlation between assertiveness and self-control subscale (r = -0.224). For students who provided negative answer on the same question, we obtained even five significant correlations with assertiveness. There were three correlations at the significance level of 1% and two at the level of 5%. We also had three positive and two negative correlations. Here, we had a correlation of assertiveness with confrontation at the level of 1%, but somewhat higher (r = 0.441). There was also a correlation with the systemic problem solving subscale at the significance level of 5% (r=0.229), and the self-control subcale at higher intensity (r = -0.387). Two additional correlations was obtained between assertiveness and distancing subscale at the level of 1% (r = -0.295), and between assertiveness and social support subscale at the level of 5% (r = 0.202).

Gender differences

The results of recent study show that men are generally more assertive than women (10) which seems logical as assertiveness is typical

and desirable for masculine behavior (11). Hovewer, our findings are consistent with contradicting studies, namely that there is no significant difference in the assertiveness of the sexes (12). Accordingly, we did not obtain a difference in the level of assertiveness or strategy of dealing with stress - but we did in their relationship. With regard to males we obtained two significant correlations: between assertiveness and confrontation at the level of 5% (r = 0.302), and between assertiveness and planned problem solving at the level of 1% (r = 0.470). As for females we obtained more significant correlations with assertiveness (all at the level of 1%). There were four negative correlations with acceptance of responsibility (r = -0.249), avoidance (r = -0.214), distancing (r = -0.362), and self-control. Confrontation subscale was in positive correlation with assertiveness (r = 0.414). It was interesting that we neglected the correlation between the assertiveness and the problem-solving subscale (r = 0.151), whereas the male gender was the highest (r = 0.470).

Limitations of the study and future research

Our study investigated correlation of the results on the Rathus scale with the whole subcalculations on the Folkman & Lazarus scale, not with the individual particles they consist of. Given that the individual particles were characterized by the authors as a measure of positive and negative coping strategies, further studies should investigate the differences between correlation with positive and negative coping strategies in order to determine the relationship with assertiveness. In this way, we created the preconditions for a more precise direct examination of the impact of the assertiveness training on stray-dealing strategies and adaptation to study. We also tried to examine differences in attitude towards religion, but in the statistical processing, there was no sufficient representation of categories of agnostics and atheists in the sample, therefore we had to reject H5 hypothesis. We believe that this important aspect should be explored in future studies. In addition, it should be noted that, due to the nature of the statistical correlation process, research on the larger sample would provide more reliable correlations, and perhaps even more significant correlations with subscales. Such results would provide deeper insight into the correlation of the concept of assertiveness with coping strategies, that is, in our case, a more comprehensive picture of the relationship between the sketches of Rathus and Folkman & Lazarus.

Implications

The research results of Rudy, Merluzzi and Henahan, 1982, and Alinčić, 2013 (13) have shown that less assertive people use strategies that reflect anxiety in social situations. Also, numerous studies which investigated the relationship between assertiveness and anxiety confirmed that the relationship between these variables was negative, those who achieved more results on the anxiety scale were generally less assertive (14). Furthermore, it has been found that more ascertainable children have more efficient stress-bearing skills and that teachers find that more ascertainable children have better skills and ways of expressing their knowledge and more self-esteem. All of these were reasons to suppose the assertiveness and the strategies of dealing with stress are significantly related - and that is what we showed. Our findings show that assertiveness is associated with stress-coping strategies, which provides us with a basis for assuming that increasing assertiveness also influences stress-bearing, that is, any eventual assertiveness training would be helpful to students in their stress-pacing. Ultimately, this can result in better adaptation to the study. Given that we have shown that, depending on sex, changing residence and rebuilding of the year, ascertainment and stress-response strategies vary, we can assume that increasing assertiveness would have different impacts on these groups, which should be remembered if planning of assertiveness. In our research, our task was to find arguments for a cheap and efficient way to help students in stress-bearing of learning experience. By exploring the literature, we came to the conclusion that the concept of assertiveness could be of great help, and our research has shown that there are grounds for such a hypothesis. Assertiveness training has proven effective (16), a relatively simple and inexpensive way to increase assertiveness, which favors the socio-economic environment prevailing in Bosnia and Herzegovina.

CONCLUSION

The association of the concept of assertiveness with the strategies of dealing with stress, which was presented in our research, gives us the opportunity to memorize the planning of possible training to increase their adaptation to specific groups, and therefore their effectiveness. Unlike teaching individual stress-bearing strategies, teaching assertiveness to an individual provides a tool that is mapped to all aspects of our lives by influencing the way we deal with the world around us. We do not want to imply that direct teaching strategies for coping with stress is worse or better than training for assertiveness, but to approach the problem of stress from another direction and to propose additional alternatives.

Conflict of interest: none declared.

REFERENCES:

1. Lazarus RS, Folkmann S. Stres, procjena i suočavanje. Jastrebarsko: Naklada Slap; 2004.
2. Spencer AR. Temelji psihologije. Jastrebarsko: Naklada Slap; 2000.
3. Bernbaum H, Connely J. The effect of stress on hedonic capacity. J Abnorm Psychol. 1993;102(3):474-481.
4. Aldwin CM. Stress, coping, and development. New York: Guilford Press; 1994.
5. Webster M. Webster's Ninth New Collegiate Dictionary. Springfield: Mass U.S.A, 1998.
6. Coyne JC, Racioppo MW. Never the Twain shall meet? Closing the gap between coping research and clinical intervention research. Am Psychol. 2000;55(6):655-664.
7. Watson MJ, Logan HL, Tomar SL. The influence of active coping and perceived stress on health disparities in a multi-ethnic low income sample. BMC Public Health. 2008;8:41.
8. Senel P, Consuelo A, Amaury N, McPherson R, Pisecco S. Relation between assertiveness, academic self efficacy and psychosocial adjustment among international graduate students. Journal of College Student Development. 2002;43:632-42.
9. Živčić-Bećirević I. Konstrukcija skale samoprocjene asertivnog ponašanja djece. Godišnjak Zavoda za psihologiju (Rijeka). 1996;4(5):85-90.
10. Maccoby EE. The Role of Parents the Socialization of Children. Developmental Psychology. 1992;28(6):1006-1017.
11. Zarevski P, Mamula M. Pobijedite sramežljivost: a djecu cijepite protiv nje. Jastrebarsko: Naklada Slap; 1998.
12. Sarkova M, Bacikova-Sleskova, M, Orosova, O, Greckova, AM, Katreniakova, Z, Jitse W. The associations between assertiveness, psychological well-being and self-esteem in adolescents. Journal of Applied Social Psychology. 2013;43:147-54.

13. Alinčić M. Osobine ličnosti i asertivnosti kao prediktori samopoštovanja i socijalne anksioznosti. Primjenjena psihologija. 2013;6(2):139-154.
14. Svetić A. Ispitivanje odnosa asertivnosti, ispitne anksioznosti i školskog uspjeha kod djece osnovnoškolske dobi. Neobjavljen diplomski rad. Zadar: Odsjek za psihologiju Sveučilišta u Zadru; 2000.
15. Tovilović S. Asertivni trening: efekti tretmana, održivost promena i udeo terapeuta u terapijskom ishodu. Novi Sad: Odjek za psihologiju Filozofski fakultet, Novi Sad; 2005.

Reprint requests and correspondence:
Alma Džubur-Kulenović, MD, PhD
Psychiatric Clinic
Clinical Center University of Sarajevo
Bolnička 25, 71000 Sarajevo
Bosnia and Herzegovina
Email: almadzuburkulenovic@yahoo.com

Bosnia and Herzegovina versions of Guidelines for Patients!
Bosanskohercegovačka verzija Vodiča za pacijente!





Social adaptation in patients with anxious and depressive disorders

Socijalno prilagođavanje osoba sa anksioznim i depresivnim poremećajima

Mira Spremo[1,2]*, Nada Veselić[3], Zihnet Selimbašić[4,5]

[1]Psychiatric Clinic, University Clinical Center Banja Luka, Branka Ćopića 15, 78000 Banja Luka, Bosnia and Herzegovina
[2]Faculty of Medicine, University of Banja Luka, Save Mrkalja 14, 78000 Banja Luka, Bosnia and Herzegovina
[3]Faculty of Philosophy, University of Banja Luka, Bulevar vojvode Petra Bojovića, 1A, 78000 Banja Luka, Bosnia and Herzegovina
[4]Psychiatric Clinic, University Clinical Center Tuzla, Ulica prof. dr. Ibre Pašića, 75000 Tuzla, Bosnia and Herzegovina
[5]Faculty of Medicine, Univerity of Tuzla, Univerzitetska 1, 75000 Tuzla, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

The aim of the study was to examine the social functioning of people with anxiety and depressive disorders and to compare the social adaptability of this group of respondents with a group of the general population. Materials and methods: the sample included 107 respondents (48.6% with a diagnosis of anxiety or depressive disorder (Group II) and 51.4% from the general population (Group I). The following standardized instruments were applied: Spielberger State-Trait Anxiety Inventory - STAI, Beck depression inventory - BDI, Social Adaptation Self-evaluation Scale (SSAS) and Socio-demographic Questionnaire. Data were subjected to descriptive analysis and comparison of groups by means of the t-test. The results are presented descriptively and in tables. There was no significant difference between the groups of respondents regarding demographic characteristics. Statistically significant differences were minor, specifically of moderate effect obtained in case of addictive disorders in parents ($\chi 2$ (1)=20.90, p<.001, φ=.-46) and the presence of fear in family members ($\chi 2$(1)=13.62, p<.001, φ=-41), where the respondents classified in the experimental group had more frequent cases of addiction and fear in their family history. Major statistically significant differences between the groups of respondents were identified with respect to depression, anxiety, both state and trait anxiety, where diagnosed respondents scored higher results. The scale of social adaptation generated negative statistically significant correlations with depression and anxiety scales as a tendency, where depression was in a somewhat higher correlation with this scale. Conclusion: adaptable social functioning and anxiety of respondents with anxiety and depression disorders is altered even in the state of solid remission, thus indicating higher sensitivity in these subjects.

Keywords: anxiety, depression, disorders, social functioning

SAŽETAK

Cilj istraživanja je ispitati socijalno funkcionisanje osoba sa anksioznim i depresivnim poremećajima i uporediti socijalnu adaptabilnost ove grupe ispitanika sa grupom iz opšte populacije. Materijali i metode: uzorkom je obuhvaćeno 107 ispitanika (48,6% ispitanika čine osobe koje imaju dijagnozu (grupa II) i 51,4 % ispitanika iz opšte populacije (grupa I). U ispitivanju su primijenjeni sljedeći instrumenti: Spielbergerov upitnik za samoprocjenu-STAI, Beckov inventar depresije-BDI, Skala samoizvještavanja socijalne adaptacije (SSAS) i Upitnik o sociodemografskim podacima. Podaci su podvrgnuti deskriptivnoj analizi i poređenju grupa putem t-testa. Rezultati su predstavljeni deskriptivno i tabelarno. Ispitivane grupe se ne razlikuju značajno prema demografskim obilježjima. Statistički značajne razlike su male, odnosno umjerene veličine efekta dobijene u slučaju bolesti zavisnosti kod roditelja ($\chi 2$(1)=20.90, p<.001, φ=.-46) i prisutnosti strahova kod članova porodice ($\chi 2$(1)=13.62, p<.001, φ=-41), pri čemu oni ispitanici koji su klasifikovani u grupu sa dijagnozom imaju češće slučajeve zavisnosti i strahova u svom porodičnom kontekstu. Identifikovane su velike statistički značajne razlike između grupa ispitanika u odnosu na depresivnost, anksioznost kao stanje i kao osobinu ličnosti, tako što ispitanici sa dijagnozom ostvaruju više rezultate. Skala socijalnog prilagođavanja ostvaruje negativne statistički značajne korelacije sa skalama depresivnosti i anksioznosti kao dispozicijom, pri čemu je depresivnost u blagoj višoj korelaciji sa ovom skalom. Zaključak: Adaptibilno socijalno funkcionisanje i anksioznost ispitanika sa anksioznim i depresivnim poremećajima se razlikuje i u stanjima dobre remisije što ukazuje na veću osjetljivost dijagnostikovanih ispitanika.

Ključne riječi: anksioznost, depresivnost, poremećaji, socijalno funkcionisanje

INTRODUCTION

Anxiety is the oldest universal human emotion and a primary emotion of each human being. These feelings achieve significant impact on the relationships of personality with outside as well as with their inner world (1). Anxiety is a reaction not only to external but also more frequently to external threatening danger (2). It most often occurs during childhood, thus the studies show that almost 6-8% of children and adolescents suffer from certain type of anxiety disorder (3). The time of onset, course and outcome differ in different anxiety disorders.

Despite everything, anxiety is a normal phenomenon in everyday life. Pathology is related to intensity, duration and number of symptoms present, and especially the degree of control which the person retains in the state of anxiety. Pathological form of anxiety occurs outside of dangerous situation, lasts long after activity of a stressor or danger, hinders the functioning of the individual, and negatively changes his/her thinking and behavior, thus emerging from the framework of his/her control (4). Anxiety can be understood as more general, fundamental and more permanent phenomenon, and is therefore determined as a feature, i.e. personality trait. In theory of personality, anxiety as a trait is commonly referred to as neuroticism or negative feeling and is one of the most commonly described personality traits. It is considered a basic emotion (like fear) and is frequently in the spotlight of interest of clinical psychology (5). Taylor (6) considers that trait anxiety relates to general tendency of a person to react anxiously to potential anxiety provoking stimuli. In the literature, a type of the affective connection is often mentioned as a cause of susceptibility to anxiety which according to Bowlby has central importance in the emotional life of people (7). Bowlby (7) indicated two models of parental or guardian behavior which can contribute to development of anxiety and depression: threatening and hostile, that is rejecting behavior. He believed that forms of threatening behavior, that is the behavior which shows the child that the parent is not emotionally or physically accessible when needed, are characteristics of affectionate relationship of particularly anxious people. Every year in Great Britain one out of five women and one out of ten men take medication to treat depression (8).

Depressive symptoms of different intensity occur in the life of almost all individuals and are most commonly reaction to stressful situations. Depression should be distinguished from normal sorrow, i.e. grief (1). Depression belongs to a group of mood disorders and is one of the most common mental disorders in the general population. Although this is the most common disorder, recognition of disorder is not always successful. Depression is accompanied by serious personal, interpersonal and social consequences. On the subjective level, depression causes significant psychological changes, reduces the quality of life, and increases the risk of cardiovascular diseases and suicide (9). More serious cases of depression may lead to material collapse and family breakdown. Depression is characterized by different clinical presentation with number of symptoms. The key features of depression include: low mood, lack of energy, loss of interests or pleasure. Other common symptoms include low concentration, low self-esteem, feelings of guilt, pessimism, suicidal thoughts, sleep disorder, loss of appetite etc. (9). Depression reduces the quality of life; the more severe the depression, the worse quality of life (10). Depression is often accompanied by the feeling of anger, helplessness and anxiety. Although it remains unknown, a large number of people who have experienced anxiety have also experienced depression. Depression occurs in women 2 - 3 times more often than in men. There is a difference in treatment and in reaction to medications (11). Depression is also linked to completeness of a family. It has been shown that in Australia depressed children living with one parent are more frequently depressed than those living with both parents (12).

It is important to clarify that depression and anxiety are not the same, although at first glance they may seem very similar. Studies have shown that depression is correlated with gender, completeness of a family and financial situation. Depression and anxiety are influenced by cognitive distortions and they usually go hand in hand. Studies have shown that 60 - 70% of people suffering from clinical depression also have the feeling of anxiety and vice versa, 50% of patients with anxiety as a main symptom also suffer from depression. Reaction to stress is excessive in both disorders. Nowadays, the fact that anxiety is frequently a precondition for development of depression provides broad possibilities in the prevention of depression (9).

Social adaptation is linked with symptoms of anxiety and depression. People with depression find it difficult to participate in social and family functioning, and work. Inability linked with depression may to a large extent worsen activities and productivity in people with chronic somatic disorders (10,13). Social adaptation can be analyzed in different frameworks and there is little consensus as to how to describe it in the best possible way. Studies report about multidimensional definition which includes behavioral and affective indicators of social functioning. Behavioral, and particularly affective indicators of social functioning, are damaged in patients with anxious and depressive disorders and in majority of patients with comorbidity. Behavioral indicators present objective and quantitative measures of social functioning which include the size of social network, frequency of social activities and social support. Affective indicators rather reflect subjective and evaluation measures of interpersonal and socio-emotional functioning including loneliness, affiliation to the group and perceived social exclusion. Social adaptation of depressive patients suggests that affective indicators are significantly more present than behavioral. Santini, et al. (14) show that perceptive support, as an affective indicator, is more important for social adaptation of depressive patients than the received support (behavioral indicator) and it increases levels of perceptive support which has a protective role in depressive episodes. Cacioppo, et al. (15) state that loneliness increases the risk of depression. McKnight, et al. (16) report that anxiety disorders such as Agoraphobia and Generalized Anxiety Disorder (GAD) have stronger bond with social isolation than Social Phobia. That is contrary to previous claims that Generalized Anxiety Disorder, Social Phobia and Panic Disorders do not differ in terms of overall social damage or with findings that people with Panic Disorder are more socially damaged than the people with other anxiety disorders (17).

Interestingly, after the affective disorder symptoms remission is achieved, residual damage remains whereas poor social adaptation may be a predictor of the future psychopathology (10).

MATERIALS AND METHODS

This study was anonymous, conducted in Banja Luka in 2017/2018 on an appropriate number of respondents. The respondents were informed about the purpose of the study and the manner of collected data usage (they were previously briefed about the aim of the study and they were explained that the collected data would be exclusively used for the purpose of the study). The questionnaires were filled in by respondents who voluntarily accepted to participate in the study. Two approximately similar groups of participants were analyzed. The clinical sample was collected during outpatient control checkups and it consisted of patients diagnosed with affective (N=28) and anxiety disorders (N=23), according to DSM IV diagnostic criteria (18). In the course of the study all respondents were in satisfactory remission, regularly taking their medication and having regular control checkups. Control group consisted of randomly selected respondents with no

diagnosis, analyzed during systematic psychiatric evaluation. The groups were approximately uniformed in respect of gender and age. Each respondent needed between 30 to 45 minutes to complete the set of questionnaires. Data was subjected to descriptive analysis and comparison of groups by means of t-test.

Instruments

1. For examination of anxiety STAI - Spielberger State- Trait Anxiety Inventory was used (20). The STAI is used to measure anxiety and it consists of two forms: "state anxiety" related to transient emotional state characterized by conscious feeling of tension and fear and "trait anxiety" related to relatively stable individual differences in tendency to anxiety. Both questionnaires are recommended for studying anxiety in medical and psychological studies and consist of 20 questions which are answered based on a 4-point frequency scale. For each statement respondents have to select one of four provided answers. In the "state anxiety" scale the respondents should focus on their present feeling whereas in the "trait anxiety" scale they are required to describe how they feel in general. The respondents were suggested to complete both questionnaires in 20 minutes so that possible changes in the anxiety level were reduced to a minimum. Validity tests showed that Spielberger scales had great validity. Reliability coefficient for "trait anxiety" scale was between .65 and .86 while for the "state anxiety" scale it varied from .16 to .62. In our study the coefficient of the internal consistency for the "state anxiety" scale was .93, and for the "trait anxiety" scale .94.

2. The Beck Depression Inventory (BDI) (21) was used for examination of depression. The instrument consists of 21 question to evaluate the severity of depression in adoloscents and adults. The score on this scale is obtained simply by adding up the score for each 21 questions used for the depression evaluation, which includes: mood, pessimism, sense of failure, self-dissatisfaction, guilt, punishment, self-dislike, self-accusation, suicidal ideas, crying, irritability, social withdrawal, hesitance, body image, work difficulties, insomnia, fatigue, appetite, weight loss, bodily preocupation and loss of libido. Each item consists of four possible answers scored from zero to three. The respondent has to select the most suitable answer, i.e. the answer which in the best way describes how he or she felt over the past two weeks, including the day of the test. The results indicate one of the following degrees of depresion: 0-9 indicates minimal depression, 10-16 indicates mild depression; 17-29 indicates moderate depression and 30-63 indicates severe depression (22). Reliability obtained from clinical populations is from .79 to .90. Beck, Steer and Garbin (22) have published the alfa coefficient of 0.86 in meta-analysis for nine psychiatric samples, and .81 for 15 non-psychiatric samples which suggests that the scale has high inner consistancy both in clinical and non-clinical population. The review of ten studies which used test-retest reability method with psychiatric patients, Beck, et al. (23) have published that Pearson's correlation, the product of moment in different time intervals, was from .48 to .86 and in non-psychiatric patients the span was from .60 and .90. In this study the BDI also showed high inner consistency and Cronbah's alfa of the overall inventary was .90.

3. Social Adaptation Self-evaluation Scale (SSAS) consists of 21 items and was design by Bosc et al., (24). SSAS measures the degree of depressive patients social functioning. Patients answer the first and the second questions related to their employment with yes or no. The other questions are answered using a four-degree scale. The scale is one dimensional. Total theoretical score is from 0 to 60, and the higher score shows the better social adaptation. Internal consistency coefficient in this study was .81.

4. Socio-demographic Questionnaire on respondents' data consists of 12 questions: gender, age, education, profession, marital status, family structure, birth order and if there have been addiction diseases or expressed fears in the family.

RESULTS

The final sample included 107 subjects, (30.8% male and 59,2% female). Out of the initial number of 131 subjects who participated in this study, 24 were excluded from analysis because the data were incomplete. Both univariate and multivariate outliers (initial sample N=131). The structure of the sample included 51.4% of respondents from general population, while 48.6% of the respondents had a diagnosis of anxiety or depressive disorder. The research data indicate that the rate of adherence to antidepressant therapy recommendations was about 40% (19).

The control group consisted of randomly selected respondents without a diagnosis. The respondents were mainly young people (37.4% of them were from 20 to 25 years of age), whereas only a small number of respondents were elderly people (6.5% of them were over 40 years of age). The rest of the respondents was in their middle ages, namely from 26 to 30 years of age (15.9%), from 31 to 35 years of age (15.0%) and from 36 to 40 years of age (15.0%) whereas 10.3% of the respondents did not declare in this regard. The majority of the respondents completed secondary school (66.4%), followed by respondents with higher education or university (30.8%) and the rest of the respondents were with completed primary school. Respondents from this study were single (52.3%), followed by those who were married (33.6%) and the lowest number related to divorced respondents (2.8%) and widows/widowers (2.8%). More than half of the respondents resided in the city (64.5%) followed by those who resided in suburbs (18.7%) and 15.9% of them who lived in the village. Out of the total number of respondents, 34.6% of them did not declare in respect of their working status, whereas 39.3% of them were employed, and 24.3% were unemployed. Majority of respondents were firstborn children (57.9%), followed by those who were second born children (29.9%) and one examinee was a third born child whereas other respondents did not declare in this regard. 72.9% of respondents originated from complete families, 9.3% of them had only mother, 7.1% had both parents but did not live with them, while 8.4% of them had only father, and 4.6% did not have any parents. The rest of the respondents (4.7%) did not provide information in this regard. 44.9% of respondents resided with their parents, 16.8% with their spouses, 7.5% of them lived alone, 6.5% of them lived with their children while the remaining 24.1% of them did provide information in this regard.

Smaller number of respondents stated addiction (19.6%) and fear (18.7%) attributed to some member of their families. Potential differences between all demographic variables and affiliation to diagnosed (Group II) and non-diagnosed group (Group I) were analyzed.

Statistically significant differences of minimum and moderate effect size respectively (25) were obtained in the case of the parental addiction ($\chi2(1)=20.90$, $p<.001$, $\varphi=.46$) and fear presence in the family members ($\chi2(1)=13.62$, $p<.001$, $\varphi=.41$), where diagnosed group respondents had more frequent presence of addiction and fear in their families.

Table 1 Results of descriptive statistics in respect of gender, t-test for independent samples and internal consistency of the used scales.

Scales	Healthy		Diagnosis						
	M	*SD*	*M*	*SD*	α	ώ	*t*	*p*	*d*
STAT	37.45	9.24	48.70	12.84	.93	.94	-5.60	<.001	-1.08
STAI	37.88	10.90	48.00	12.84	.95	.95	-4.41	<.001	-.86
BDI	25.49	6.54	34.58	8.97	.90	.91	-6.05	<.001	-1.17
SASS	43.50	6.10	41.64	7.24	.81	.82	.85	.40	.21

Table 1 shows that all used scales have satisfactory internal consistency according to Cronbach's alfa and McDonald's omega coefficient (26).

Statistically significant differences were found between the respondents from diagnosed and non-diagnosed groups in respect of state anxiety (25). Dominant statistically significant difference was determined in the trait depression and anxiety scale (27). It was evident in all three cases that respondents with official diagnoses achieved higher scores. Gender differences were tested in respect of all used scales with statistically significant differences.

Table 2 Correlations of the used scales in respect of group affiliation.

	STAT	STAI	BDI	SASS
STAT		.76**	.51**	-.50*
STAI			.51**	-.32
BDI				-.55*
SASS				
STAT		.76**	.68**	-.50*
STAI			.69**	-.53**
BDI				-.71**
SASS				

*Note: Correlation coefficients over the horizontal line are related to respondents from healthy population and those below the line are related to diagnosed respondents; * p<.05 **p<.01*

Results in Table 2 relate to correlations between all used scaled in respect of the two groups of respondents - those with a diagnosis of anxiety or depressive disorder and those without a diagnosis. With regard to the first group of respondents, statistically significant correlation of the anxiety scale and depression and anxiety trait is evident, whereas both anxiety scales are identically correlated with depression. Social adaptation scale has negative statistically significant correlations with depression and anxiety scales where depression is in slightly higher correlation with this scale. Identical correlations between state and trait anxiety as in the first group, and there is almost identical correlation between these two scales and depression scale, where state anxiety, although to less significant extent, is however strongly correlated with depression scale. Also, in comparison to the first group of respondents, negative correlation of state anxiety on the one hand and social adaptation on the other hand was found, besides negative correlations of social adaptation and trait anxiety and depression (the strongest link with this scale).

DISCUSSION

This study dealt with understanding the social adaptation of people suffering from anxiety and depressive disorders in relation to the group of respondents from general population, specifically respondents without mentioned disorders. The results show that there are no significant differences between the groups in respect of sociodemographic features, although data from other studies show that material status, birth order, separation, place of residence and marital status are linked with anxiety and depression symptoms, and anxiety is particularly linked with gender and socioeconomic status (28, 29, 30). Although majority of respondents from this study were first born child (57.9%), statistically significant difference between the examined groups was not established, which is contrary to the results of similar studies showing that anxiety and depression are linked with birth order. First-borns are somewhat "damaged" in their socio-emotional relation with mother who pays more attention to the later born child (28, 31). Out of the total number of respondents from this study over 50% (52.3%) of them were single, and only 7 (5%) of them lived alone which was to a large extent linked with their economic situation, but also with personal and traditional family attitudes. Significant number of respondents resided in the same household with their parents, and according to cultural and traditional family beliefs they were less socially involved and majority of their needs satisfied indirectly over their parents and family members. The study data show that married people have the lowest degree of depression, and single or divorced people have the highest depression prevalence (32). It was hypothesized that unemployment would be more frequent in anxious and depressive people, which was not the case. We assume that the reason thereof was in low number of unemployed respondents, weak form of the disorder and well-maintained remission of the respondents. One of the possible explanations is that the respondents were of younger age and functionally well preserved. Rizvi (33) examined anxiety in depressive patients and concluded that there was a higher level of anxiety in unemployed, depressive respondents. Lower level of education, unemployment and job loss contribute to psychological and economic stress which could be predisposition or risk factor for depression, and on the other hand depression can prevent sick people from finding and keeping a job (32). In similar studies employed respondents with depressive disorders were of younger age, with few hospitalizations and the first depressive episode occurring at younger age. Groups of employed and unemployed depressive respondents differed in respect of illness duration, expressed anxiety and anhedonia, and overall reduced social functioning (33). In the examined sample (Group I and Group II), the majority of respondents were from complete families (72.9%), although the trend of increased number of divorces, one-parent families and singles is evident in our society. There was no significant difference between male and female respondents included in the study, although majority of studies show the higher presence of the mentioned symptoms in women. Data obtained from earlier studies point out that women show more symptoms of psychological pain than men (somatization, depression, anxiety, phobia) (34). In the study of Despot-Lucina (35) women also had more psychosomatic difficul-

ties, they showed higher level of depression and less satisfaction with life. One of the possible explanations is that women are more inclined to internalized forms of coping with stress (sadness, loneliness, concern, self-rejection) (36). Besides, women are more inclined to self-blame and social support requests (37). The risk of illness is higher in women, from 1.4:1 to 2.7:1 in all countries and has persisted during the time (19). Also, anxiety is frequently found in female respondents (38). The highest depression prevalence has been observed in early adulthood, from 15 to 44 years, decreasing with age, and is less than 5% in elderly population (32). Our study also included respondents of younger age (20-40 years), but also due to the fact that old age required significantly more variables that are more difficult to control (physical illness, loneliness due to the death of a spouse, hormonal changes, CNS organic changes, retirement and other), which may have significant influence on symptomatology (occurrence or deterioration of the existing symptoms). More than half of the respondents resided in the city (64.5%), 18.7% of them resided in suburbs and 15.9% in the village. Although there was no difference between the groups of respondents in respect of place of residence, other studies point to lower prevalence of depression and depressive symptoms in rural in respect of urban areas (39). Guida and Ludlow (40) have discovered that lower socioeconomic status is associated with higher levels of anxiety and vice versa.

Respondents with anxiety and depressive disorders do not differ significantly from general population in terms of education, employment, marital status and age. The results of this study indicate that affective and anxiety symptoms are more associated with respondents having symptoms of addiction and fear present among the members of their families (41). Significantly higher number of female relatives suffers from anxiety disorders in respect of male relatives, whereas there is no gender difference in the prevalence of mood disorders and abuse of psychoactive substances (42).

Significant difference between the examined groups in terms of anxiety and depression was expected. Although Group II respondents were in satisfactory remission, on the self-assessment scale, however, they showed significantly higher scores in respect of anxiety and depression in respect of Group I respondents, which was not noticeable in the social adaptation scale where there was no significant difference between these two groups of respondents (Table 2). This can be explained by features of the clinical sample consisting of people with anxiety and depressive disorders which last longer, with no psychiatric comorbidity, and milder overall clinical picture. Most likely, Group II respondents made additional effort in social adjustment and functioning, regardless of the symptoms which could be associated with fear of stigmatization and social rejection. Given that the scale measures social adaptation, it is questionable to which extent respondents from the clinical sample were exposed to new social situations particularly due to partly limiting effect of symptoms.

Data from similar studies show that in comparison with the control group, patients with anxiety disorders have lower quality of life, especially in the field of social interaction and patients and feelings of subjective well-being (43).

The social adaptation scale was significantly correlated with the depression ($p < .01$) and state anxiety scale ($p < .01$), which was not recorded in Group I. Significantly higher scores regarding anxiety and depression of Group II respondents indicate to lower stress resistance in respect of Group I respondents, which was confirmed by the study results showing that low levels of depression were predictors of better psychological resistance (44).

The study deficiencies relate to the size of the sample, and use of self-reporting questionnaires which may trigger a large number of socially desirable responses not corresponding to actual situation. Respondents were examined in the state of satisfactory remission during outpatient checkups, which provided a positive and current picture of social functioning, not reflecting their general condition. Longitudinal monitoring of respondents would contribute to objectivity of data.

CONCLUSION

It is known that anxiety and depression interfere with social functioning and adaptive behavior, but the results of this study show that there is no statistically significant difference between the study groups, regardless of statistically significantly higher representation of anxiety and depression in the clinical sample (Group II). Also, the results show that depressive respondents had weaker social functioning than anxious respondents, although it was not statistically significant at the time of the study, which was probably due to satisfactory recovery. Satisfactory social functioning of the respondents with anxiety and depression disorders is to a large extent associated with adherence to prescribed medication regimen and regular checkups which support and encourage development of specific mental skills reducing and preventing anxiety and depression.

Conflict of interest: none declared.

REFERENCES

1. Kaličanin P. Klinička psihijatrija. Beograd: Institut za mentalno zdravlje, 2003.
2. Golubović ZG. Osnovi opšte psihopatologije. Niš: Unigraf, 2008.
3. Bogels SM, Zigterman D. Dysfunctional cognitions in children with social phobia, separation anxiety disorder, and generalized anxiety disorder. Journal of Abnormal Child Psychology. 2000;28,205-211.
4. Caster JB, Inderbitzen HM, Hope D. Relationship between youth and parent perceptions Beck, R. C. Motivacija: Teorija i načela. Jastrebarsko: Naklada Slap; 2003.
5. Taylor, S. Issues in the conceptualitazion and measurement of anxiety sensitivity, Journal of Anxiety Disorders. 1995b; 9, 163-174.
6. Taylor S. Issues in the conceptualitazion and measurement of anxiety sensitivity. Journal of Anxiety Disorders. 1995b;9:163-174.
7. Bowlby J. Attachment and Loss: Separation. New York: Basic Book, 1973.
8. Kendler KS, Neale MC, Kessler RC, Heath AC, Eaves LJ. Generalized anxiety disorder in women: A population - based twin study. Arch Gen Psychiatry. 1992;49(4):267-272.
9. Baldwin DS, Birtwistle J. An Atlas of Depression. Southampton: University of Southampton 2002.
10. Wells KB, Stewart A, Hays RD, Burnam MA, Rogers W, Daniels M, et al. Defuncioning and weing-being of depressed patient: result from the medical outcome study. JAMA. 1989;262(7):914-919.
11. Wade TJ, Cairney J, Pevalin DJ. Emergence of gender differences in depression during adolescence: national panel results from three countries. J Am Acad Child Adolesc Psychiatry. 2002;41(2):190-8.
12. Cairney J, Thorpe C, Rietschlin J, Avison WR. 12-month prevalence of depression among single and married mothers in the 1994 National Population Health Survey. Can J Public Health. 1999;90(5):320-4.
13. Hays RD, Wells KB, Sherbourne CD,Rogers W, Spritzer K. Functioning and well-being outcomes of patients with depression compared with chronic general medical illnesses. Arch Gen Pschyatry 1995; 52(1):11-9.
14. Santini ZI, Koyanagi A, Tyrovolas S, Mason C, Haro JM. The association betwen social relationship and depression: a systematic review. J Affect Disod. 2015;175:53-65.
15. Cacioppo JT, Hughes ME, Waite LJ, Hawkley LC, Thisted RA. Loneliness as a specific

risk factor for depressive symptoms: Cross-sectional and longitudinal analyses. Psychol Aging. 2006;21(1):140-151.
16. McKnight PE, Monfort SS, Kashdan TB, Blalock DV, Calton JM. Anxiety symptoms and functional impairment: a systematic review of the correlation between the two measures. Clin Psychol Rev. 2016;45:115-30.
17. Markowitz J, Weissman M, Ouellette R, Lish J, Klerman G. Quality of life in panic disorder. Arch Gen Psychiatry. 1989;46:984-992.
18. American Psychiatric Association. DSM-IV Diagnostic and Statistical Manual of Mental Disoders-4th Ed. American Psychiatric Assocation, Washington DC, 1994.
19. Janković S. Depression. In: Atanasković-Marković Z, Bjegović V, Janković S et al. Serbian burden of disease and Injury in Serbia. Belgrade: Ministry of Health of the Republic of Serbia. 2003;109-113.
20. Spielberger CD, Gorssuch RL, Lushene PR, Vagg PR, Jacobs GA. Manual for the State-Trait Anxiety Inventory. Palo Alto, CA: Consulting Psychologists Press, 1983.
21. Beck AT, Steer RA, Garbin MG. Psychometric proporties of the Beck Depression Inventory: Twenty - five years of evolution. Clinical Psychology Review. 1988;8(1),77 -100.
22. Beck AT, Ward C, Mendelson,M. Beck Depression Inventory (BDI). Arch Gen Psychiatry. 1961;4(6):561-571.
23. Beck AT, Steer RA, Garbin MG. Psychometric proporties of the Beck Depression Inventory: Twenty - five years of evolution. Clinical Psychology Review. 1988;8(1):77-100.
24. Bosc N, Dubini A, Polin V. Development and validation of a social functioning scale, the Social Adaptation Selfevaluation Scale. Eur Neuropsychopharmacol. 1997;7(Suppl 1):57 -70.
25. Cumming G. Understanding the new statistics: Effect sizes, confidence intervals and meta-analysis. New York: Taylor and Francis Group, 2012.
26. Dunn TJ, Baguley T, Brunsden V. From alpha to omega: a practical solution to the pervasive problem of internal consistency estimation. Br J Psychol. 2014;105(3):399-412.
27. Sawilowsky, S. New effect size rules of thumb. Journal of Modern Applied Statistical Methods. 2009;8(2):597- 599.
28. Ornstein R. The roots of our "I". Zagreb: Educa, 2001.
29. Stanojević D, Jaredić B, Manić S. Depression in Adolescents in Kosovo and Metohija. Timočki Medicinski glasnik. 2012;37(4):223-228.
30. Zajonc RB, Markus GB. Birth order and Intellectual Development. Psychological Review. 1978;82(1):74-88.
31. Jeremić V, Todorović J, Vidanović S. Depression and play about self-adolescents without parental care. Yearbook for Psychology. 2006;4:173-188.
32. Horwath E, Cohen RS, Weissman MM. Epidemiology of depressive and anxiety disorders. In: Tsuang MT, Tohen M (eds). Text book in psychiatric epidemiology. New York: Wiley-Liss; 2002:389-426.
33. Rizvi SJ, Cyriac A, Grima E, Tan M, Lin P, Gallaugher LA, et al. Depression and Employment in Primary and Tertiary Care Settings. Can J Psychiatry. 2015;60(1):14-22.
34. González de Rivera, JL. de las Cuevas, C, Gracia-Marco, R, Henry-Benítez, M. Age, Sex, and Marital Status Diffferences in Minor Psychiatric Morbidity. Europ J Psychiatry. 1991;5(3):166-176.
35. Despot - Lučanin J. Experience of aging. Jastrebarsko: Edition Slap, 2003.
36. Lacković-Grgin, K. Stress in Children and Adolescents. Jastrebarsko: Edition Slap, 2000.
37. Whitbourne SK. Adaptation. In Birren JE (Ed.), Encyclopedia of Gerontology (pp. 29 - 36). San Diego: University of California, Elsevier Inc., 2007.
38. Cizek GJ, Burg SS. Addressing test anxiety in a high-stakes environment. Thousand Oaks, CA: Corwin Press, 2006.
39. Ayuso-Mateos JL, Vázquez-Barquero JL, Dowrick C, Lehtinen V, Dalgard OS, Casey P, et al. Depressive disorders in Europe: prevalence figures from the ODIN study. Br. J. Psychiatry. 2001;179:308-16.
40. Guida FV, Ludlow LH. A cross-cultural study of test anxiety. Journal of Cross-Cultural Psychology. 1989;20(2):178-190.
41. Center for Substance Abuse Treatment. Substance Abuse Treatment and Family Therapy. Rockville (MD): Substance Abuse and Mental Health Services Administration (2004).
42. Skre I, Onstad S, Edvardsen J, Torgersen S, Kringlen E. A family study of anxiety disorders: familial transmission and relationship to mood disorder and psychoactive substance use disorder. Acta Psychiatrica Scandinavica. 1994;90(5):366-74.
43. Cramer V, Torgersen S, Kringlen E. Quality of life and anxiety disorders: a population study. J Nerv Ment Dis. 2005;193:196-202.
44. Tugade MM, Fredrickson BL, Barrett FL. Psychological resilience and positive emotional granularity: Examining the benefits of positive emotions on coping and health. Journal of personality. 2004;72(6):1161-1190.

Reprint requests and correspondence
Mira Spremo, MD
Psychiatric Clinic
University Clinical Center Banja Luka
Branka Ćopića 15, 78000 Banja Luka
Bosnia and Herzegovina
Phone: + 387 65 611 898
Email: spremom@gmail.com

Immune parameters as reliable biomarkers during the treatment of patients with renal complications infected with beta-hemolytic streptococci

Imunološki parametri kao pouzdani biomarkeri tokom liječenja pacijenata sa renalnim komplikacijama inficiranim sa beta hemolitičkim streptokokom

Mevludin Mekić[1]*, Đemo Subašić[2], Edmira Isak[2], Majda Hadžiabulić[2], Alen Džubur[1], Sanja Mišeljić[1]

[1]Clinic of Heart Disease, Blood Vessels and Rheumatism, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina
[2]Faculty of Natural Sciences and Mathematics, University of Sarajevo, Zmaja od Bosne 33-35, 71000 Sarajevo, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

Objective: the main goal of this research was to define specific quantitative values of immune parameters in cases of post-streptococcal renal complications (APSGN-Acute post-streptococcal glomerulonephritis and ARPG - Acute post-streptococcal rheumatic fever) in purpose of early diagnostic separation these two diseases and to define parameters leading to acute progressive glomerulonephritis for the purpose of effective and timely treatment. Materials and methods: in the laboratory of Insitute for Immunology of the Clinical Center University of Sarajevo (CCUS), in the period from 2015 to 2016y, a total of 350 samples taken from patients with post-streptococcal complications were analyzed. The majority of patients had APSGN, and a lower percentage indicated on ARPG. Quantitative measurements of ASPGN markers (C1qCIC, C3, C4, and IgG) and ARPG markers (ASTO and CRP) were performed by ELISA and Nephelometry methods. Results: laboratory analyses showed that in 57.5% of cases have diagnosed ASPGN. Seven samples showed a higher value of C1qCIC parameters. Those patients had an active disease of high intensity. The mild intensity of disease had 16 patients which are coincided with a quantitative value of C1qCIC. Characteristic values of some parameters for patients suspected of showed 17 samples (42.5%). Values of analyzed ARPG parameters showed an acute phase of disease for 6 samples and 11 samples of the mild intensity of the disease. Conclusion: by this analytic approach, it was possible to fast and early determination of disease activity and clinical separation of ASPGN and ARPG patients, and practical clinical and therapeutic improvement of these diseases. For future researches, it is necessary to perform PCR-CFHR5 gene polymorphism and flow cytometric immunophenotyping analyses of expression D8/17markers. The main goal of these investigations should be the determination of possible specific genetic predisposition for these diseases and selective Anti-StreptInCor vaccine introduction in specific clinical cases.

Keywords: ASPGN markers (C1qCIC, C3, C4 IgG), ARPG markers (ASTO and CRP), S. Pyogenes, ELISA, nephelometry

SAŽETAK

Cilj: glavni cilj ovog istraživanja je definisanje specifične kvantitativne vrijednosti imunih parametara u slučajevima poststreptokoknih renalnih komplikacija i akutnom reumatskom groznicom (APSGN-Akutni poststreptokokni glomerulonefritis i ARPG-Akutna poststreptokokna reumatska groznica) s ciljem rane dijagnostičke separacije ove dvije bolesti i poboljšanja kliničkog menadžmenta pacijenata sa tim komplikacijama. Također je važno pravovremeno definirati kvantitativne vrijednosti određenih biomarkera, koje bi mogle ukazati na mogućnost indukcije akutnog progresivnog glomerulonefritisa, s ciljem pravovremenog i efikasnog liječenja. Materijali i metode: u laboratoriju Instituta za imunologiju KCU Sarajevo tokom 2015. i 2016. godine analizirano je 350 uzoraka pacijenata sa sumnjom na poststreptokokne komplikacije. Većina pacijenata je imala APSGN, i manji broj pacijenata ARPG. Analizirani su ASPGN markeri (C1qCIC, C3, C4 IgG), kao i ARPG markeri (ASTO i CRP), pomoću ELISA testa i nefelometrijskih mjerenja. Rezultati laboratorijskih analiza su pokazali da je kod 57,5% slučajeva dijagnosticiran ASPGN. Sedam uzoraka je pokazalo visoke vrijednosti C1qCIC parametera. Ti pacijenti imaju aktivnu bolest sa visokim intenzitetom. Blagi intenzitet bolesti je imalo 16 pacijenata, što se podudara sa kvantitativnim vrijednostima C1qCIC. Karakteristične vrijednosti nekih parametara dokazano je kod 17 uzoraka (42,5%) pacijanta suspeknih na bolest. Vrijednosti analiziranog parametra ARPG su ukazale na akutnu fazu bolesti kod 6 uzoraka, dok je kod 11 uzoraka tokom iste analize ukazalo na bolest blagog intenziteta. Zaključak: ovakav analitički pristup pokazao se korisnim u smislu određivanja intenziteta ASPGN i ARPG oboljenja (akutna faza i bolest blagog intenziteta), što je od izuzetnog značaja za poboljšanje terapijskog tretmana pacijenata sa ovim komplikacijama. Za buduća istraživanja neophodno je uključiti i PCR analizu polimorfizma CFHR5 gena, imunofenotipizacijske analize ekspresije D8/17 markera, ustanoviti specifičnu genetičku predispoziciju za indukciju poststreptokoknog glomerulonefritisa, te razmotriti mogućnost selektivnog korištenja Anti-StreptInCor vakcine.

Ključne riječi: ASPGN markeri (C1qCIC, C3, C4 IgG), ARPG markeri (ASTO i CRP), S. Pyogenes, ELISA, nefelometrija

INTRODUCTION

Strepococcus pyogenes (GAS-Group A streptococcus) is bacteria which causes infections resulting in various clinical manifestations such as pharyngitis, glomerulonephritis, impetigo and autoimmune diseases, rheumatic fever and rheumatic heart disease. Specific biomarkers monitoring in the renal complications cases of post-streptococcal etiology (APSGN-Acute post-streptococcal glomerulonephritis) and acute post-streptococcal rheumatic fever (APRF), has great importance regarding their clinical management improvement of therapy and disease activity.

Recently, in the worldwide developed countries, the incidence of ASPGN is lower than usually (less than 9:100.000). The main clinical feature of glomerulonephritis is abnormal glomerular deposition of circulating immune complexes (C1Q-CIC, C3, IgG, and IgM) which can lead to obesity, chronic renal failure, diabetes and albuminuria. Also, this pathological state is specific by leukocyte infiltration and kidney endocapillary proliferation as well.

Accumulation C1q-CIC is important differential parameter in case of renal complication autoimmune or post-streptococcal etiology, especially in case of glomerulonephritis. Active phase of APSGN disease is indicated by very high C1Q-CIC values and it means that this accumulation is very important pathogenic event. Often, the high values of this biomarker are followed by severe proteinuria, hypertension, and lower concentrations of C3 and C4 complement components. Therefore, the main feature of APSGN is renal deposition of circulating immune complexes in over 70% of cases. Also it is characterized by higher values of IgM and IgG antibodies in 90% of cases. Globally, each year, this pathogen cause over than 700 million of new infections and about half million of deaths (3.5%) (1-6).

Immunological and epidemiological scientific studies showed, beyond any doubt, that GAS or group A beta hemolytic streptococcus is ethological causative agent of acute rheumatic fever (APRF) and subsequently, rheumatic heart disease (RHD). They are represents together, the important global health problems. The main role in this disease induction and inflammation, play streptococcal M protein. Chronic APRF often cause RHD and corresponding cardiovascular complications. The RHD incidence is greater in women than males, but APRF incidence is almost equal in comparison between males and females. There are more than 30 million new cases each year and about 250 thousands deadly cases (7-10).

In clinical sense, it is very important early and timely ASPGN and APRF separation according their specific biomarkers. For this reason, in this study we estimated quantitative values variations of biomarkers specific for ASPGN (C1qCIC, C3, C4 IgG) and APRF (ASTO i CRP) respectively.

MATERIALS AND METHODS

ELISA

For quantitative ELISA analyzes of C1Q-CIC, IgG, CRP and ASTO biomarkers, we used Hycor Biomedical Autostat TM-II engine according to the manufacturer's instructions (11-12).

Nephelometry

For quantitative measurements of C3 and C4 complement components we used Dade Behrin nephelometer according to corresponding protocol instructions. The intensity of scattered light was proportional to the concentration of relevant proteins in serum. Results are evaluated by comparison with a standard of known concentration (13-14).

RESULTS

In the laboratory of Insitute for Immunology of the CCUS, during the period from 2015 to 2016, a total of 40 serum samples, taken from patients suspected to post-streptococcal complications, was clinicaly selected for quantitative analyzes of ASPGN and APRF biomarkers (C1qCIC, C3, C4, IgG and ASTO and CRP) using ELISA and Nephelometry methods. Obviously, all results were considered as part of other clinical parameters. The main reason for that was early and timely seperation of these diseases, according to precise quantitative determined specific biomarkers. By comparison to clinical findings it was possible to determinate disease activity level for both of them. In majority of cases, laboratory data coincided with the clinical disease findings.

Table I Obtained quantitative values of analyzed ASPGN and APRF laboratory biomarkers.

	ASTO	CRP	C3	C4	IgG	C1q/CIC
Refferal Values	<200 IU/ml	<3 mg/l	0,9-1,8 g/l	0,1-0,4 g/l	7,0-16.0	>50µg/ml
Sample 1.	423,0	95,2	0,81	0,10	9,43	35,60****
Sample2.	1400	36,3	0,77	0,09	13,0	18,50***
Sample 3.	1170	90,0	0,84	0,09	17,6	29,31***
Sample 4.	400	<3	0,97	0,04	17,70	86,07**
Sample 5.	527	<3	0,73	<0,01	19,90	93,67*
Sample 6.	945	<3	0,85	0,07	8,12	55,81**
Sample 7.	622	<3	0,66	0,06	10,30	45,44**
Sample 8.	545	99,5	0,59	0,07	8,73	49,39****
Sample 9.	1690	113	0,90	0,05	6,69	44,44***
Sample 10.	893	112	0,75	0,12	7,65	41,62****
Sample 11.	1310	<3	0,86	0,20	15,90	62,27**
Sample 12.	1330	<3	0,90	0,08	27,80	174,14*
Sample 13.	1060	<3	0,75	0,16	7,65	48,04**
Sample 14.	632	<3	0,86	0,18	10,02	50,61**
Sample 15	744	<3	0,82	0,09	15,70	99,47*
Sample 16.	2940	<3	0,86	0,15	21,06	64,35**
Sample 17.	859	243	0,77	0,21	7,02	27,21***
Sample 18.	1652	<3	0,80	0,06	29,03	211,7*
Sampel 19.	938	<3	0,50	0,05	12,40	87,95**
Sampel 20.	763	81	0,88	0,11	8,47	44,79****
Sample 21.	1632	<3	0,75	0,26	11,13	66,08**
Sample 22.	1414	<3	0,88	0,11	10,09	62,20**
Sample 23.	803	<3	0,55	0,11	27,08	209,6*
Sampel 24.	485	169	0,48	0,02	8,02	22,95****
Sample 25.	375	<3	0,86	0,13	13,80	67,23**
Sample 26.	908	213	0,85	0,10	14,90	32,34****
Sample 27.	290	<3	0,88	0,15	12,30	57,06**
Sample 28.	313	96	0,89	0,14	9,41	31,70****
Sample 29.	624	<3	0,45	0,08	8,56	50,11**
Sample 30.	808	<3	0,61	0,11	14,90	74,36**
Sample 31.	1013	173	0,73	0,12	9,47	34,38***
Sample 32.	502	58	0,83	0,17	10,34	24,11****
Sample 33.	667	<3	0,81	0,02	7,33	33,67****

Sample 34.	380	<3	0,92	0,04	7,33	52,60**
Sample 35.	451	<3	0,87	0,16	22,01	135,07*
Sample 36.	659	212	0,78	0,11	16,30	130,03***
Sample 37.	360	77	0,40	0,17	8,57	34,88****
Sample 38.	399	51	0,78	0,16	9,70	33,89****
Sample 39.	401	<3	0,89	0,12	16,50	131,03*
Sample 40.	1140	<3	0,98	0,17	7,33	51,43**

*APSGN (Acute post-streptococcal glomerulonephritis) - active disease of high intensity
**APSGN (Acute post-streptococcal glomerulonephritis) - mild intensity disease
***APRF (Acute post-streptococcal rheumatic fever) - acute phase of disease
****APRF (Acute post-streptococcal rheumatic fever) - moderate intensity disease

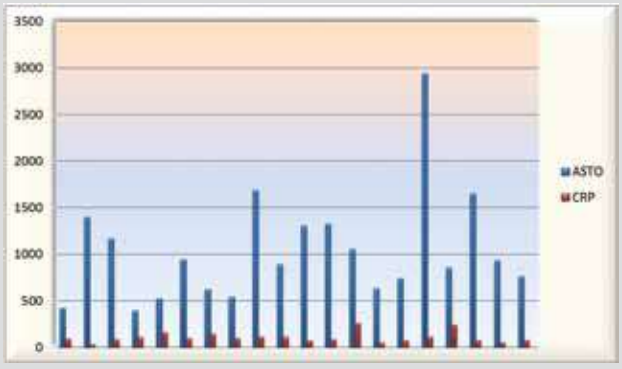


Figure 1 Quantitative comparison of obtained values between ASTO and CRP (Specimens 1-20).

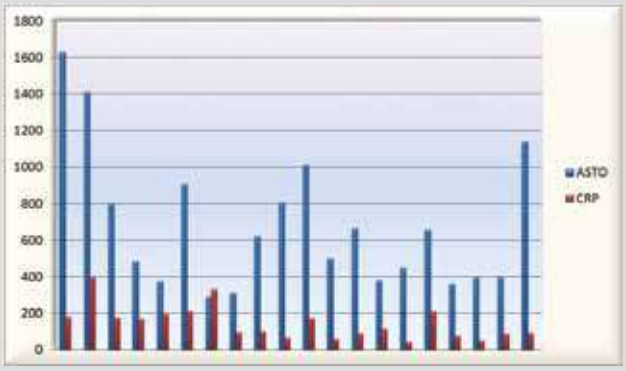


Figure 2 Quantitative comparison of obtained values between ASTO and CRP (Specimens 21-40).

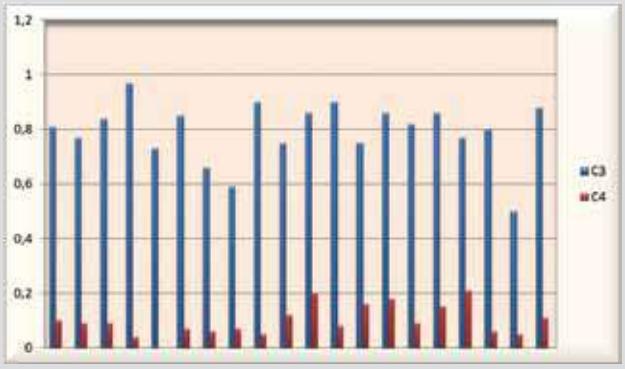


Figure 3 Quantitative comparison of obtained values between C3 and C4 (Specimens 1-20).

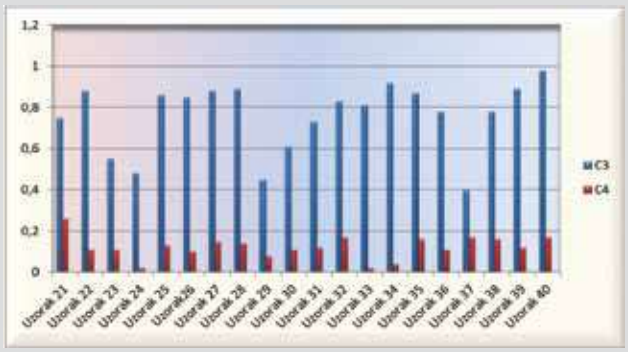


Figure 4 Quantitative comparison of obtained values between C3 and C4 (Specimens 21-40).

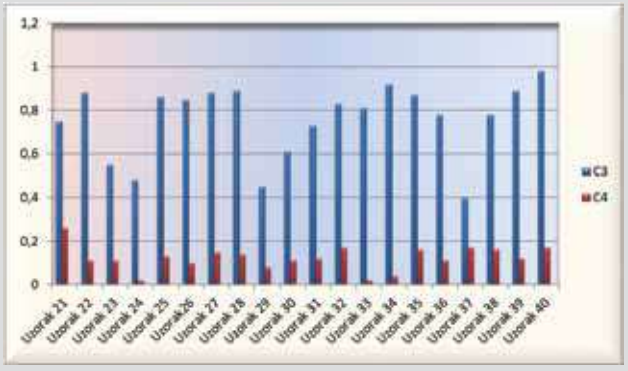


Figure 5 Quantitative comparison of obtained values between IgG and C1q/CIC (Specimens 1-20)

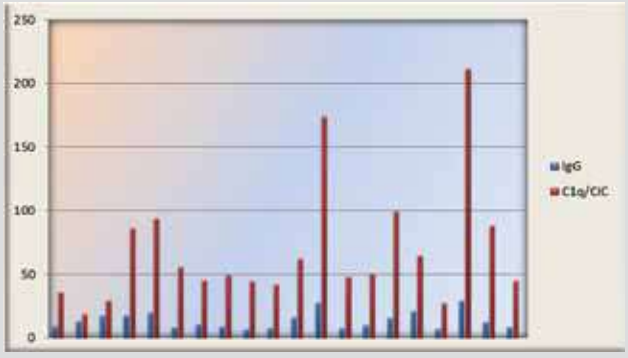


Figure 6 Quantitative comparison of obtained values between IgG and C1q/CIC (Specimens 21-40).

Table 2 Comparative study of obtained quantitative values an percentage.

	Number of specimens with normal values	Percentage (%)	Number of specimens with values above referrals	Percentage (%)	Number of specimens with values below referrals	Percentage (%)
ASTO	/	/	40	100%	/	/
CRP	/	/	40	100%	/	/
C3	4	10%	36	90%	/	/
C4	24	60%	/	/	16	40%
IgG	29	72,5%	10	25%	1	2,5%
C1q/CI	22	55%	/	/	18	45%

Results of laboratory analyses showed that in 57.5% of cases ASPGN was diagnosed. Seven samples showed higher value of C1qCIC parameters. Those patients had an active high intensity disease. Mild intensity of the disease was recorded in 16 patients which coincided with quantitative value of C1qCIC. Also, reduced level of C3 and C4 parameters was determined, under reference limits. Characteristic value of some parameters (ASTO, CRP, tropomyosine, sedimentation) for patients suspected for ARPG which were in corelation with some clinical parameters, were showen in 17 samples (42.5%). Values of analysed ARPG parameters showed acute phase of disease in 6 samples, and mild intensity disease in 11 samples.

DISCUSSION

Among the most dangerous bacterial infection are definitely β-hemolytic streptococci group A. Gram positive, nonsporogenic, immovable and their cells after division stay together, building shorter or longer chains. Aerobic or facultative aerobic, and there are a large number of different types immportant for human medicine. B-hemolytic streptococci group A can cause a primary group of disease as: streptococci inflammation of the pharynx, tonsillitis, scarlatina, skin and wounds infection, pneumonia and etc. In addition to the primary group of disease, as complication of streptococci infection can cause sinusitis, otitis media, inflammation of the meninges and serious complications as rheumatic fever and glomerulonephritis. The gold standard for determining strain of this bacteria is hypervariable amino-acid sequence at the amino terminus of the M protein.

The infections caused by this bacteria often has serious complication like and kidney complications such as (Acute post-streptococcal rheumatic fever (APRF) and acute post-streptococcal glomerulonephritis (APSGN). In clinical sense, it is very important early and timely separation of these diseases, regarding disease activity monitoring and therapy effectiveness. All this improves the clinical management of affected patients.

Sick men or carriers are the most immportant source of infection for the environment. Confirmation of the streptococci infection is done by streptococci presence in throat swabs, wound swab, CSF, blood. Also, the streptococci can be found in healthy people. Determination of their specific microscopic characteristics is important for final confirmation of this infective agents (15-20).

Pathogenic effect S. pyogenes exercised by means of many virulence factors which participating in adhesion processes and colonization, avoiding immune response, cellular internalization, tissue invasion and production of toxins. Virulence factors are divided on somatic and extracellular. The most immportant somatic virulence factors that are expressed on the surface of bacterial cell are M protein, capsule, fimbriae and other adhesion molecules.

For future research is needed to include PCR analysis of polymorphism CFHR5 gene, and determine for our territory, specific genetic predisposition on induction of post-streptococal glomerulonephritis. Mutation of this gene causes dysfunction of CFHR5 protein which eventualy causes CFHR5 deficiency and result of it is progresive disease glomerulonephritis (21-26).

For further research it would be useful to involve immunephenotypization analyses of expression D8/17markers on B lymphocyte, by flow cytometry. Recent researches showed increased expression of this molecules, over 95% at patients with reumatic fever post-streptococcal ethiology.

The main goal of prevention of this diseases, shoud be considered introduction of Anti-StreptInCor vaccine, because it contains antibodies which are very efficient in neutralization of S. pyogenes. In this way is achieved the necessary organizm protection from further pathological infections and complications.

Also, it should bi useful to introduce anti-ASO,anti-DNAse B, anti-AHase i anti-NAD tests, because they alow dignostic laboratory differentiation of accute autoimmune reumatic fever (joint problems, skin rash, carditis) from accute glomerulonephrytis (exhaustion, reduced urination, bloded urin, joint problems) (27-31).

CONCLUSION

Presented findings of investigations showed, beyond any doubt, that on this approach is possible to fast and timely separation and confirmation of ASPGN and APRF, which has a practical clinical significance for their clinical management improvement regarding disease activity monitoring and effectiveness of therapy. Used quantitative methods certainly justified their purpose.

Conflict of interest: none declared.

REFERENCES

1. Cunningham MW. Pathogenesis of group A streptococcal infections. Clin Microb Rev. 200;13(3):470-511.
2. Tejani A, Ingulli E. Poststreptococcal glomerulonephritis. Current clinical and pathologic concepts. Nephron. 1990; 55(1):1-5.
3. Rodriguez-Iturbe B, Musser JM. The current state of poststreptococcal glomerulonephritis. J Am Soc Nephrol. 2008;19(10):1855-64.
4. Fries JW, Mendrick DL, Rennke HG. Determinants of immune complex-mediated glomerulonephritis. Kidney Int. 1988;34(3):333-45.
5. Lewis EJ, Carpenter CB, Schur PH. Serum complement component levels in human glomerulonephritis. Ann Intern Med. 1971;75(4):555-60.
6. Cameron JS, Vick RM, Ogg CS, et al. Plasma C3 and C4 concentrations in management of glomerulonephritis. Br Med J. 1973;3(5882):668-672.
7. Barash J, Mashiach E, Navon-Elkan P, Berkun Y, Harel L, Tauber T, Padeh S, Hashkes PJ, Uziel Y. Differentiation of post-streptococcal reactive arthritis from acute rheumatic fever. J Pediatr. 2008;(153):696-9.
8. Bisno AL. Nonsuppurative Poststreptococcal sequelae: rheumatic fever and glomerulonephritis. In: Mandell GI, Bennett JE, Dolin R, ur. Principles and practices of infections diseases. 7th Ed. Philadelphia: Churchil Livingstone. 2010; 2611-12.
9. Guilherme L, Cury P, Demarchi LM, Coelho V, Abel L, et al. Rheumatic heart disease: proinflammatory cytokines play a role in the progression and maintenance of valvular lesions. Am J Pathol. 2004;165(5):1583-91.
10. Hahn RG, Knox LM, Forman TA. Evaluation of Poststreptococcal illiness. Am Fam Physician 2005;71(10):1949-54.
11. Hycor Biomedical Autostat TM-II C1q-CIC enzyme linked immunosorbent assay (ELISA) specific for C1q-containing circulating immune complexes. Hycor Bic Medic al Inc. USA. 2006.
12. Subašić Đ, Karamehić J, Gavrankapetanović F, Hodžić H, Kasumović M, Delić-Šarac M, Prljača-Zečević L. ELISA subtypization of anti-ENA autoantibodies in clinical management of autoimmune diseases in Bosnia and Herzrgovina. Medical Archive. 2009; 63(6):312-316.
13. Okumura N, Nomura M, Tada T, Effects of sample storage on serum C3 assay by immunonephelometry. Clin Lab Sci. 1990;3:54-57.
14. Procedure for C3/C3c and C4/C4c quantitative determination. Dade Behring. Germany; 2005.

15. Carapetis JR, Steer AC, Mulholland EK, Weber M. The global burden of group A streptococcal diseases. Lancet Infect Dis. 2005;5(11):685-94.
16. Habeš S. Rasprostranjenost i specifična grupna pripadnost beta-hemolitičkih streptokoka kod preškolske djece u Sarajevu. Med Arh.1983;37(3):155-158.
17. Rantala S, Vähäkuopus S, Siljander T, Vuopio J, Huhtala H, Vuento R, et al. Streptococcus pyogenes bacteraemia, emm types and superantigen profiles. Eur J Clin Microbiol Infect Dis. 2012;31(5):859-65.
18. Guilherme L, Alba MP, Ferreira FM, Oshiro SE, Higa F, Patarroyo ME, et al. Anti-group A streptococcal vaccine epitope: structure, stability, and its ability to interact with HLA class II molecules. J Biol Chem. 2011;286(9):6989-98.
19. De Amicis KM, Freschi de Barros S, Alencar RE, Postól E, Martins CO, Arcuri HA, et al. Analysis of the coverage capacity of the StreptInCor candidate vaccine against Streptococcus pyogenes. Vaccine. 2014;32(32):4104-10.
20. Guilherme L, Oshiro SE, Fae KC, Cunha-Neto E, Renesto G, et al. T cell reactivty against streptococcal antigens in the periphery mirrors reactivity of heart infiltrating T lymphocytes in rheumatic heart disease patients. Infect Immun. 2001;69(9):5345-51.
21. Bisno AL. Nonsuppurative Poststreptococcal sequelae: rheumatic fever and glomerulonephritis. In: Mandell GI, Bennett JE, Dolin R, ur. Principles and practices of infections diseases. 7th Ed. Philadelphia: Churchil Livingstone. 2010;2611-12.
22. Guilherme L, Alba MP, Ferreira FM, Oshiro SE, Higa F, Patarroyo ME, et al. Anti-group A streptococcal vaccine epitope: structure, stability, and its ability to interact with HLA class II molecules. J Biol Chem. 2011;286(9):6989-98.
23. De Amicis KM, Freschi de Barros S, Alencar RE, Postól E, Martins CO, Arcuri HA, et al. Analysis of the coverage capacity of the StreptInCor candidate vaccine against Streptococcus pyogenes. Vaccine. 2014;(32):4104-10.
24. Freschi De Barros S, De Amicis KM, Alencar R, Smeesters PR, Trunkel A, Postól E, et al. Streptococcus pyogenes strains in Sao Paulo, Brazil: molecular characterization as a basis for StreptInCor coverage capacity analysis. Freschi de Barros et al. BMC Infectious Diseases. 2015;15:308.
25. Smeesters PR, Dramaix M, Van Melderen L. The emm-type diversity does not always reflect the M protein genetic diversity - is there a case for designer vaccine against GAS. Vaccine. 2010;28(4):883-5.
26. Guerino MT, Postol E, Demarchi LM, Martins CO, Mundel LR, Kalil J, et al. HLA class II transgenic mice develop a safe and long lasting immune response against StreptInCor, an anti-group A streptococcus vaccine candidate. Vaccine. 2011;29(46):8250-6.
27. Ramasawmy R, Faé KC, Spina G, Victora GD, Tanaka AC, Palacios SA, et al. Association of polymorphisms within the promoter region of the tumor necrosis factor-alpha with clinical outcomes of rheumatic fever. Mol Immunol. 2007;44(8):1873-1878.
28. Azevedo PM, Bauer R, Caparbo VF, Silva CA, Bonfa E, Pereira RM. Interleukin-1 receptor antagonist gene (IL1RN) polymorphism possibly associated to severity of rheumatic carditis in a Brazilian cohort. Cytokine. 2010;49(1):109-13.
29. Settin A, Abdel-Hady H, El-Baz R, Saber I. Gene polymorphisms of TNF-alpha(-308), IL-10(-1082), IL-6(-174), and IL-1Ra(VNTR) related to susceptibility and severity of rheumatic heart disease. Pediatr Cardiol. 2007;28(5):363-71.
30. Ferreti J, Stevens D, Fischetti V. Streptococcus pyogenes-Basic Biology to Clinical Manifestations. University of Oklahoma Health Sciences Center (USA). 2016.
31. Maness D, Martin M, Mitchell G. Poststreptococcal Illnes - Recognation and Management. American Fam. Physician. 2018;97(8):517-522.

Reprint requests and correspondence
Mevludin Mekić, MD, PhD
Clinic of Heart Diseases, Blood Vessels and Rheumatism
Clinical Center University of Sarajevo
Bolnička 25, 71000 Sarajevo
Bosnia and Herzegovina
Email: mevludinmekic@yahoo.com



Symptoms of depression in patients with coronary heart diseases following myocardial revascularization

Depresivni simptomi kod pacijenata s koronarnom bolešću nakon tretmana revaskularizacije miokarda

Dženana Hrustemović[1]*, Ermina Mujičić[2], Alma Džubur-Kulenović[3]

[1]Clinic of Heart Diseases, Blood Vessels and Rheumatism, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina
[2]Clinic of Cardiovascular Surgery, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina
[3]Psychiatry Clinic, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina

*Correponding author

ABSTRACT

Introduction: cardiovascular diseases (CVDs) are the leading cause of disability and death worldwide. Treatment modalities of patients with coronary heart diseases entail medicamentous treatment, percutaneous coronary intervention (PCI) and coronary artery bypass grafting (CABG). Numerous studies have described the prevalence of depression symptoms in patients with cardiovascular diseases even after the treatment. Aim of the study: to determine the presence of depression symptoms in patients after coronary artery bypass grafting (CABG) and after a percutaneous coronary intervention (PCI). Materials and methods: this is a prospective, cohort, clinical-descriptive study. It included 120 patients with coronary heart disease treated at Cardiology and Cardio Surgery Clinics of the Clinical Center University of Sarajevo (CCUS), divided into two groups. Study group (n = 60) included patients with coronary heart disease treated by percutaneous coronary intervention (PCI) divided into three subgroups: Subgroup 1 (n = 20) before and after 1 months, Subgroup 2 (n = 20) before and after 3 months and Subgroup 3 (n = 20) before and after 6 months. Control group (n = 60) included patients with coronary heart disease treated with coronary artery bypass grafting (CABG) divided into three subgroups: Subgroup 1 (n = 20) before and after 1 months, Subgroup 2 (n = 20) before and after 3 months and Subgroup 3 (n = 20) before and after 6 months. Data were collected from patients' histories of illness, anamnesis and clinical interviews conducted in the period from 12 August 2016 to 12 February 2017. The study used the Hamilton Depression Rating Scale (HDRS) and a socio-demographic questionnaire designed especially for this purpose. Results: regarding patients treated by PCI there was no significant deterioration of mental health following the procedure (p1 = 0,265; p3 = 0,122; p6 = 0,531). Statistically significant deterioration of mental health following the surgery was observed in all patients after 1 and 3 months (p1 = 0,012; p3 = 0,001), whereas there was no significant difference in their mental health after 6 months (p6 = 0,234). Conclusion: the presence of depression symptoms in patients after PCI was not significant and with no differences regardless of the time of the survey (p= 0.056). Presence of depression symptoms in patients after the surgery was significant after 1 month (p1= 0,012) and after 3 months (p3= 0,001), whereas there was no difference after 6 months (p6= 0,234). Conclusion: the presence of depression symptoms in patients after PCI was not statistically significant and there was no difference regardless of the time of the interviews. However, the presence of depression symptoms in patients after the surgery (CABG) was significantly expressed after 1 (p1= 0,012) and after 3 months (p3= 0,001), wheres there was no difference after 6 months (p6= 0,234).

Keywords: coronary heart disease, mental health, PCI, CABG

SAŽETAK

Uvod: kardiovaskularne bolesti (KVB) su vodeći uzrok oboljevanja i umiranja u svijetu. Modaliteti liječenja bolesnika sa koronarnom bolešću obuhvataju medikamentozno liječenje, perkutanu koronarni intervenciju (PCI) i operativni način liječenja (CABG). Učestalost depresivnih simptoma u osoba s kardiovaskularnim poremećajem i nakon tretmana opisan je u brojnim istraživanjima. Cilj istraživanja: da se utvrdi prisustvo depresivnih simptoma kod pacijenata nakon tretmana ugradnje premosnica na srcu hirurškom metodom (CABG) i nakon tretmana ugradnje stenta perkutanom koronarnom intervencijijom (PCI). Ispitanici i metod istraživanja: istraživanje je prospektivna, kohortna, kliničko - deskriptivna studija. Istraživanje obuhvata 120 pacijenata sa koronarnom bolešću Klinike za kardiologiju i Klinike za kardiohirurgiju KCUS-a, podjeljeni u dvije grupe: ispitivana grupa pacijenata (n = 60) - pacijenti sa koronarnom bolešću tretirani perkutanom koronarnom intervencijom (PCI) podjeljena u tri podgrupe: 1. podgrupa (n = 20) prije i nakon 1 mjeseca, 2. podgrupa (n = 20) prije i nakon 3 mjeseca i 3. podgrupa (n = 20) prije i nakon 6 mjeseci; kontrolna grupa pacijenata (n = 60) - pacijenti sa koronarnom bolešću tretirani hirurškom metodom (CABG) podjeljeni u tri podgrupe: 1. podgrupa (n = 20) prije i nakon 1 mjeseca, 2. podgrupa (n = 20) prije i nakon 3 mjeseca i 3. podgrupa (n = 20) prije i nakon 6 mjeseci. Podaci su se prikupljali iz istorije bolesti pacijenata, anamneze pacijenta te putem kliničkog intervjua u periodu od 12.08.2016. godine do 12.02.2017. godine. U istraživanju je korištena standardizirana skala za procjenu simptoma depresije Hamiltonova skala (HDRS) i sociodemografski upitnik koji je kreiran specifično za ovo istraživanje. Rezultati: za pacijente rađene PCI metodom nema signifikantnog pogoršanja psihičkog stanja nakon procedure (p1 = 0,265; p3 = 0,122; p6 = 0,531). Može se uočiti statistički

značajno pogoršanje psihičkog stanja pacijenta nakon operacije kod svih pacijenata nakon 1 i 3 mjeseca (p1 = 0,012; p3 = 0,001), dok nakon 6 mjeseci nema signifikantne razlike (p6 = 0,234). Zaključak: Prisustvo depresivnih simptoma kod pacijenata nakon PCI nije značajno i nema razlike bez obzira na vrijeme anketiranja (p = 0,056). Prisustvo depresivnih simptoma kod pacijenata nakon operacije je značajno nakon 1 mjeseca (p1= 0,012) i nakon 3 mjeseca (p3= 0,001), dok nakon 6 mjeseci nema razlike (p6= 0,234).

Ključne riječi: koronarna bolest, psihičko zdravlje, PCI, CABG

INTRODUCTION

Cardiovascular diseases (CVDs) are the leading cause of disability and death worldwide. The most common cardiovascular system diseases are heart and blood vessel disorders, hearth rhythm disorders and heart muscle and heart valve diseases and the most significant are myocardial infarction and angina pectoris (1). Myocardial infarction is permanent damage to a portion of the heart muscle, where size of the damage determines course of the disease and its prognosis. Cardiovascular diseases, specifically heart and blood vessel diseases, have become more common and it is alarming that they also affect young people with no most expressed risk factors. Especially susceptible are those with family history of the disease, specifically those with genetic predispositions. Therefore, treatment of cardiovascular patients, including coronary heart disease, is aimed at reducing the risk of future events, improving quality of life and life prolongation. Contemporary coronary heart diseases treatment, including contemporary revascularization methods, has significantly influence all aspects of human life - both prolongation and quality of life. Overall assessment of the disease influence and treatment methods is nowadays based on traditional outcome measuring, such as rate of survival and quality of life measuring (2,3). Incidence of depression in patients with cardiovascular diseases has been described in number of studies showing that in 200 patients with acute myoradial infarction the prevalence of depression was registered in 28% of them, and in half of the patients depression developed within the first month following myocardial infarction (4,5).

Quality of life of patients with coronary artery disease following myocardial revascularization

Quality of life is defined as perception of an individual about his/her own position in life in the context of culture and system of value in which he/she lives, and also in the context of his/her aims, expectations, standards and interests. Nowadays, quality of life and survival present main criteria for evaluation of a certain therapeutic procedure. Mental health involves mental stability, self-satisfaction and satisfaction with environment, maintenance of social contacts and adequate sleep and rest. Cardiovascular diseases are examples of somatic disorders to which psychological factors have decisive influence. Symptoms of depression after myocardial infarction may be somatic (fatigue), cognitive (concentration, slowness or agitation) and psychological (mood swings, feelings of worthlessness) which may obstruct the rehabilitation treatment and become an obstacle in lifestyle changes necessary in the post myocardial infraction period (6,7).

Aim of the study

The aim of this study was to determine the presence of depression symptoms in patients after coronary artery bypass grafting (CABG) and after percutaneous coronary intervention (PCI).

MATERIALS AND METHODS

The study included 120 patients with coronary heart disease treated at Cardiology and Cardio Surgery Clinics of the Clinical Center University of Sarajevo (CCUS), divided into two groups.

Study group (n = 60) included patients with coronary heart disease treated by percutaneous coronary intervention (PCI) divided into three subgroups: Subgroup 1 (n = 20) before and after 1 months, Subgroup 2 (n = 20) before and after 3 months and Subgroup 3 (n = 20) before and after 6 months;

Control group (n = 60) included patients with coronary heart disease treated by coronary artery bypass grafting (CABG) divided into three subgroups: Subgroup 1 (n = 20) before and after 1 months, Subgroup 2 (n = 20) before and after 3 months and Subgroup 3 (n = 20) before and after 6 months.

Data for the study was collected from patients' history of illness, anamnesis and through clinical interviews conducted in the period from 12 August 2016 to 12 February 2017. The study used the standardized Hamilton Depression Rating Scale (HDRS) and sociodemographic questionnaire designed especially for this purpose.

Statistical processing

The obtained data was entered into special charts and statistically processed.

Descriptive group statistics, analysis and statistical processing was done within the groups with the application of appropriate statistical methods for the obtained parameters. Parametric data was tested by the Student's T-test whereas nonparametric data were processed using the Pearson's chi-square test, Fisher's exact test and Mann-Whitney U test and proportions. Correlations were examined using the Spearman's correlation. Level of significance was $p < 0.05$. The obtained results were presented in texts, numbers, tables and figures.

RESULTS

Demographic statistics

Table 1 Patients undergoing different types of treatments, classified by gender.

	Gender of patients						
	Male		Female		Total		
	n	%	n	%	n	%	
Type of treatment	Surgery	50	83.3%	10	16.7%	60	100.0%
	PCI	50	83.3%	10	16.7%	60	100.0%
	Total	100	83.3%	20	16.7%	120	100.0%

Binomial tests showed significant deviation in male and female ratio with respect to the population ($p < 0.001$). There was no significant differences in gender distribution between the groups, Fisher's exact test ($p = 1.000$).

Table 2 Mean value ($\bar{x}$) and standard deviation (σ) of age of patients undergoing different types of tretments, classified by gender (age).

Age of patients							
	Male		Female		Total		
	$\bar{x}$	σ	$\bar{x}$	σ	$\bar{x}$	σ	
Type of treatment	Surgery	59.9	8.1	60.5	7.8	60.0	8.0
	PCI	58.6	10.1	59.7	11.5	58.8	10.2
	Total	59.2	9.1	60.1	9.5	59.4	9.1

The Student's T-test did not show significant difference in age structure of different gender patients ($p = 0{,}703$), or between patients subjected to different types of treatments ($p = 0.487$). Based on the Kolmogorov-Smirnov Test the overall sample of patients followed the normal division ($p > 0.200$).

Table 3 Results of survey on mental health of patients based on all responses (Hamilton Depression Rating Scale) after surgery (CABG) and PCI (poorer mental health characterized by the higher grade) described by median ($\tilde{x}$) response score, arranged according to the survey period.

	No. of months after the treatment															
	1				3				6				All patients			
	Type of treatment				Type of treatment				Type of treatment				Type of treatment			
	Surgery		PCI		Surgery		PCI		Surgery		PCI		Surgery		PCI	
	Survey		Survey		Survey		Survey		Survey		Survey		Survey		Survey	
	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po
	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$
1. Depression	0	0	0	0	4	2	0	0	1	0	0	0	1	1	0	0
2. Feeling of guilt	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
3. Suicide	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4. Initial insomnia	0	1	0	0	1	1	0	1	1	1	0	0	1	1	0	0
5.Middle-of-the-night insomnia	0	1	0	1	1	1	0	0	2	1	0	0	1	1	0	0
6. Severe insomnia	1	1	0	1	1	1	0	0	0	0	0	0	1	1	0	0
7. Work releted problems	1	1	0	1	1	2	0	0	0	1	1	1	1	1	0	0
8. Slow thinking	0	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0
9. Anxiety	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0
10. Fear	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0
11. Digestion related problems	1	1	1	0	1	2	0	1	1	0	0	1	1	1	0	0
12. Loss of apetite	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
13. Heavy limbs	1	1	1	0	1	2	0	1	1	1	0	1	1	1	0	1
14. Low libido	0	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0
15. Hypochondria	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
16. Weight loss	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0
17. Accepting health related problems	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0
HDRS sum total	9	15	6	8	13	20	5	9	10	15	6	9	11	17	6	8
P	0.012		0.265		0.001		0.122		0.234		0.531		<0.001		0.056	

The Table presents median ($\tilde{x}$) sum of related responses and total sum, where poorer mental health was characterized by the higher grade. Data was arranged by the survey period (in number of months after the treatment) and type of survey (before and after the treatment). Statistically significant differences between the distribution of certain grades are highlighted, according the Wilcoxon signed rank test ($p < 0.05$), and relate to comparison between grades before and after certain treatment. Statistically significant differences in grading depending on the type of treatment are presented in bold, according to the Mann-Whitney U test ($p < 0.05$). The last row presents results of the Wilcoxon signed-rank test of paired samples in comparison of the Hamilton scale rating before and after the treatment.

Table 4 Results of survey on mental health of patients, arranged by related responses, (Hamilton Depression Rating Scale) after the surgery (CABG) and PCI (poorer mental was caracterized by the higher grade) described by median ($\tilde{x}$) of the response grade, arranged according to the survey period.

	No. of months after the treatment															
	1				3				6				All patients			
	Type of treatment				Type of treatment				Type of treatment				Type of treatment			
	Surgery		PCI		Surgery		PCI		Surgery		PCI		Surgery		PCI	
	Survey		Survey		Survey		Survey		Survey		Survey		Survey		Survey	
	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po	Pr	Po
	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$	$\tilde{x}$
Bad mood	0	2	0	0	4	4	0	1	2	2	0	0	2	2	0	0
	0.024		0.944		0.171		0.699		0.037		0.478		0.075		0.939	
Insomnia	2	3	0	2	4	4	0	2	4	2	1	1	3	3	0	2
	0.232		0.301		0.855		0.119		0.414		0.634		0.632		0.078	
Psycho-physical problems	1	3	0	1	2	4	1	0	1	3	1	2	1	3	0	1
	0.131		0.114		0.007		0.892		0.021		0.573		<0.001		0.100	
Anxiety and fear	1	1	0	0	0	2	0	0	1	1	0	0	0	2	0	0
	0.390		0.357		0.000		0.273		0.198		0.918		0.001		0.250	
Physical problems	2	3	1	1	2	5	0	2	2	2	2	2	2	3	1	1
	0.008		0.721		0.002		0.029		0.033		0.977		<0.001		0.156	
Accepting disease	1	1	1	1	1	2	1	0	0	1	0	1	1	1	1	1
	0.053		0.716		0.080		0.361		0.340		0.873		0.005		0.509	

The Table presents median ($\tilde{x}$) sum of related responses and total sum, where poorer mental health was characterized by the higher grade. Highlighted spots show significant changes in group of questions for particular time periods after the surgery. The only significant difference was noted in the group of questions related to physical problems of patients 3 months after the PCI.

Statistically significant deterioration could be noted in physical health of all patients after 1 and 3 months following the surgery ($p1 = 0.012$; $p3 = 0.001$), whereas there was no significant difference after 6 months ($p6 = 0.234$). For patients treated with the PCI method there was no significant deterioration of physical health following the procedure ($p1 = 0.265$; $p3 = 0.122$; $p6 = 0.531$)

Table 5 Significant symptoms of depression after the surgery (CABG) and PCI.

	Type of treatment						
	Surgery		PCI		Total		
	n	%	n	%	n	%	
Categorization according to the HDRS	Normal	10	16.7%	26	43.3%	36	30.0%
	Mild depression	10	16.7%	22	36.7%	32	26.7%
	Moderate depression	15	25.0%	10	16.7%	25	20.8%
	Severe depression	10	16.7%	2	3.3%	12	10.0%
	Very severe depression	15	25.0%	0	0.0%	15	12.5%

Pearson's chi-square test shows significant difference between the two groups of patients, where better physical state after the PCI procedure was noted regardless of the survey period ($p < 0.001$). The table clearly shows that a very small number of patients subjected to the PCI procedure suffered from severe (3.3%), and not a single patient suffered from very severe depression. On the other hand, 25% of patients suffered from severe and 16.7% from very severe depression after the surgery (CABG).

DISCUSSION

Myocardial revascularization, performed either surgically (CABG) or by coronary angioplasty (PCI), is a modern approach in the treatment of patients with angiographically proven coronary artery disease. In the past 10 years, secondary prevention of ischemic heart disease by percutaneous coronary procedures and surgical methods has undergone significant changes both in technical and vital and biological sense. However, it is well known that myocardial infarction results in significant reduction of patient's quality of life. The study included the total of 120 patients of which 60 (50%) patients were subjected to surgical procedure (CABG) and 60 (50%) to percutaneous coronary intervention (PCI). Out of the total number of patients there were 100 (83.3%) male and 20 (16.7%) female patients. Binomial test showed that male and female ratio in the selected sample significantly deviates from the proportional ration in the overall population (Table 1). On the other hand, Fisher's exact test did not show significant difference in sex distribution between the groups of patients subjected to the surgery (CABG) and the group treated by PCI ($p = 1.000$), which means that comparison of the survey results between the two interventions was not influence by differences in gender of patients. Results of the Redžek A, et al. (2007) study showed that gender and age were independent predictors of survival and quality of life following surgical myocardial revascularization. Statistically significant differences in quality of life in respect to gender were noted. In their study, better survival and quality of life was registered in male patients under 65 years of age. Statistically significant improvement of quality of life in respect to preoperative values in the group of patients over 65 years of age despite lower survival rate confirms that surgical myocardial revascularization is also a justified therapeutic option for these patients (11,8).

Mental state of patients assessed based on the Hamilton Depression Rating Scale (HDRS) (after 1, 3 and 6 months in both groups of patients) showed that mental state of patients deteriorated after the surgery, especially in patients interviewed in the first and third month after the intervention. Statistical processing of median differences (with a confidence interval of 95%) in mental state rating scale of patients (HDRS), showed significant statistical mental state deterioration after the surgery in all patients after 1 and 3 months ($p1 = 0.012$; $p3 = 0.001$), whereas there was no significant differences after 6 months ($p6 = 0.234$) (Table 3). The study conducted by Cohen at al. (2011) in the New England Journal of Medicine showed that patients who underwent triple surgical myocardial revascularization had slightly better quality of life after 6 and 12 months in respect to patients treated by the PCI (18). Significant mental health deterioration in three groups of patients interviewed at different times was not noted, but analysis of all patients showed significant scoring difference. Median score increased from 6 to 9 ($p = 0.009$). For patients treated by PCI there was no significant deterioration of mental health after the procedure ($p1 = 0.265$; $p3 = 0.122$; $p6 = 0.531$) (Table 4).

In the study conducted by Larsena K in Denmark (2013) it was established that in patients who after myocardial infarction developed depression, cognitive and affective symptoms were less expressed in respect to patients who were treated by psychiatric services. Given that depression is related to cardiovascular mortality, treatment of depressive disorder would reduce the related mortality (13).

Following comparison of the Hamilton Depression Rating Scale after the surgery and after PCI, statistically significantly better mental health of patients treated by PCI was noted (for all three groups and in total). The total sum of median score after the surgery was higher than the score obtained after PCI ($p < 0.001$). Pearson's chi-squared test showed significant difference between the two groups of patients, where better mental health was observed after PCI regardless of the interview time ($p < 0.001$). Kancez Fatima, et al., in their study published in March 2016 in cardiology journal under the title Cardiology Research and Practice, after reviewing 447 articles related to comparison of PCI and surgical myocardial revascularization concluded that larger number of studies suggest that after angioplasty, as a less invasive procedure, patients has better quality of life (14). Results of this research mainly correspond to the results of other studies and clearly point to the importance of depressive symptoms recognition in patients treated for coronary diseases, enabling their early and adequate treatment.

CONCLUSION

Presence of depression symptoms in patients with coronary artery diseases after PCI is not statistically significant and there is no difference regardless the time of the interviews. However, the presence of depression symptoms in patients with coronary artery diseases after the surgery (CABG) is significantly expressed after 1 ($p1 = 0,012$) and after 3 months ($p3 = 0,001$), where there is no difference after 6 months ($p6 = 0,234$).

REFERENCES

1. Ivanović J, Milan S, Veljković A. Kvalitet života bolesnika sa koronarnom bolešću i revaskularizacijom miokarda. Studentski Medicinski glasnik. 2011;II(1-4):21-25.
2. Tasic I, Lazarevic G, Kostic S, Djordjevic D, Simonovic D, Rihter M, et al. Administration and effects of secondary prevention measures in coronary heart disease patients from Serbia according to gender and cardiometabolic risk. Acta Cardiol. 2010;65(4):407-14.

3. Kotseva K, Wood D, De Backer G, De Bacquer D, Pyorala K, Keil U. EUROASPIRE Study Group. Cardiovascular prevention guidelines in daily practice: a comparison of Euroaspire I, II, and III surveys in eight European countries. Lancet. 2009;373(9667):929-40.
4. Thygesen K, Alpert JS, White HD. Universal Definition of Myocardial Infarction.Task force for the Redefinition of Myocardial infarction of the Europen Society of Cardiology. Eur Heart J. 2007;50(22):2173-95.
5. Hoen PW, Whooley MA, Martens EJ, Na B, van Melle JP, de Jonge P. Differential associations between specific depressive symptoms and cardiovascular prognosis in patients with stable coronary hearth disease. J Am Coll Cardiol. 2010;56(11):838-44.
6. Smajkic A. Social, medical and epide-1. miological aspects of vascular diseases (CVD). 4 th Congress of the Associaation of Cardiologists and An-giologists of Bosnia and Herzegovina, Book of Abstracts, Mostar, 17-19 May 2007: 8-10.
7. Greenland P, Alpert JS, Beller GA, Benjamin EJ, Budoff MJ, Fayad ZA, et al. 2010 ACCF/AHA guideline for assessment of cardiovascular risk in asymptomatic adults: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. Circulation. 2010;122(25):584-636.
8. Gopalakrishnan P, Ragland MM, Tak T. Gender differences in coronary artery disease: review of diagnostic challenges and current treatment. Postgrad Med. 2009;121(2):60-8.
9. Abdić D. Suvremeni pristup bolesniku nakon aortokoronarnog premoštenja. Undergraduated thesis. University North. University Centre Varaždin. Department of Nursing. 2015.
10. Mesarić J, Kaić-Rak A. Bolesnikova sigurnost, bolesnik u središtu i programi Svjetske zdravstvene organizacije. Medix. 2010;86(16):111-114.
11. Redžek A, Šušak S, Zečević D. Long term survival and quality of life after myocardial revascularization with respect to age and sex distribution. Medicinski pregled. 20017;60(7-8):317-321.
12. Cohen DJ, Van Hout B, Serruys PW, Mohr FW, Macaya C, den Heijer P, et al. Quality of Life after PCI with Drug-Eluting Stents or Coronary-Artery Bypass Surgery. N Engl J Med. 2011;364(11):1016-26.
13. Larsen KK. Depression following myocardial infarction. Dan Med J. 2013;60(7):B4689.
14. Kancez F, Mohammad Y, Mehreen A, Faizan IB, Muhammad S, Akash K, et al. Comparison of the Postprocedural Quality of Life between Coronary Artery Bypass Graft Surgery and Percutaneous Coronary Intervention: A Systematic Review. Cardiol Res Pract. 2016;2016:7842514

Reprint requests and correspondence:
Dženana Hrustemović, RN, PhD
Clinic of Heart Diseases, Blood Vessels and Rheumatism
Clinical Center University of Sarajevo
Bolnička 25, 71000 Sarajevo
Bosnia and Herzegovina
Email: dzenana.hrustemovic@gmail.com

Our contribution to the reduction of cardiovascular diseases in Bosnia and Herzegovina!
Naš prilog redukciji kardiovaskularnih bolesti u Bosni i Hercegovini!



Incidence of Surgical Site Infection (SSI) in a six month sample of patients treated at Clinic of General and Abdominal Surgery of the Clinical Center University of Sarajevo

Incidenca infekcije hirurškog polja (SSI) u šesto-mjesečnom uzorku pacijenata Klinike za opštu i abdominalnu hirurgiju Kliničkog centra Univerziteta u Sarajevu

Salem Bajramagić*, Adi Mulabdić, Edin Hodžić, Samir Muhović, Jusuf Šabanović, Adnan Kulo

Clinic of General and Abdominal Surgery, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

Introduction: nosocomial or hospital infection occurs during hospital treatment of patients, appearing within 48 hours following the admission and last up to 30 days after discharge in cases of surgical interventions, and may last up to one year in cases of surgical interventions requiring foreign material implantation. The third most common type of nosocomial infection with an incidence of 3-15% relates to surgical site infections (SSI). Problem definition: surgical site infection accounts for major complications after surgical treatment, resulting in a considerable increase in morbidity, mortality and overall treatment cost. Significant efforts have been made worldwide to reduce the incidence of surgical site infections through a number of monitoring, prevention and treatment protocols. Problem formulation: the SSI prevention protocols have contributed to reducing surgical site infections incidence, but they still remain the main problem of each surgical institution. The question is which percentage of our surgically treated patients has signs of SSI and whether our data corresponds to related studies and statistics. Aim: based on medical records of our surgically treated patients with signs of infections, and microbiological sample analysis, to perform statistical analysis and compare the results of related studies. Materials and methods: this was a retrospective analysis of medical records and data of patients treated at Clinic of General and Abdominal Surgery of the Clinical Center University of Sarajevo (CCUS) with evident signs of SSI, conducted over the period of six months (from 1 February to 31 August 2018). We analyzed the age and sex structure, surgical procedure requirements, and microbiological tests and results provided by the CCUS Institute of Clinical Microbiology. Data was collected from the history of the disease, outpatient protocol of patients monitored through a specialized outpatient service, and review of the Institute for Clinical Microbiology database. Data were analyzed using descriptive statistics methods and presented in tables and figures. Results: out of the total of 1486 patients surgically treated at the CCUS Clinic of General and Abdominal Surgery, 168 (11.3%) had apparent clinical signs of infection (skin redness, swelling, secretion, pain), whereas microbiological data processing confirmed infection in 144 (9.69%) of them. Negative microbiological findings were obtained in 24 patients. Out of 144 patients, 83 (57.6%) were male and 61 (42.3%) were female patients. In 95 (65.9%) patients, signs of infection were apparent in the first five days following surgical intervention, with an average occurrence on day four. Significance: indicate the incidence of surgical site infections and emphasize drafting and use of appropriate protocols aimed at their prevention.

Keywords: nosocomial infection, surgical site infections, abdominal surgery

SAŽETAK

Uvod: nozokomijalna ili hospitalna infekcija je ona koja se javi u toku hospitalnog tretmana pacijenta, a najmanje 48 sati od prijema na bolničko odjeljenje do 30 dana od momenta otpusta sa bolničkog liječenja kod operativnih zahvata, a do godinu dana kod operativnih zahvata koji zahtijevaju implantaciju stranog materijala. Treći najčešći oblik intrahospitalne infekcije sa incidencom od 3-15% su infekcije hirurškog polja (Sugical Site Infections - SSI). Definicija problema: SSI predstavljaju ozbiljnu komplikaciju hirurškog liječenja, a rezultiraju značajnim povećanjem morbiditeta, mortaliteta te povećanjem ukupnih troškova liječenja. Širom svijeta, ulažu se veliki napori kako bi se smanjila incidenca infekcija hirurškog polja kroz brojne protokole nadzora, prevencije i liječenja istih. Formulacija problema: Protokoli prevencije SSI doprinjeli su padu incidence infekcija hirurškog polja, ali su one i dalje veliki problem svake hirurške ustanove. Postavlja se pitanje koji procenat naših hirurških pacijenata ima znakove SSI, te da li naši podaci odgovaraju

relevantnim studijama i statistici. Cilj: uvidom u medicinsku dokumentaciju hirurški tretiranih pacijenata naše klinike kod kojih su se javili znaci infekcije, registrovati podatke i uvidom u rezultate mikrobiološke analize uzoraka, učini statističku obrada i uporediti rezultate sa relevantnim studijama. Materijali i metode: retrospektivno je učinjena analiza medicinske dokumentacije i podataka za pacijente Klinike za opštu i abdominalnu hirurgiju Kliničkog centra Univerziteta u Sarajevu (KCUS), za period od 01.02.2018. do 31.08.2018. godine koji su imali evidentne znakove infekcije hirurškog polja. Analizirana je starosna i spolna struktura, uslovi izvođenja operativnog zahvata, te nalazi mikrobiološke obrade OJ Klinička mikrobiologija KCUS-a. Podaci su prikupljeni iz historija bolesti, ambulantnog protokola za pacijente koji su kontrolisani kroz specijalističku ambulantu, te uvidom u bazu podataka OJ Klinička mikrobiologija. Podaci su obrađeni metodom deskriptivne statistike i tabelarno i grafički prikazani. Rezultati: od ukupno 1486 hirurški zbrinutih pacijenata na Klinici za opštu i abdominalnu hirurgiju KCUS njih 168 ili 11.3% je imalo evidentne kliničke znakove infekcije (crvenilo, otok, sekreciju, bolove), ali se mikrobiološkom obradom infekcija potvrdi kod njih 144 ili 9.69%. Negativnih nalaza mikrobiološke obrade je bilo 24. U navedenom uzorku od 144 pacijenta, njih 83 je muškog spola ili 57.6%, a 61 pacijent je ženskog spola ili 42.3% od ukupnog broja. Znaci infekcije su kod 95 pacijenata (65.9%) bili evidentni u prvih pet dana po učinjenoj hirurškoj intervenciji, sa prosjekom u četvrtom danu. Značaj rada: ukazati na incidencu javljanja infekcije hirurškog polja te potencirati izradu i upotrebu adekvatnih protokola za prevencije iste.

Ključne riječi: nozokomijalna infekcija, infekcija hirurškog polja, abdominalna hirurgija

INTRODUCTION

Nosocomial (Greek: nosokomeion = hospital) or hospital infection is each localized or general infection developed in hospital, not present or in incubation upon the patient's admission (1). In the majority of studies it has also been referred to as Hospital Acquired Infection - HAI. According to different data, 5-15% of patients hospitalized for various reasons, acquire hospital infection. In undeveloped and countries in transition that percentage is even higher, amounting to 25% (1,2).

Hospital infections present a huge problem in every healthcare system since they significantly increase morbidity, overall mortality, disability, and overall cost of such patients' treatment. The possibility of taking clinical strain out into wider population is not less significant. 5-10% of these infections develop as smaller or bigger epidemics within one medical institution.

The most common nosocomial infections are:

1. Urinary tract infection (up to 40%);
2. Respiratory tract infection (15-20%);
3. Surgical site infection (SSI) (0.5-15%); and
4. Sepsis (5-10%).

Prevalence of infections differs at various departments or clinics. While urinary and respiratory tract infections predominantly develop at internal medicine clinics, surgical site infections mainly develop at surgical departments. The most severe types of infections, such as sepsis, develop at intensive care units, pediatric neonatal care departments and in institutions treating immunocompromised patients.

Patients, hospital and assisting medical staff, visitors, and training medical students are all exposed to infections.

There are various risk factors for the development of infections. Immunocompromised patients are mainly affected due to the disease or therapy, their condition following the surgery, use of urinary catheters and other types of drainage impairing natural epithelial barrier, longer hospitalization, prolonged and often irrational use of antibiotics, and the incidence of multiresistant nosocomial microorganisms.

Nosocomial infections can develop due to influence of endogenous microorganisms, normal flora, acting pathogenically only under specific conditions. Other infections involve exogenous microorganisms, originating from other patients, hospital stuff, visitors or other. The third category relates to MDR or Multi Drug Resistant organisms developing inside the institution as a consequence of long term use of the same antibiotic therapy resulting in the persistent antibiotic resistance.

The most common way of microorganisms transfer is directly by touch or indirectly by equipment, air, water, food and similar.

The third place among the nosocomial infections is reserved for surgical site infections. They are on average accounted for 15% of all surgical patients (1). Taking into account that approximately 40% of all hospitalized patients relate to surgically treated patients, SSI becomes a significant problem. Recent studies have shown that in the developed countries SSI occurs in 2% of patients in elective program with preventive measures, and even 40-60% of patients in emergency surgical program have apparent signs of SSI (1,3).

Infection in surgery most commonly develop during the very intervention as a consequence of direct contamination of endogenous flora or exogenous pathogen. Infection may develop immediately after surgical intervention, most commonly through surgical wound healing slowly or not healing at all, but also weeks after surgical intervention from a distant focal point through blood. Risk factors for development of SSI are as follows (Table 1):

Table 1 Risk factors for development of SSI.

Risk factors specific for patients - intrinsic factors	Procedural factors - extrinsic factors
Patient's age	Inadequate preoperative showering
Diabetes (or other metabolic disease)	Preoperative shaving the night before surgical intervention
Perioperative hyperglycemia	Prolonged intervention
Nicotine	Inappropriate antiseptic measures
Existing, remote infection	Inadequate antibiotic prophylaxis
Obesity	Disturbance in maintaining surgical asepsis
Malnutrition and immunocompromised patients	Hypothermia and hypoxia
Low pre-operative levels of serum albumin	Transfusion of blood and blood components
Steroid therapy	Poor surgical technique (tissue trauma, hematomas, detritus, necrosis and coagulation)
Prolonged hospitalization prior to surgical intervention	
Neoadjuvant radiotherapy of surgical site	
S. aureus colonization	

Guidelines for Prevention of Surgical Site Infection from 2017 confirmed by CDC, lists the most common pathogens of surgical site infection, and their prevalence during two relevant study periods (Table 2):

Table 2 Most common pathogens of surgical site infection.

Pathogen	% of all isolates in the 1986-1989 period	% of all isolates in the 1990-1996 period
S. aureus	17	20
Enterococcus	13	12
Coagulase negative staphylococcus	12	14
E. coli	10	8
Pseudomonas aeruginosa	8	8
Enterobacter spp	8	7
Proteus mirabilis	4	3
Klebsiella pneumoniae	3	3
Streptococcus	3	3
Candida albicans	2	3
Second G+ aerobic bacteria	0	2
Bacteroides fragilis	0	2

The same Protocol also lists the most common specific surgical intervention pathogens (Table 3):

Table 3 Most common specific surgical intervention pathogens.

Surgical interventions	The most possible pathogens
Implantation all types of prosthesis	Staphylococcus aureus, KoNS
Cardiosurgery, neurosurgery, plastic surgery, traumatology and vascular surgery	Staphylococcus aureus, KoNS
Ophthalmology	Staphylococcus aureus, KoNS Streptococci, gram negative bacilli
Orthopedics	Staphylococcus aureus, KoNS
Thoracic surgery	S.pneumoniae, gram negative bacilli
Appendectomy, bilirary tract, colorectal surgeries	Gram negative bacilli, anaerobes
Gynecological surgeries	Gram- negative bacilli, anaerobes bacterias enterococci, BHS-B
Urological surgeries	Gram neg bacilli

Regarding the type of wound, the most common aerobic and anaerobic pathogens are specified (Table 4):

Table 4 Most common aerobic and anaerobic pathogens based on type of wound.

Type of wound	Pathogens
Acute infection of soft tissue (skin abscess, trauma, necrotizing infection)	S.aureus, S.pyogenes, E.coli, B.fragilis, C.perfringens, Prevotella spp., Peptostreptococcus spp.
Infections in bite wounds	S.aureus, Bacteriodes spp., Peptostreptococcus spp., Pasteurella multocida, Capnocytophaga canimorsus, Eikenella corrodens
Infection in burns	P.aeruginosa, S.aureus, E.coli, K. pneumoniae, Enterococcus spp., Candida spp., Peptostreptococcus spp. Bacteriodes spp., Propionibacterium acnes
Diabetic foot infection (chronic wound)	S.aureus, S.epidermidis, Streptococcus spp., P.aeruginosa, Enterococcus spp., Enterobacteriaceae, Prevotella spp., Peptostreptococcus spp., Bacteriodes spp.
Bedsore infection and infected ulcus cruris (chronic wound)	S.aureus, P.aeruginosa, Peptostreptococcus spp., Bacteriodes fragilis and spp.

Clinically, SSI defined according to Guidelines approved by Centers for Disease Control and Preventions (CDC) and National Nosocomial Surveillance System (NNSI) is classified into three categories:

1. Superficial incisional SSI with incidence of 63%;
2. Deep incisional SSI with incidence of 27%; and
3. Organ/space SSI with incidence of 10%.

Superficial incisional surgical site infection involves skin and subcutaneous tissue, deep incisional surgical site infection involves fascia and muscles, and so - called organ/space surgical site infection involves any parts of the anatomy, e.g. organs and spaces (Figure 1).

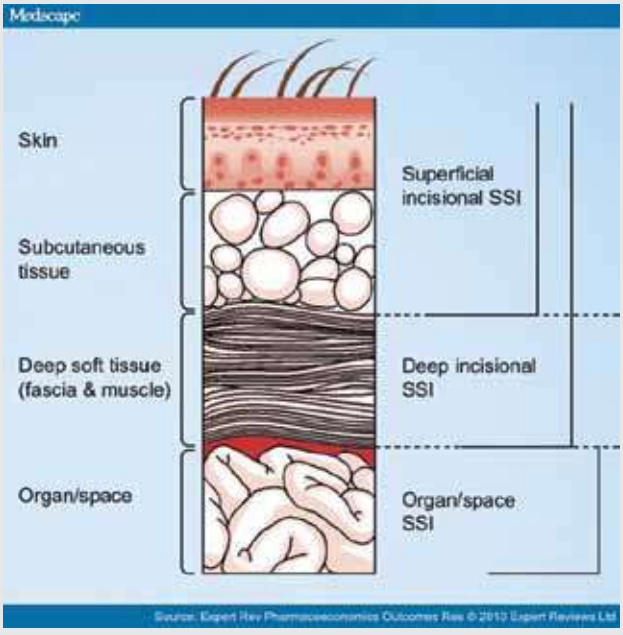


Figure 1 Classification of SSI according to NNSI.

Clinical infection of surgical site is manifested by development of redness, swelling, pain, different quality secretions in the surgical wound projection, signs of wound dehiscence and anastomosis in abdominal surgery, general signs of infections such as increased body temperature, fever and shivering, and serious complications such as bacteremia and sepsis.

Clinically apparent infection requires microbiological patogens confirmation, which in everyday work can be managed by taking a swab, by aspiration and tissue sample taking or by a blood culture test. Microbiological analysis of the samples was performed at the CCUS Institute for Microbiology.

Confirmed infections require a series of measures, primarily local surgical measures and adequate antibiotic therapy. We at the Clinic of General and Abdominal Surgery routinely use the combination of Cefasoline and Metronidazole for patients in the emergency program. As for the election program we have recently started using the Cefasoline application scheme immediately before the surgery and once after the surgery, whereas in microbiologically confirmed SSI we administer antibiotics according to antibiogram.

MATERIALS AND METHODS

Analysis of medical documentation and data of patients with evident signs of surgical site infection treated at Clinic of General and Abdominal Surgery of the CCUS was done retrospectively in the period from 1 February to 31 August 2018. The analysis involved age and gender structure, conditions under which surgical interventions were performed, and microbiological findings of the CCUS Institute for Microbiology. Data was gathered from history of diseases, clinic protocols of patients monitored through specialist outpatient clinic, and review of the Institute for Clinic Microbiology data. Data was analyzed using descriptive statistics and displayed in figures and tables.

RESULTS

Based on medical documentation, 1489 patients were surgically treated at Clinic of General and Abdominal Surgery of the CCUS in the period from 1 February 2018 to 31 August 2018. Out of that number, 168 (11.3%) of patients had apparent clinical signs of infection (redness, swelling, secretion, pain), but microbiological analysis confirmed infection in 144 (6.99%) patients. Microbiological analysis also confirmed negative findings in 24 patients.

Table 5 Presentation of patients with clinical signs of infection and microbiologically confirmed infection.

Clinical signs of infections	168	11.3 %
Microbiologically confirmed infection	144	9.69 %
Microbiologically negative samples	24	1.61 %
TOTAL	1486	100%

The percentage of confirmed SSI in all criteria corresponds to mean values quoted by majority of European and American sources, and it also corresponds to CDC data.

In the above mentioned sample and based on microbiological findings, we registered recurrence of findings in nine patients, who had previously had positive findings.

Out of the total of 144 patients, 83 (57.6%) were males and 61 (42.3%) were females.

Table 6 Representation of SSI in respect of gender.

Male patients	83	57.6%
Female patients	61	42.3%
TOTAL	144	100%

Our sample registered somewhat higher representation of male gender which corresponds to majority of studies, except those which analyzed data related to patients subjected to gynecological surgical treatments ($X2 = 13.766$, $p = 0.0002$).

Average age structure of patients in our group was 53.34 years. The youngest patient was 16 and the oldest one was 84. Data set out below corresponds to the average age of patients treated at our Clinic (Figure 2).

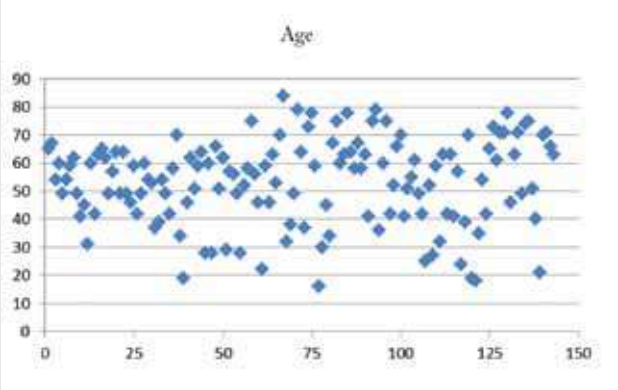


Figure 2 Average age structure of patients treated at our Clinic.

Out of the total of 144 patients with subsequently confirmed surgical site infection, 87 (60.4%) were surgically treated in emergency procedure (outpatient clinic), and 57 (39.5%) in an elective procedure program (Figure 3).

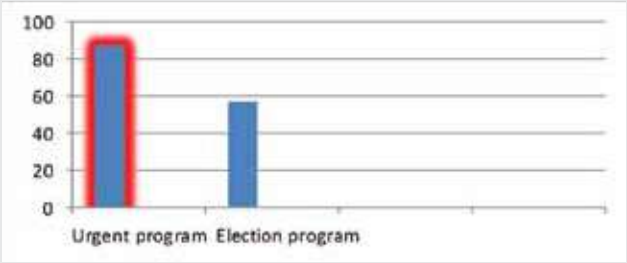


Figure 3 Patients surgically treated in emergency and elective procedure program.

This data differs from our total semi-annual impact due to the fact that the CCUS Clinic of General and Abdominal Surgery performs over 40% of surgical interventions in the emergency procedure program. However, this data corresponds to the majority of studies which in the incidence of SSI quoted the domination of patients surgically treated in the emergency procedure and in acute stages (up to 40%).

In 95 patients (65.9%) signs of infection were apparent in the first five days after surgical intervention, with an average on day four. In the 5-10 days period signs of infection developed in 31 patients (21.5%), with an average on day seven. In the up to thirty days period signs of infection developed in 18 patients (12.5%), which was recorded during outpatient checkups and based on swabs taken for microbiological analysis.

Based on microbiological analysis the following microorganisms were isolated (Table 7).

Table 7 Isolated microorganisms.

Aerobic microorganisms	
Escherichia coli	63
Enterococcus faecalis	40
Staphylococcus aureus MSSA	26
Klebsiella pneumoniae	16
Enterobacter cloacae	16
MRSA	16
Proteus mirabilis	14
Pseudomonas aeruginosa	14
Acinetobacter baumanii	13
Streptococcus alpha hemolyticus	7
Maltophilia	6
Klebsiella oxytoca	3
Streptococcus species	3
Serratia marcensens	2
Morganella morgani	2
Streptococcus agalactiae	2
Staphylococcus epidermidis	1

Anaerobic microorganisms	
Bacteroides spp	14
Fusobacter spp	8
Clostridium spp	6
Prevotella melaninogenica	4
Peptococcus spp	4

Table 7 shows representation of microorganisms and their recurrence. In the majority of patients bacteria of normal intestinal flora were isolated, which corresponds to surgical intervention site. Out of fourteen samples anaerobic positive on Bacteroids spp, B. Fragillis was isolated in five samples.

DISCUSSION

Incidence of surgical site infection established in our study (9.7%) was higher in respect to the results of studies conducted in developed countries such as the USA (1.9%) (6), France (1.0%) (7) and Italy (2.6%) (8). Brazilian study, which investigated surgical site infections in general surgery, confirmed higher incidence of SSI in comparison to our study (11.0%) (9). Diversity of data obtained from our and other studies can be reflected in different interpretation and monitoring of surgical infections, possible lack of their reporting, and in the fact that in majority of studies surgical infections were monitored in general surgery, unlike our study which primarily included abdominal surgery patients. Furthermore, relatively high SSI incidence in our study can also be justified by the fact that patients included in the study were from secondary and tertiary level of healthcare protection.

Out of the total of 144 patients with confirmed microbiological infections, 57.6% were male, and 42.3% were female. In the research dealing with gender differences as the cause of SSI, Corrina L, et al., concluded that incidence of SSI was significantly lower in women with the rate of 2.92/100, whereas SSI was developed in 4.37/100 male patients ($p < 0.001$) (10). Romana-Souza, et al. showed that gonadal hormones differently modulate skin wound healing, in the way that androgens contribute to slowing down the healing process whereas estrogens have adverse effect. Furthermore, it is proved that androgens have pro-inflammatory effect on the wound, which reduces repeated epithelialization, while estrogens have anti-inflammatory effect. Finally, they state that men and women differently react to stress (11).

Average age of patients in our study was 53.43 years, which corresponds to data from similar available studies (12).

Higher incidence of surgical site infection develops in emergency cases as opposed to interventions in the election abdominal surgery program due to poorly designed preoperative preparation and ,,dirty operations", as was evident in the results of a comprehensive study conducted in Japan. This study showed that contamination of incision was a strong risk factor for development of SSI in emergency abdominal surgery (12). Out of the total of 144 patients from our study with subsequently confirmed surgical site infection, 87 (60.4%) was surgically treated in the emergency procedure program and 57 (39.5%) in the elective procedure program, which corresponds to the results of the above mentioned study. Irrigation of subcutaneous tissue and peritoneal lavage are effective in prevention of SSI development (13). Published meta-analysis of 24 randomized controlled studies which included 5004 patients subjected to intraoperative washing, showed that use of povidon-iodide significantly reduced the rate of SSI (14). Protection of incisional wound significantly reduced development of SSI in elective open colorectal surgery (15,16). It is considered that protection preserves moisture and prevents the damage of tissue at incision site which could contribute to prevention of incisional SSI (17). Consequently, usage of providon-iodide and protection of incision could contribute to reducing SSI even in emergency surgical procedures. Therefore, we consider it should become a customary practice in our clinic.

Infection of surgical wound most frequently occurs thirty days after surgical intervention (18). In 98 patients (65.9%) from our study signs of infections were evident in the first five days after surgical intervention, with an average on day four. Accordingly, special attention in SSI prevention should be given to this particular period which in our study proved to be significant. We believe that this conclusion sets a new research ground, specifically in the sense of multivariate analysis which would determine independent risk factors SSI development in this early postoperative period.

Microbiological profile found in patients included in the study supports the claim that endogenous factors are the most responsible for infections. In our case E.Coli was in the first place with the incidence of 37.5% which corresponds to the results of study conducted in Turkey (22.8%), which also included abdominal surgery patients (19). In a similar study which included patients treated in the field of general surgery, Staphylococcus aureus (24,3%) and E. Coli (15,3%) were primary identified pathogens (20).

Nurse/medical technician and other members of healthcare team may perform particular activities aimed at reducing the SSI incidence (20). These activities include preoperative washing (21,8,22), better preoperative control of glucose in patients diagnosed with Diabetes Mellitus, preoperative preparation of surgical site, control of operating room environment factors, regular wound management and other postoperative procedures, which they can perform independently and in cooperation with respective physicians (22).

CONCLUSION

Surgical field infection does not avoid hospital institutions in the world most developed countries, with our region not being and exemption. The sample tested in our study has shown that the incidence of SSI corresponds to currently published international percentage. Given that SSI cannot be avoided, it is necessary to make additional efforts in the monitoring and prevention of SSI, which is a joint task of a surgical institution, competent infectology and microbiological services, public health institutions and competent health ministries of a particular region. The key solution is in regular cooperation between the mentioned institutions and adoption of recent protocols derived from more comprehensive, multiethnic studies.

Conflict of interest: none declared

REFERENCES

1. Smyth ETM, Emmerson AM. Surgical site infection surveillance. J Hosp Infect. 2000; 45:173-184.
2. Mangram JA, Horan CT, Pearson LM, Silver LC, Jarvis RW. The Hospital Infection Control Practices Advisory Committee: Guidelines for Prevention of Surgical Site Infection; 1999.
3. Nichols RL. Postoperative infections in the age of drug-resistant Gram-positive bacteria. Am J Med. 1998;104:11-19.
4. Garner JS. The CDC Hospital Infection Control Practices Advisory Committee. Am J Infect Control. 1993;21:160-2.
5. Berríos-Torres SI, Umscheid AC, Bratzler WD, Leas B, Stone CE, Kelz R, et al Centers for Disease Control and Prevention Guideline for the Prevention of Surgical Site Infection. JAMA Surg. 2017;152(8):784-791.
6. Mu Y, Edwards JR, Horan TC, Berrios-Torres SI, Fridkin SK. Improving risk-adjusted measures of surgical site infection for the National Healthcare Safety Network. Infect Control Hosp Epidemiol. 2011;32(10):970-986.
7. Saunders L, Perennec-Olivier M, Jarno P, L'Heriteau F, Venier A, Simon L, et al. Improving prediction of surgical site infection risk with multilevel modeling. Plos One. 2014;9(5):e95295.
8. Marchi M, Pan A, Gagliotti G, Morsillo F, Parenti M, Resi D, et al. The Italian national surgical site infection surveillance programme and its positive impact, 2009 to 2011. Euro Surveill. 2014;19(21):1-7.
9. Agência Nacional de Vigilância Sanitária (ANVISA) Critérios diagnósticos de infecção relacionada à Assistência à Saúde [Internet] Brasília (DF): Ministério da Saúde; 2013.
10. Langelotz C, Mueller-Rau C, Terziyski S, Rau B, Krannich A, Gastmeier P, Geffers C. Gender-Specific Differences in Surgical Site Infections: An Analysis of 438,050 Surgical Procedures from the German National Nosocomial Infections Surveillance System. Viszeralmedizin. 2014;30:114-117.
11. Romana-Souza B, Assis de Brito TL, Pereira GR, Monte-Alto-Costa A. Gonadal hormones differently modulate cutaneous wound healing of chronically stressed mice. Brain Behav Immun. 2014;36:101-110.
12. Masanori W, Hideyuki S, Satoshi N, Kentaro M, Naoto C, Osamu K, et al. Risk Factors for Surgical Site Infection in Emergency Colorectal Surgery: A Retrospective Analysis. Surgical Infections. 2014;15(3):256-261.
13. Horiuchi T, Tanishima H, Tamagawa K, Sakaguchi S, Shono Y, Tsubakihara H, et al. A wound protector shields incision sites from bacterial invasion. Surg Infect. 2010;11(6):501-3.
14. Fournel I, Tiv M, Soulias M, Hua C, Astruc K, Aho Glélé LS. Meta-analysis of intraoperative povidone-iodine application to prevent surgical-site infection. Br J Surg. 2010;97(11):1603-13.
15. Horiuchi T, Tanishima H, Tamagawa K, Matsuura I, Nakai H, Shouno Y, et al. Randomized, controlled investigation of the anti-infective properties of the Alexis retractor/protector of incision sites. J Trauma. 2007;62(1):212-5.
16. Reid K, Pockney P, Draganic B, Smith SR. Barrier wound protection decreases surgical site infection in open elective colorectal surgery: A randomized clinical trial. Dis Colon Rectum. 2010;53(10):1374-1380.
17. Horiuchi T, Nakatsuka S, Tanishima H, et al. A wound retractor/protector can prevent infection by keeping tissue moist and preventing tissue damage at incision sites. Infect Dis. 2007;3:17-23.
18. Holihan LJ, Flores-Gonzalez RJ, Mo J, Ko CT, Kao SL, Liang KM. How Long Is Long Enough to Identify a Surgical Site Infection? Surgical Infections. 2017;18(4):419-423.
19. Isik O, Kaya E, Dundar HZ, Sarkut P. Surgical site infection: re-assessment of the risk factors. Chirurgia. 2015;110(5):457-461.
20. Carvalho RLR, Campos CC, Franco LMC, Rocha AM, Ercole FF. Incidence and risk factors for surgical site infection in general surgeries. Rev Lat Am Enfermagem. 2017;25:e2848.
21. Franco LMC, Ercole FF, Mattia A. Infecção cirúrgica em pacientes submetidos a cirurgia ortopédica com implante. Rev SOBECC. 2015;20(3):163-170.
22. Lindblom RPF, Lytsy B, Sandstrom C, Ligata N, Larsson B, Ransjo U, et al. Outcomes following the implementation of a quality control campaign to decrease sternal wound infections after coronary artery by-pass grafting. BMC Cardiovasc Disord. 2015;15(154):1-9.

Reprint requests and correspondence
Salem Bajramagić, MD
Clinic of General and Abdominal Surgery
Clinical Center University of Sarajevo
Bolnička 25, 71000 Sarajevo
Bosnia and Herzegovina
Email: salem.bajramagic@gmail.com

Complication of chickenpox in immunocompetent patients hospitalized at the Clinic of Infectious Diseases of the Clinical Center University of Sarajevo

Komplikacije varičela kod imunokompetentnih pacijenata hospitaliziranih na Klinici za infektivne bolesti Kliničkog centra Univerziteta u Sarajevu

Rusmir Baljić[1]*, Hadžan Konjo[2], Bekir Rovčanin[3], Mirsada Hukić[4]

[1]Clinic of Infectious Disease, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina
[2]Clinic of Orthopedics and Traumatology, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina
[3]Faculty of Medicine, University of Sarajevo, Čekaluša 90, 71000 Sarajevo, Bosnia and Herzegovina
[4]Burch International University, Francuske revolucije bb, 71210 Ilidža, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

Chickenpox is a childhood disease and generally considered as self-limited with favorable outcomes. In some cases, mostly in immunocompromised hosts, complications are possible. Severe complications have also been reported in previously healthy persons, even with lethal outcome. In this prospective study, we observed 35 immunocompetent patients with chickenpox who developed a complication during the clinical course. Most of them had a bacterial superinfection of skin lesions. Around 80% required antibiotic therapy. There were no lethal cases.

Keywords: chickenpox, immunocompetency, complications

SAŽETAK

Varičele predstavljaju dječije oboljenje koje se smatra samolimitirajućim uz povoljan ishod. U nekim slučajevima, većinom kod imunokompromitiranih domaćina, moguće su komplikacije. Ozbiljne komplikacije zabilježene su i kod prethodno zdravih osoba, čak i sa smrtnim ishodom. U ovoj prospektivnoj studiji posmatrali smo 35 imunokompetetnih pacijenata sa varičelama, kod koji su se tokom hospitalizacije javile komplikacije. Većina je imala bakterijske superinfekcije kožnih promjena. Oko 80% njih zahtjevalo je antibiotiski tretman. Nismo zabilježili smrtne slučajeve.

Ključne riječi: varičele, komplikacije

INTRODUCTION

Varicella zoster virus is a highly contagious alphaherpesvirus that causes a primary infection known as varicella (chickenpox). Reactivation of the infection after few years causes herpes zoster (shingles) (1). Chickenpox is a disease of childhood because in 90% of cases it occurs in children under 13 year of age. It is generally considered to be self-limited disease, but in some cases, especially in immunocompromised hosts, bacterial complications are possible. Severe complications have been reported in previously healthy persons, and some of them with lethal outcome (2). Most complications in cutaneous lesions relate to secondary bacterial infection, often associated with gram-positive organisms. The most frequent noncutaneous site of involvement is central nervous system, which can be manifested as acute cerebellar ataxia or encephalitis (3). A serious and life-threatening complication is varicella pneumonitis, a complication that occurs more commonly in adults and in immunocompromised persons, approximately 1 in 400 cases (4).

MATERIALS AND METHODS

This prospective, descriptive study was conducted in the period from July 2014 to January 2016 at the Clinic of Infectious Diseases of the Clinical Center University of Sarajevo (CCUS). We observed hospitalized, immunocompetent patients, with confirmed chickenpox diagnosis. The study was approved by the CCUS Ethics Committee. We examined general characteristics such as age, sex, occupation, epidemiology, symptoms presented at admission, complications, therapy, clinical course and outcome of illness.

RESULTS

In the18-month period (from 07.07.2014 to 07.01.2016) we had 416 patients diagnosed with chickenpox. Out of that number we hospitalized 44 patients, and observed 35, given that other patients did not meet the inclusion criteria. There were 25 (71%) male and 20 (29%) female pa-

tients. The youngest patient was three years old, whereas the oldest one was 54, with median and standard deviation 24.57 and 14.14 respectively.

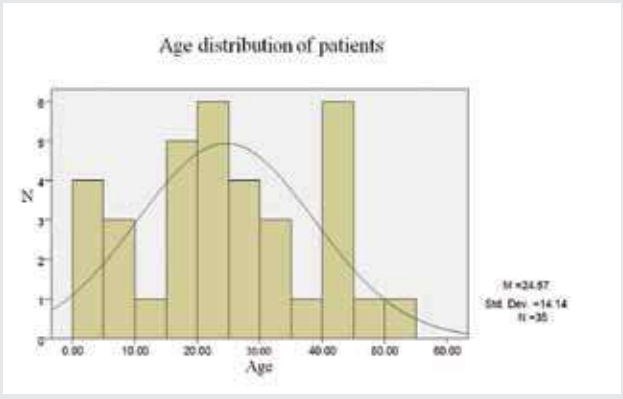


Figure 1 Age distribution of patients.

Table 1 Presented complications.

Complications	N	%
Bacterial skin superinfection with Pnemonitis	8	22.85
Bacterial skin superinfection	7	20
Bacterial skin superinfection with Conjunctivitis	5	14.28
Pneumonitis	4	11.43
Extensive rash	3	8.57
Conjunctivitis	2	5.71
Bacterial skin superinfection + Pneumonitis +Conjunctivitis	2	5.71
Pneumonitis+Hepatitis	2	5.71
Abscess	1	2.86
Bacterial skin superinfection + Hepatitis	1	2.86

According to anamnesis, in 26 (74%) patients positive contact with chickenpox was confirmed, whereas in 9 (26%) that contact could not be confirmed. We had 25 (71%) adults and 10 (29%) children. Among them 13 (37%) were with high school education and 8 (23%) with high degree education, while others were students or pupils. All patients had characteristic rash and high temperature at admission to hospital, while headache was registered in 17 (48%), myalgies in 11 (31%) and respiratory tract symptoms in 14 (40%) patients. The patients developed from one to three complications during the course of their hospitalization. Only one complication was presented in 23(63%) cases, two in 10 (28%) and three in 3 (9%) cases. Most of the complications were due to bacterial superinfection and infection of lower respiratory tract. Antivirotic therapy (aciclovir) was indicated for all patients, whereas 28 (80%) of them also required antibiotic therapy, as presented in Table 2. One patient with facial and neck abscess required surgical procedure and multidisciplinary approach. Average duration of antibiotic therapy was 10.39 days. All patients were discharged from hospital as „recovered" or „healed".

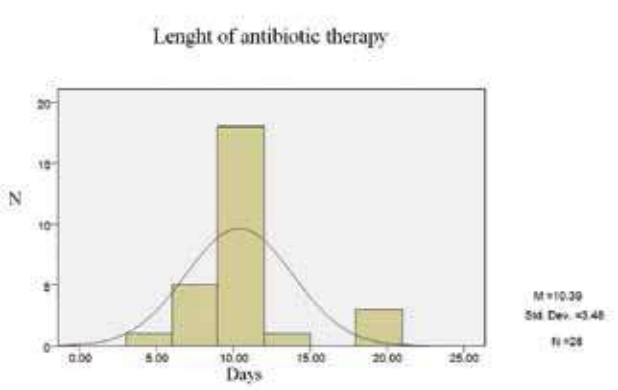


Figure 2 Lenght of antibiotic therapy.

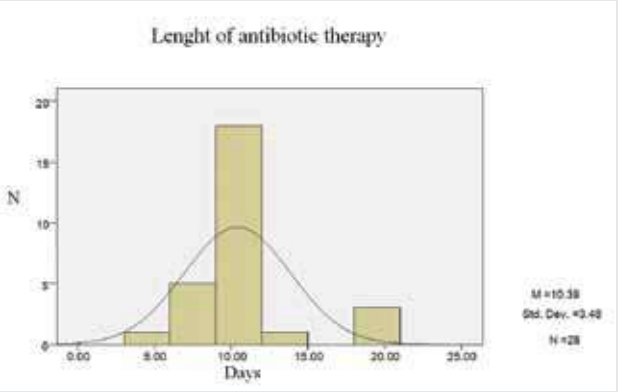


Figure 3 Lenght of hospitalization.

Table 2 Antibiotics used in therapy.

Antibiotics	N	%
Cefasolin	9	32.14
Cefasolin+Tobramycin	6	21.42
Claritromycin	4	14.28
Amoxicillin/Klav.acid	3	10.71
Amoxicillin/Klav.acid +Tobramycin	1	3.57
Ampicillin	1	3.57
Ceftriaxone+Clindamycin	1	3.57
Ceftriaxone+Vankomycin	1	3.57
Clindamycin	1	3.57
Tobramycin	1	3.57
TOTAL	28	100

DISCUSSION

Chickenpox is highly contagious disease, with benign or mild clinical course, but in some cases it can be severely complicated. It mostly affects adults, but in rare cases even children may develop wide range of complications. Also, males have higher risk to develop complica-

tion during the clinical course, and we had 2:1 male/female ratio. Main reasons underlying diseases in male population, like smoking, adiposity, cardiovascular and chronic respiratory diseases, were also described in Bovill and Bannister study (5). In our study, large number of patients developed more than one complication during the clinical course, almost 50%. Bacterial superinfection of the skin comes on the first place, both in children and adults, and in over 60% of our patients some complications were also registered. More than 40% of patients had bacterial superinfection of the skin accompanied with other complication. Some other authors described this complication as dominant in chickenpox (6,7), whereas some other studies described pneumonia as a primary complication. In a study of Dulovic et al, pneumonia was dominantly presented as a complication, while we had it in around 40% cases, alone or with other complication (8). Neurological complications, like cerebellitis were dominant in a study conducted by Liesse et al, but they observed only children (9). Cerebellitis comes as a post-varicellous complication, and while we observed only patients in acute phase of disease, we had no similar cases.

Other registered complications included conjunctivitis, extensive rash, hepatitis, and the most serious complication was facial and neck abscess, developed in 3-yrs old kid. Treatment of this patient also required local surgical procedures, with prolonged antibiotic therapy. Similar complications in children were also described earlier (10). Therapy of these patients included antiviral (acyclovir) and antibiotic, with supportive treatment. All patients received acyclovir, 5-7 days, and 28 patients required antibiotic. In nine (32%) cases we used combination of two or even three antibiotics, while others considered monotherapy as appropriate. Duration of antibiotic therapy vary from four to 21 days, depending on clinical course. Length of hospitalization was 4-21 days. A patient who was hospitalized for four days left the clinic earlier, on his own will. Outcome of the disease was good in all patients. We had no lethal cases.

CONCLUSION

Bacterial superinfection of skin lesions and infection of respiratory tract are the most significant complications that can be expected in immunocompetent patients with chickenpox. In case of bacterial etiology, appropriate antibiotic therapy lead to positive outcome of the disease.

Conflict of interest: none declared.

REFERENCES

1. Kim SR, Khan F, Ramirez-Fort MK, Downing C, Tyring SK. Varicella zoster: an update on current treatment options and future perspectives. Expert Opin Pharmacother. 2014;15(1):61-71.
2. Pollard A, Isaacs A, Lyall EG, Curtis N, Lee K, Walters S, et al. Potentially lethal bacterial infection associated with varicella zoster virus. BMJ. 1996;313:283-285.
3. Preblud SR, Orenstein WA, Bart KJ. Varicella: clinical manifestations, epidemiology and health impact in children. Pediatr Infect Dis. 1984;3:505-509.
4. Weber DM, Pellecchia JA. Varicella pneumonia: Study of prevalence in adult men. JAMA. 1965;192:572-573.
5. Bovill B, Bannister B. Review of 26 years' hospital admissions for chickenpox in North London. J Infect. 1998;36(1):17-23.
6. Abro AH, Ustadi AM, Das K, Abdou AM, Hussaini HS, Chandra FS. Chickenpox: presentation and complications in adults. J Pak Med Assoc. 2009;59(12):828-31.
7. Dinleyici EC, Kurugol Z, Turel O, Hatipoglu N, Devrim I, Agin H, et al. The epidemiology and economic impact of varicella-related hospitalizations in Turkey from 2008 to 2010: a nationwide survey during the pre-vaccine era. Eur J Pediatr. 2012;171(5):817-825.
8. Dulović O, Gvozdenović E, Nikolić J, Spurnić AR, Katanić N, Kovarević-Pavićević D. Varicella complications: is it time to consider a routine varicella vaccination? Vojnosanit Pregl. 2010;67(7):523-529.
9. Ziebold C, von Kries R, Lang R, Weigl J, Schmitt HJ. Severe complications of varicella in previously healthy children in Germany: a 1-year survey. Pediatrics. 2001;108(5):E79.
10. Cameron JC, Allan G, Johnston F, Finn A, Heath PT, Booy R. Severe complications of chickenpox in hospitalised children in the UK and Ireland. Dis Child. 2007;92(12):1062-1066.

Reprint requests and correspondence:
Rusmir Baljić, MD, PhD
Clinic of Infectious Disease
Clinical Center University of Sarajevo
Bolnička 25, 71000 Sarajevo
Bosnia and Herzegovina
Email: rusmir.ba@gmail.com

Evaluation of patients with life-threatening injuries at Clinic of Emergency Medicine of the Clinical Center University of Sarajevo

Evaluacija pacijenata sa životno ugrožavajućim povredama na Klinici urgentne medicine Kliničkog centra Univerziteta u Sarajevu

Amela Tuco*, Zoran Hadžiahmetović

Clinic of Emergency Medicine, Clinical Center University of Sarajevo, Bolnička 25, 71000 Sarajevo, Bosnia and Herzegovina

*Corresponding author

ABSTRACT

Trauma is one of the leading causes of mortality in the World, which puts it in the focus of researches aimed at preventing injuries, better medical care of patients and reducing the mortality rate. The severity of trauma is calculated with many scores, but ISS (Injury Severity Score) is mostly used. The mortality risk is in direct correlation with injury severity, but also with the mechanism of injury and age of the injured person. Thanks to the introduction of scoring, the term „life-threatening trauma" (major trauma) got its quantitative definition, instead of many previous based only on qualitative characteristics. Nowadays it relates each and every injury (poly- or isolated trauma) with total ISS ≥16 and at least one physiological disorder. This study evaluated patients corresponding to this definition. Materials and methods: retrospective analysis included a medical database of 256 patients with life-threatening injuries treated at our Clinic in the period from 01. January to 31. December 2017y. The study excluded patients with incomplete injury-related documentation and those who died before completing diagnostic procedures. The aim was to present a quantitative analysis of injury severity using ISS and expected mortality rate based on sex and age of patients, to prove the adequacy of including monotrauma in the class of trauma major and give recommendations for further analyses and actions for advancing the severely injured patients' care. The majority of patients were men with 73.83 % and only 26.17% were women. The mean ISS value was 33.23 (+/- 9.92), and majority of patients, specifically 82.03% of them had ISS ≥25 (critical trauma). The mean age of patients was 47, and about one-third of all patients or 34.90% of them were over 60 years of age (age of very high mortality risk). About one-third of patients (33.20%) had an injury of only one body region, and 83.53% of them had ISS ≥25. It was obvious that the majority of patients had a high expected mortality rate. Therefore, it is important to improve performances in the treatment of this category of patients and to achieve coordination with world trends. Furthermore, it is very important to pay attention to isolated injuries due to their high percentage and severity. It is important to implement and adhere to the Clinical Pathway for life-threatening injured patients and to maintain regular statistics on quality of injuries and hospitalization outcomes in order to establish actual mortality rates and arrange work procedures for the purpose of minimizing thereof. Registration should be taken on the hospital institution-level due to lack of a unified trauma register on the state level which establishes is inevitable in the future for adjustment with the world standards. Having in mind that registering trauma on the state or regional level is a general trend, the EU countries have developed their unique register.

Keywords: life-threatening injury, polytrauma, ISS-score, mortality risk, clinical pathway, trauma register

SAŽETAK

Trauma je jedan od vodećih uzroka mortaliteta u svijetu što je stavlja u fokus istraživanja u cilju preventivnog djelovanja, boljeg zbrinjavanja povrijeđenih i samim time smanjenja mortaliteta. Težina traume procjenjuje se skoriranjem, pri čemu je ISS skor najviše u upotrebi. Rizik mortaliteta je u direktnoj korelaciji sa težinom povrede, ali i sa mehanizmom povrede i starosti povrijeđenih. Zahvaljujući uvođenju skoriranja pojam „životno ugrožavajuće traume" (trauma major) je nakon nekoliko opisnih konačno dobio kvantificiranu definiciju kojom se obuhvataju politraume i monotraume sa ISS ≥16 i poremećajem najmanje jednog fiziološkog parametra. U ovom radu je napravljena evaluacija pacijenata koji odgovaraju ovoj definiciji. Materijal i metode: retrospektivno je analizirano je 256 pacijenata sa životno ugrožavajućim traumama koji su zbrinuti na Klinici urgentne medicine Kliničkog centra Univerziteta u Sarajevu (KCUS) u jednogodišnjem intervalu (od 01.01 do 31.12.2017. godine) prema podacima iz medicinske dokumentacije. Iz analize su isključeni pacijenti sa nepotpunom dokumentacijom o povredama i pacijenti koji su preminuli prije završetka dijagnostičkih procedura. Ciljevi su bili prikazati kvanitativnu analizu težine povreda prema ISS skoru i očekivanu stopu mortaliteta u odnosu na spol i starost pacijenata, dokazati ulogu izoliranih trauma kao životno ugrožavajućih, dati preporuke za daljnje analize i djelovanje u cilju unapređenja zbrinjavanja životno ugroženih traumatiziranih pacijenata. Preovladavali su pacijenti muškog spola njih 73,83%, dok je žena bilo 26,17%. Prosječna vrijednost ISS skora je bila 33,23 (+/- 9.92), a najveći broj pacijenata, njih

82,03%, je imao ISS ≥25 (veoma jaka trauma). Prosječna starost pacijenata je bila 47 godina, a oko trećine pacijenata, njih 34,9% je bilo starije od 60 godina (dob veoma visokog rizika mortaliteta). Oko trećine pacijenata (33,20 %) je imalo povredu samo jednog organskog sistema, od kojih 83,53% sa ISS skorom ≥25. Prikazana analiza je pokazala da preovladavaju pacijenti sa visokom očekivanom stopom mortaliteta. Usavršavanje rada i usklađivanje sa svjetskim trendovima je stoga neophodno. Važno je pored politrauma posvetiti pažnju i izoliranim traumama zbog njihovog visokog udjela i težine. Neophodna je implementacija i pridržavanje Kliničkog puta teško traumatiziranog pacijenta, kao i vođenje redovne statistike o kvaliteti povreda i bolničkih ishoda, kako bi se utvrdile stvarne stope mortaliteta i modelirale radne procedure u cilju smanjenja istog. Evidencija bi se zasada trebala provoditi na nivou hospitalnih ustanova zbog nedostatka jedinstvenog trauma-registra na nivou države, iako je njegovo kreiranje neminovno u budućnosti zbog usklađivanja sa europskim standardima. Naime, evidencija ne samo na nivou država, nego i regija je generalni trend, stoga zemlje EU imaju jedinstveni registar.

Ključne riječi: životno ugrožavajuća povreda, politrauma, monotrauma, ISS skor, rizik mortaliteta, klinički put,trauma registar

INTRODUCTION

Definition and classification of life-threatening trauma presented a problem for physicians in emergency and trauma centers for many years. This term was considered to be synonym to polytrauma or multiple trauma. First attempts in defining polytrauma emerged in 1970. However, one of the most common definitions was created in 1984 by H. Tscherne who considered polytrauma as two or more injuries, among which at least one was life threatening (1). Tscherne's definition was the most popular and used, but in everyday practice there is a possibility of subjective interpretation of „life threatening injury" and making wrong diagnoses and prognostic presumptions. Many other definitions were evaluated through the years, but none of them was appropriate. That instigated the development of a new idea about quantifying the severity of injuries in correlation with expected complications, consequences and mortality rate. For this purpose many scoring systems have been developed (1,2). The first step in grading severity of individual injuries was made by the Association for the Advancement of Automotive Medicine (AAAM). They developed the Abbreviated Injury Scale (AIS) which was introduced in 1971. It has been frequently revised and updated according to expectable outcomes, and nowadays it has wide implementation due to its predictive validity. It is anatomically based, consensus-derived, internationally approved system for individual injuries grading (2,3). It classifies individual injuries on a six point scale according to its relative severity:

1. minor
2. moderate
3. serious
4. severe
5. critical
6. maximal (untreatable)

Every injury has a specific seven-number code which describes type, location and severity. The first number codes one of nine body regions:

1. Head
2. Face
3. Neck
4. Thorax
5. Abdomen
6. Spine
7. Upper extremity
8. Lower extremity
9. Unspecified

Numbers 2-6 describe detailed location, specific organ, structure and type of injury. Severity of injury is presented by 7th number, which is actually AIS value (1-6) (3,4,5). Head injuries make exception, because of grading a loss of consciousness, not only anatomic structures (5,6).

AIS value can predict relative risk of fatal outcome of some individual injuries. However, that is not sufficient when observing organ systems or complete body. That's why it is used mostly as a basis for making more advanced scores.

Injury Severity Score - ISS was introduced in 1974 by Baker et al. It is anatomically based, global implemented system for summarizing multiple injuries in a single patient and calculating total severity for predicting mortality risk. Basis for calculating this score is AIS value. For this purpose, six regions of the body are observed: 1. Head and neck (including cervical spine), 2. Face 3. Chest (including thoracic spine) 4. Abdomen with pelvic content (including lumbar spine) 5. Extremities and pelvic bones 6. External (including skin injuries and burns).

From each body region, only the most severe injury is taken, according to AIS value. For measuring the overall severity of injured patients, three most affected body regions are used. ISS - score is the sum of squares of AIS values in three most severely injured body regions: ISS = AIS1^2 +AIS2^2 +AIS3^2.

ISS - score takes values from 0 to 75. If there is any injury in the body with AIS value 6, ISS score automatically takes maximal value 75 (2,7). ISS scoring system has many limitations. It takes maximal three body regions and only one injury from each of them. It does not have to grade all the injuries, but only some of them, not even necessary the hardest. One region can involve more than one injury with the same AIS value, but only one is squared. All injuries are important for the final outcome of polytrumatized patients because of interaction, reason more for ISS limitations in prediction (2,8).

Anatomically-based scores have many disadvantages. They ignore the role of the vital parameters and the influence of the patient's condition on the final outcome. Given the importance of these factors in predicting mortality, many functionally based scored are developed. Most common are Trauma score (TS), and its revised form RTS (Revised Trauma Score).

Trauma Score is physiological scoring system based on a numeric grading of vital signs of traumatized patient for estimating his condition and trying to predict the outcome. Observed parameters are: 1. GCS, 2. Systolic blood pressure, 3. Respiratory rate, 4. Respiratory effort, 5. Capillary refill. Trauma score was revised for everyday routine use. Revised trauma score (RTS) observes and grades values

of three physiological parameters: GCS, systolic blood pressure and respiratory rate. Final RTS is calculated by consensus based formula: RTS = 0.9368 GCS + 0.7326 SBP + 0.2908 RR Values of RTS range from 0 to 7.8408 (2,9).

Trauma Injury Severity Score - TRISS combines anatomical and physiological parameters. It includes ISS, RTS and age of traumatized patient to estimate probability of surviving. Expected probability of surviving is different depending on type of the injury, blunt of penetrating. For patients under 15, regardless of injury type, it is calculated using the blunt injury formula (2). Physiological and combine scores also have disadvantages. They are not appropriate for intubated patients because of the impossibility to observe pulmonary expansion. Sedated or patients on drugs and patients with some comorbidities are problematic in evaluating their state of consciousness.

Rearranging the present scores and creating new ones are attempts to annul the limitations, so today there are many different scoring systems, more or less acceptable in different conditions, depending on injury type and patient's condition. Still, there is no „perfect" scoring system without limitations and appropriate for each traumatized patient. Despite imperfections, injury scores are spread worldwide and improved in measuring and comparing severity of injuries and predicting mortality. To facilitate and speed the calculating process in emergency departments and trauma centers, special calculators are created and set on the web sites of many traumatology associations, for example http://www.trauma.org/js/trisscalc.html

Idea about quantifying and categorization of injuries and development of scoring systems show lack of validation of life threatening injury definitions which were in use in that time. It was inescapable to find unique definition set on exact parameters. New term was installed - „trauma major" (or severe trauma) for signing the life-threatening trauma, term considered equal to polytrama.

For decades, there were different opinions and interpretations about which ISS value should be taken as critical for survival of injured patient. To stop having disagreements, representatives of greatest traumatology associations in the world (German Trauma Society-DGU, European Society for Trauma and Emergency Surgery, British Trauma Society, American Association for the Surgery of Trauma, Australian and New Zealand Association for the Surgery of Trauma) established International Working Group for Polytrauma. Results of work and preliminary definition were presented in 2012 in Aachen at the 12th International Polytrauma Course. The new unique and consensus definition was accepted. It was agreed that injuries from at least two body regions with individual AIS ≥2 (ISS ≥ 16) and at least one physiological disorder presented polytrauma (1). However, it was evident, that even isolated injury could have ISS score ≥16. Monotrauma could also be categorized as severe, so equalizing term „trauma major" with polytrauma wasn't correct. Today, trauma major, as life-threatening trauma is considered as trauma with ISS score ≥16 with at least one physiological disorder despite the number of injured body regions or organs (10,11)

Problem with definition is not completely solved, because it is not specified which physiological parameters and values should be taken. There is still no worldwide consensus (1). As result of many consensus meetings, international experts are developing the new concept known as 'New Berlin definition'. It proposes five independent parameters which cut off values were calculated based on a set mortality rate of 30%: hypotension (systolic blood pressure ≤ 90 mm Hg), level of consciousness (Glasgow Coma Scale [GCS] score ≤ 8), acidosis (base excess ≤ -6.0), coagulopathy (international normalized ratio ≥ 1.4/partial thromboplastin time ≥ 40 seconds), and age (≥70 years). Researchers are still in progress (12).

Scoring systems for injuries were primarily created for predicting mortality of traumatized patients, but in time they have become important indicator of complexity of traumatized patient's population, and have been implemented in organization of working process in health institutions. For this use, the most appropriate ISS score is the one intended for hospital centers. Researches showed that mortality rate of injured patients with same ISS value rised proportionally with age (13,14), which resulted in different approach to traumatized patients in each age population and modulation of procedures and work organization (15,16). ISS score enabled not only registration of total number of injuries and deaths in countries and regions, but also registration of correlation between injury severity and rate of permanent invalidity and mortality. Nowadays, developed countries maintain complete statistical analyses related to number, types, causes and outcomes of all injuries for defining and observing risk group of patients and making preventive actions aimed at reducing the amount of traumatism and enhancing medical treatment. One of the first statistical projects of that kind was MTOS register (Major Trauma Outcome Study), activated by the American College of Surgeon's Committee on trauma. It was frequently revised until establishment of the national USA database known as National Trauma Data Bank (NTDB) (17,18). For this purpose, the European Union IDB register (Injury Database) is in use (19). Those are the greatest registers containing data from each country.

Preventing and safety actions are planned and implemented on national and regional level. The role of health system is organization and improvement of methodology related to caring for and treatment of traumatized patients in emergency departments and implementation of damage control surgery strategy (20).

MATERIALS AND METHODS

Analysis included 256 patients with life-threatening injuries (trauma major) treated at Clinic of Emergency Medicine of the CCUS in period from 1 January to 31 December 2017. Information about patients was taken retrospectively from medical database of the Clinic. For estimating severity of injuries ISS was used, as specific for hospital health care. Criteria for including patients were injuries of one or more organ systems with ISS value ≥16 and disorder of at least one physiological parameter. Patients with incomplete documentation about injuries and patients who died before completing diagnostic procedures were excluded.

The aim of the study was to present quantitative analysis of severity of injuries using ISS and expected mortality rate according to sex and age of the patients, to prove the adequacy of including monotrauma in the class of trauma major, and to give recommendations for further analyses and actions for advancing the severely injured patients' care.

Statistical analysis was done in MS Excel and the results were presented in tables by number of cases, percentage, mean (X) with a standard deviation (SD), maximal and minimal value.

RESULTS

Table 1 Sex structure of the patients according to ISS value (n=256).

Sex	ISS 16-24	ISS 25-49	ISS ≥50	ISS 75	Total
M	35 (13.67%)	139 (54.30%)	13 (5.08%)	2 (0.78%)	189 (73.83%)
F	11 (4.30%)	49 (19.14%)	8 (3.13%)	0.0%	67 (26.17%)
Total	46 (17.97%)	188 (73.44%)	20 (7.81%)	2 (0.78%)	256 (100.00%)

Table 1 shows that most of patients with life threatening injuries were men, 189 of them or 73.83%. There were 67 women or 26.17 % (Figure 1).

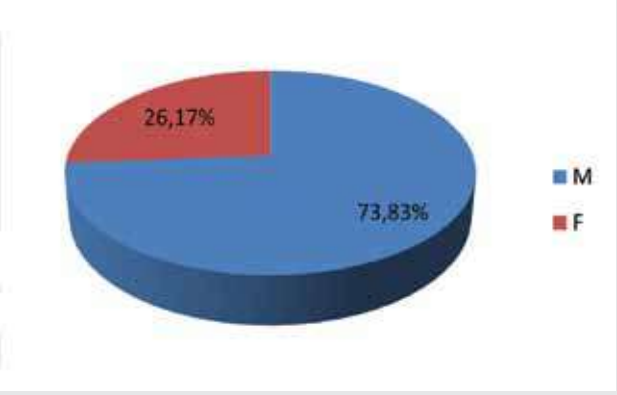


Figure 1 Sex structure of the patient (n=256).

Table 2 Values of ISS-score of the patients.

Variable	Mean	St. dev.	Max.	Min.
ISS value	33.23	+/- 9.92	75	16

According to the American College of Surgeons' Committee on Trauma injuries can be classified in four groups depending on the ISS value:

Minor - ISS 1-8
Moderate - ISS 9-15
Severe - ISS 16-24
Critical - ISS >24

Severe and critical injuries are considered life threatening with expected mortality rate of 7 and 35 %, respectively (21).

Table 1, Table 2, and Figure 2 show that majority of patients or 83.03% of them had the ISS value ≥ 25, which means that they had critical injuries.

Patients with the ISS score ≥ 50 covered only 7.81% of the total number of patients, and just 2 patients or 0.78% had ISS 75 which is the maximal value incompatible with survival.

Mortality rate rises with the ISS value (22), and low percent of very high ISS value patients to the CUM, as referral hospital centre, is expected, because of the time needed for their transportation.

Patients with the ISS 16-24 cover 17.97% of the total number of patients, which is expected in this type of hospital, given that less severely injured patients can be treated in lower level hospitals and units.

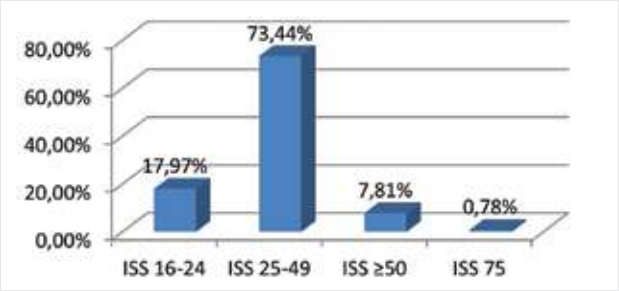


Figure 2 Injury severity according to the ISS value (n=256).

Table 3 Age structure of patients with life threatening injuries (n=252) according to injury severity.

Age	Sex	ISS 16-24	ISS 25-49	ISS ≥ 50	ISS 75	Total
0-19	M	4	20	1		40(15.9%)
	F	5	5	5		
20-39	M	8	35	2	2	53(21%)
	F		6			
40-59	M	8	36	5		71(28.2%
	F	4	16	2		
60-79	M	14	36	3		71(28.2%)
	F	1	17			
≥80	M	1	10	1		17(6.7%)
	F	1	4			

Out of the total of 256 observed patients, 252 had date of birth in medical documentation, and they were observed in the age structure analysis. Table 3 shows that 56.4% of those patients were between 40 and 79 years of age.

Table 4 Age structure of the patients.

Variable	Mean	Max.	Min.
Age	47	98	2

Table 4 provided data related to serious condition of analyzed patients. The average age of patients was 47 years. Mortality rate rises proportionally with age, but according to American College of Surgeons' Committee for patients with ISS ≥ 16 over 45 years of age it is much bigger than it could be expected from proportional rise (13,21). Most of the available literature consider persons older than 60 years as a high risk population because of lower chance for surviving (23,24) they have even twice higher mortality rate than younger persons with same severity of trauma (25). Among the observed patients, 34.9 % of them were in this age group, which means more than one third of all patients with life threatening injuries treated at the Clinic of Emergency Medicine in 2017.

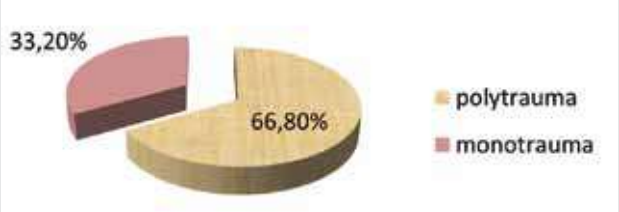


Figure 3 Proportion of isolated injuries and multiple injuries.

Out of the total number of analyzed patients (n=256) 171 (66.80%) were polytraumatized and 85 (33.20%) had isolated injury.

Table 4 Presentation of ISS values of patients with one organ system injury (n=85).

ISS	Number	%
16-24	13	15.29%
25	71	83.53%
75	1	1.18%

Data from Figure 3 and Table 4 prove the importance of scoring isolated injuries and putting them in group of life threatening conditions (10,11). About one third of all observed patients were those with monotrauma (33.20%). Number of them is not the only reason of their importance, but also their severity. Although injuries of only one organ or body region, even 83.53% of them had ISS 25, which is maximal value for this kind of injuries (maximal AIS score inside one body region is 5, it's square is 25 what is final ISS value) if ISS 75 is excluded (another possible option). According to American College of Surgeons' Committee on Trauma /ASC COT, injuries with ISS ≥25 qualifies as critical injuries with mortality rate of about 35 % (21).

DISCUSSION

Our analysis showed that majority of patients had very severe injuries (ISS ≥25), mostly elderly people, with high expected mortality rate which points to difficult and challenging work with this kind of patients and place an obligation to maximize dedication and continue training and development. Deficiency of this analysis results in absence of real mortality and hospital outcomes rates, so present programs of performance improvement are solely based on the expected results.

Currently Bosnia and Herzegovina does not have systematic approach to trauma. There is a terminology mess regarding injuries and lack of registration in the country. Unique trauma register with quantitative and qualitative data and real outcomes and mortality rates is required. The respective data should reveal how good we have been and what can we do to make it better. Producing complete statistics of trauma is general trend today which presents basis for planning prevention, protection, enhancement of health system and institutions, selection and delivery of resources for better care of traumatized patients (17,26).

CONCLUSION

Due to a large number of severely traumatized patients, keeping up with trends in scoring and terminology of life threatening injuries is very important for function of hospital emergency center such as Clinic of Emergency Medicine of the CCUS. Lack of the unique register should encourage the CCUS to make its own internal statistics of hospital outcomes and mortality rate, and based on such analyses and following general patterns to improve work organization and procedures. The main step is the implementation and maintenance of the Clinical Pathway for severely traumatized patients. This pathway will enable description of patients' condition, scoring and categorization of injuries, definition (poly - or monotrauma), demonstration of diagnostic and therapy procedures and final outcomes.

Conflict of interest: none declared.

REFERENCES

1. Butcher NE, Balogh ZJ. Update on the definition of polytrauma. Eur J Trauma Emerg Surg. 2014 Apr;40(2):107-11. doi: 10.1007/s00068-014-0391-x. Epub 2014 Mar 19
2. Pohlman T, Bjerke H, Offner P. Trauma Scoring Systems. Emedicine.medscape.com. 2014.
3. https://www.aaam.org/abbreviated-injury-scale-ais/
4. https://en.wikipedia.org/wiki/Abbreviated_Injury_Scale#cite_note-Abbreviated_injury_scale-4
5. The Abbreviated Injury Scale (AIS) a brief introduction. The Trauma Audit & Research Network (TARN). Available at : https://www.tarn.ac.uk/content/downloads/72/coding.pdf
6. Injury Scoring & Assigning an ISS. The Trauma Audit & Research Network (TARN). Available at : https://www.tarn.ac.uk/content/downloads/19/6.%20AIS%20and%20ISS.pptx
7. http://www.trauma.org/archive/scores/iss.html
8. Samin OA, Civil ID. The New Injury Severity Score Versus the Injury Severity Score in Predicting Patient Outcome : A Comparative Evaluation on Trauma Service Patients of the Auckland Hospital. Annu Proc Assoc Adv Automot Med. 1999;43:1-15
9. https://www.sciencedirect.com/topics/medicine-and-dentistry/revised-trauma-score
10. Paffrath T, Lefering R, Flohe S. Trauma Register DGU. How to define severely injured patients? - an Injury Severity Score (ISS) based approach alone is not ssufficient. Injury. 2014;45(Suppl 3):S64-9.
11. Palmer C. Major Trauma and the Injury Severity Score - Where Should We Set the Bar? Annu Proc Assoc Adv Automot Med. 2007;51:13-29.
12. Pape HC, Lefering R, Butcher N, Peitzman A, Leenen L, Marzi I, et al. The definition of polytrauma revisited: An international consensus process and proposal of the new 'Berlin definition'. J Trauma Acute Care Surg. 2014;77(5):780-786.
13. Morris JA Jr, MacKenzie EJ, Damiano AM, Bass SM. Mortality in trauma patients: the interaction between host factors and severity. J Trauma. 1990;30(12):1476-82
14. Ottochian M, Salim A, DuBose J, Teixeira PG, Chan LS, Margulies DR. Does age matter? The relationship between age and mortality in penetrating trauma. Injury. 2009;40(4):354-7.
15. Jacobs DG, Plaisier BR, Barie PS, Hammond JS, Holevar MR, Sinclair KE, et al. Practice management guidelines for geriatric trauma: the EAST Practice Management Guidelines Work Group. J Trauma. 2003;54(2):391-416.
16. Soreide K, Kruger AJ, Vardal AL, Ellingsen CL, Soreide E, Lossius HM. Epidemiology and contemporary patterns of trauma deaths: changing place, similar pace, older face. World J Surg. 2007;31(11):2092-103.
17. Champion HR, Copes WS, Sacco WJ, Lawnick MM, Keast SL, Bain LW Jr, Flanagan ME, Frey CF. The Major Trauma Outcome Study: establishing national norms for trauma care. J Trauma. 1990;30(11):1356-65.
18. https://www.facs.org/quality-programs/trauma/ntdb/docpub
19. Bauer R, Steiner M, Kisser R, Macey SM, Thayer D. Accidents and injuries in the EU. Results of the EuroSafe Reports. Bundesgesundheitsblatt Gesundheitsforschung Gesundheitsschutz. 2014;57(6):673-80.
20. Hadžiahmetović Z, Dedović-Halibegović G. The strategy of damage control surgery in treatment of life threatenig - traumatized patients. Medical Journal. 2015;21(3):200-204.
21. National Trauma Data Bank Annual Report 2016. Available at : https://www.facs.org
22. Nikolić S, Micić J, Mihailović Z. Correlation between survival time and severity of injuries in fatal injuries in traffic accidents. Srp Arh Celok Lek. 2001;129(11-12):291-5.
23. Campbell-Furtick M, Moore BJ, Overton TL, Laureano Phillips J, Simon KJ, Gandhi RR, et al. Post-trauma mortality increase at age 60: a cutoff for defining elderly? AM J Surg. 2016;212(4):781-785.
24. Braun BJ, Holstein J, Fritz T, Veith NT, Herath S, Morsdorf P, Pohlemann T. Polytrauma in the elderly: a review. EFORT Open Rev. 2016;1(5):146-151.
25. Alberdi F, Garcia I, Atutxa L, Zabarte M. Trauma and Neurointensive Care Work Group of the SEMICYUC. Epidemiology of severe trauma. Med Intensiva. 2014;38(9):580-8.
26. TraumaRegister DGU. 20 years TraumaRegister DGU: development, aims and structure. Injury. 2014t;45(Suppl 3):S6-S13.

Reprint requests and correspondence:
Amela Tuco, MD
Clinic of Emergency Medicine
Clinical Center University of Sarajevo
Bolnička 25, 71000 Sarajevo
Bosnia and Herzegovina
Email: amelatuco@gmail.com

INSTRUCTIONS TO AUTHORS

Journal **"Medical Journal"** publishes original research articles, professional, review and educative articles, case reports, criticism, reports, professional news, in the fields of all medical disciplines. Articles are written in-extenso in English, with the abstract and the title in English and Bosnian/Croatian/Serbian language.
Authors take responsibility for all the statements and attitudes in their articles. If article was written by several authors, it is necessary to provide full contact details (telephone numbers and email addresses) of the corresponding author for the cooperation during preparation of the text to be published.
Authors should indicate whether the procedures carried out on humans were in accordance with the ethical standards of medical deontology and Declaration of Helsinki.
Articles that contain results of animal studies will only be accepted for publication if it is made clear that ethics standard were applied.
Measurements should be expressed in units, according to the rules of the SI System.

Manuscript submission should be sent to Editorial Board and addressed to:
"MEDICINSKI ŽURNAL"
Disciplina za nauku i nastavu Kliničkog centra Univerziteta u Sarajevu
Bolnička 25
71000 Sarajevo
Bosna i Hercegovina
e-mail: institutnir@bih.net.ba; bibliotekanir@kcus.ba

COVER LETTER

Apart from the manuscript, the authors should enclose a cover letter, with the signed statements of all authors, to the Editorial Board of "Medical Journal" stating that:
1. the work has not been published or accepted for publication previously in another journal,
2. the work is in accordance with the ethical committee standards,
3. the work, accepted for publication, becomes ownership of "Medical Journal".

PREPARATION OF MANUSCRIPT

Article should be no longer than 10 computer pages, including figures, graphs, tables and references. The article may be submitted as a CD disk (Word Windows), or e-mail.
Spacing: 1,5: left margin: 2,5 cm; right margin: 2,5 cm; top and bottom margin: 2,5 cm.
Graphs, tables, figures and drawings should be incorporated in the text, precisely in the text, where these will be published, regardless of the program in which they are prepared. Articles are written in-extenso in English language.
The manuscript should be submitted on a good quality CD disc, or by e-mail, together with two printed copies (if possible). Sent CD disks will not be returned to the authors.

ARTICLE CONTAINS:

TITLE OF THE ARTICLE IN ENGLISH LANGUAGE
TITLE OF THE ARTICLE IN BOSNIAN/SERBIAN/CROATIAN (B/S/C) LANGUAGE
First and last name of the author/co-author(s)

Name and address of the institution in which author/co-authors is employed (same for all authors) in B/S/C and English language as well as the address of corresponding author at the end of the article.

Summary in B/S/C language with the precise translation in English. Abstract of approximately 200-250 words should concisely describe the contents of the article.
Key words (in B/S/C and in English language): up to five words should be listed under the Abstract.

ARTICLE BODY

The main body of the article should be systematically ordered under the following headings:
- **INTRODUCTION**
- **MATERIALS AND METHODS**
- **RESULTS**
- **DISCUSSION**

- **CONCLUSION**
- **REFERENCES**

INTRODUCTION

Introduction is a concise, short part of the article, and it contains purpose of the article relating to other published articles with the same topic. It is necessary to quote the main problem, aim of investigation, and/or main hypothesis which is investigated.

MATERIALS AND METHODS

This part should contain description of original or modification of known methods. If there is a method that has previously been described, it would be sufficient to include it in the reference list. In clinical and epidemiological studies the following should be described: sample, protocol and type of clinical investigation, place and period of investigation. Main characteristics of investigation should be described (randomization, double-blind test, cross test, placebo test), standard values for tests, time framework (prospective, retrospective study), selection and number of patients – criteria for inclusion and exclusion from the study.

RESULTS

Main results of investigation and level of its statistical significance should be quoted. Results should be presented in tables, graphs, figures, and directly incorporated in the text, at the exact place, with ordinal number and concise heading. Table should have at least two columns and explanation; figures clean and contrasted, graphs clear, with visible text and explanation.

DISCUSSION

Discussion is concise and refers to own results, in comparison with the other authors' results. Citation of references should follow Vancouver rules. Discussion should be concluded by the confirmation of the stated aim or hypothesis, or by its negation.

CONCLUSION

Conclusion should be concise and should contain most important facts, which were obtained during investigation and its eventual clinical application, as well as the additional studies for the completed application. Affirmative and negative conclusions should be stated.

REFERENCES – Instructions for writing references

References should follow the format of the requirements of **Vancouver rules**.

Every statement, knowledge and idea should be confirmed by reference. Each reference in the text is given its own sentence case in Arabic number in parenthesis at the end of the sentence according to the order of entering. Every further referring to the same reference, number of the first referring in the text should be stated. References are to be placed at the end of the article, and are to be numbered by ordinal numbers in the order of entering in the text (entering reference number). Journal's title is abbreviated using Index Medicus abbreviations. The names of the first six authors of each reference item should be provided, followed by "et al.".

It is very important to properly design references according to instructions that may be downloaded from addresses National Library of Medicine Citing Medicine http://www.ncbi.nlm.nih.gov/books/bv.fcg?rid=citmed.TOC&depth=2,

or International Committee of Medical Journal Editors Uniform Requirements for Manuscripts Submitted to Biomedical Journals: Sample References http://www.nlm.nih.gov/bsd/uniform_requirements.html.

UPUTSTVA AUTORIMA

Časopis "Medicinski žurnal" objavljuje originalne naučne radove, stručne, pregledne i edukativne, prikaze slučajeva, recenzije, saopćenja, stručne obavijesti i drugo iz područja svih medicinskih disciplina. Rad *in-extenso* (cjelokupan) piše se na engleskom jeziku, uz sažetak i naslov rada koji uz engleski trebaju biti napisani i na našim jezicima (bosanski, hrvatski i srpski). Autori su odgovorni za sve navode i stavove u njihovim radovima. Ukoliko je rad pisalo više autora, potrebno je navesti tačnu adresu (uz telefonski broj i e-mail adresu) onog autora s kojim će uredništvo sarađivati pri uređenju teksta za objavljivanje.
Ukoliko su u radu prikazana istraživanja na ljudima, mora se navesti da su provedena u skladu s načelima medicinske deontologije i Deklaracije iz Helsinkija.
Ukoliko su u radu prikazana istraživanja na životinjama, mora se navesti da su provedena u skladu s etičkim načelima. Prilikom navođenja mjernih jedinica, treba poštovati pravila navedena u SI sistemu.
Radovi se šalju Redakciji na adresu:

"MEDICINSKI ŽURNAL"
Disciplina za nauku i nastavu Kliničkog centra Univerziteta u Sarajevu
Bolnička 25
71000 Sarajevo
Bosna i Hercegovina
e-mail: institutnir@bih.net.ba; bibliotekanir@kcus.ba

POPRATNO PISMO

Uz svoj rad, autori su dužni Redakciji "Medicinskog žurnala" dostaviti popratno pismo, koje sadržava vlastoručno potpisanu izjavu svih autora:

1. da navedeni rad nije objavljen ili primljen za objavljivanje u nekom drugom časopisu,
2. da je istraživanje odobrila Etička komisija,
3. da prihvaćeni rad postaje vlasništvo "Medicinskog žurnala".

OPSEG I OBLIK RUKOPISA

Radovi ne smiju biti duži od deset stranica na računaru, ubrajajući slike, grafikone, tabele i literaturu. CD zapis teksta je obavezan (Word of Windows), ili e-mail.
Prored: 1,5: lijeva margina: 2,5 cm; desna margina: 2,5 cm; gornja i donja margina: 2,5 cm.
Grafikone, tabele, slike i crteže unijeti/staviti u tekst rada, tamo gdje im je mjesto, bez obzira u kojem programu su rađene. Cijeli rad obavezno napisati na engleskom jeziku, a sažetak i naslov još i na našem jeziku.
Rad se dostavlja na CD-u, i/ili e-mailom, uz dva štampana primjerka (ako je moguće). CD se ne vraća.

RAD SADRŽI:

NASLOV RADA NA ENGLESKOM JEZIKU
NASLOV RADA NA NAŠEM JEZIKU

Ime i prezime autora i koautora

Naziv i puna adresa institucije u kojoj je autor-koautor/i zaposlen/i (jednako za sve autore), na engleskom jeziku, te na kraju rada navedena adresa kontakt-autora.

Sažetak na našem jeziku, kao i na engleskom - max. 200–250 riječi, s najznačajnijim činjenicama i podatcima iz kojih se može dobiti uvid u kompletan rad.
Ključne riječi - Key words, na našem jeziku i na engleskom, ukupno do pet riječi, navode se ispod Sažetka, odnosno Abstracta.

SADRŽAJ

Sadržaj rada mora biti sistematično i strukturno pripremljen i podijeljen u poglavlja i to:

- **UVOD**
- **MATERIJAL I METODE**
- **REZULTATI**
- **DISKUSIJA**
- **ZAKLJUČAK**
- **LITERATURA**

UVOD

Uvod je kratak, koncizan dio rada i u njemu se navodi svrha rada u odnosu na druge objavljene radove sa istom tematikom. Potrebno je navesti glavni problem, cilj istraživanja i/ili glavnu hipotezu koja se provjerava.

MATERIJAL I METODE

Potrebno je da sadrži opis originalnih ili modifikaciju poznatih metoda. Ukoliko se radi o ranije opisanoj metodi dovoljno je dati reference u literaturi. U kliničko-epidemiološkim studijama opisuju se: uzorak, protokol i tip kliničkog istraživanja, mjesto i vrijeme istraživanja. Potrebno je opisati glavne karakteristike istraživanja (npr. randomizacija, dvostruko slijepi pokus, unakrsno testiranje, testiranje s placebom itd.), standardne vrijednosti za testove, vremenski odnos (prospektivna, retrospektivna studija), izbor i broj ispitanika – kriterije za uključivanje i isključivanje u istraživanje.

REZULTATI

Navode se glavni rezultati istraživanja i nivo njihove statističke značajnosti. Rezultati se prikazuju tabelarno, grafički, slikom i direktno se unose u tekst gdje im je mjesto, s rednim brojem i konciznim naslovom. Tabela treba imati najmanje dva stupca s obrazloženjem što prikazuje; slika čista i kontrastna, a grafikon jasan, s vidljivim tekstom i obrazloženjem.

DISKUSIJA

Piše se koncizno i odnosi se prvenstveno na vlastite rezultate, a potom se nastavlja upoređivanje vlastitih rezultata s rezultatima drugih autora, pri čemu se citiranje literature navodi po važećim Vankuverskim pravilima. Diskusija se završava potvrdom zadatog cilja ili hipoteze, odnosno njihovim negiranjem.

ZAKLJUČAK

Treba da bude kratak, da sadrži najbitnije činjenice do kojih se došlo u radu tokom istraživanja i njihovu eventualnu kliničku primjenu, kao i potrebne dodatne studije za potpuniju aplikaciju. Obavezno navesti i afirmativne i negirajuće zaključke.

LITERATURA - Upute za citiranje - pisanje literature

Literatura se obavezno citira po **Vankuverskim pravilima**.

Svaku tvrdnju, saznanje ili misao treba potvrditi referencom. Reference u tekstu treba označiti po redoslijedu unošenja arapskim brojevima u zagradi na kraju rečenice. Ukoliko se kasnije u tekstu pozivamo na istu referencu, navodimo broj koji je referenca dobila prilikom prvog unošenja/pominjanja u tekstu. Literatura se popisuje na kraju rada, rednim brojevima pod kojim su reference unesene u tekst (ulazni broj reference), a naslov časopisa se skraćuje po pravilima koje određuje Index Medicus. Ukoliko je citirani rad napisalo više autora, navodi se prvih šest i doda "et al.".

Vrlo je važno ispravno oblikovati reference prema uputama koje se mogu preuzeti na adresama National Library of Medicine Citing Medicine http://www.ncbi.nlm.nih.gov/books/bv.fcgi?rid=citmed.TOC&depth=2 , ili International Committee of Medical Journal Editors Uniform Requirements for Manuscripts Submitted to Biomedical Journals:

Sample References http://www.nlm.nih.gov/bsd/uniform_requirements.html.